# Role of business formation and exit for unemployment and vacancy dynamics

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**Abstract**

Vacancies comove positively with establishment entry and negatively with exit; states with higher exit rates accumulate larger labor market gaps during recessions. We develop a model of endogenous business formation and exit with labor market frictions to explain these patterns. Monopolistically competitive firms rent labor from specialized recruiters and face exit risk from product obsolescence and stochastic fixed costs. Both business creation and vacancy posting require sunk costs, making them partially predetermined. Three results follow. First, the business formation margin affects labor markets through love of variety and the endogenous exit rate. Second, because vacancy posting is a sunk cost, exiting firms simultaneously eliminate jobs and posted vacancies, so destruction shocks shift the Beveridge curve outward–even if vacancies can respond elastically. Third, match separation shocks destroy only filled positions; firms optimally repost, generating positive unemployment-vacancy comovement. Both technology and destruction shocks produce second-round amplification through further exit and reduced product variety. We estimate by Bayesian methods targeting conditional moments—impulse responses to a Bartik-identified destruction shock—and unconditional labor market moments. Technology and destruction shocks account for the predominant share of labor market volatility.

## 1. Introduction

Labor market variables do not recover at equal speed, and the disparity does not just depend on the severity of the initial shock. The 2020 recession produced the highest peak unemployment rate of any postwar episode — 13.0 percent — yet labor markets had recovered within two years,

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accumulating only 26.6 percentage-point quarters of total unemployment excess over twenty quarters from recession onset. The 2008–09 recession generated a smaller peak unemployment rate of 9.9 percent yet produced 76.6 percentage-point quarters of cumulative damage and a recovery stretching across six years.

The standard Diamond–Mortensen–Pissarides (DMP) framework, in which all separations are identical and firms repost vacancies freely after any job loss, offers no structural account of why two recessions of comparable peak unemployment can generate such different cumulative damage: the composition of job loss — not just its volume — has no influence on persistence. We argue that a key missing margin is firm destruction — the permanent elimination of business opportunities and their associated vacancy slots — which ran at 7.8 percent of base employment in 2008–09 but only 2.9 percent in 2020.[2] When a firm exits, its vacancies vanish permanently and must be rebuilt from scratch at sunk cost. Recovery requires not merely a rebound in profitability but a costly reconstruction of the vacancy stock, generating a persistent outward shift of the Beveridge curve. Provided vacancy continuation is sufficiently cheap relative to initial creation, surviving firms repost after separation, so the aggregate vacancy stock partially self-corrects even while unemployment remains elevated, producing a transitory displacement along the Beveridge curve rather than a persistent outward shift.

Coles and Moghaddasi Kelishomi (2018) show that extending the DMP model with finitely elastic vacancy creation enables separation shocks to generate a downward-sloping Beveridge curve and, jointly with technology shocks, fit a broad set of labor market moments well. But CK assume a single separation shock, conflating firm destruction and match dissolution into one margin, and separations are entirely exogenous. Gabrovski and Silva (2025) build on CK by distinguishing between destruction of a product line $\delta$ and breakdown of a match $s$, and show that under technology shocks alone, for empirically relevant values of $\delta$, the model generates mild unemployment–labor productivity comovement and matches other key labor market moments, including the Beveridge curve. In both frameworks, sunk entry costs make the vacancy stock predetermined: $v_{t-1}$ enters the equilibrium alongside $u_t$, and surviving firms repost vacancies freely following match dissolution, substantially weakening the link between unemployment and

[2]Financial frictions, nominal rigidities, and other margins absent from the benchmark model have each received substantial attention in the literature; we focus on firm destruction and its interaction with endogenous business formation.

contemporaneous productivity.

This paper opens additional channels related to business formation and exit. First, we introduce love-of-variety preferences over differentiated product lines, so that the number of active product lines $N_t$ becomes a third independent state variable: greater product diversity raises the return to posting vacancies, while greater employment raises the resources available for product-line formation. The two-way feedback between employment and product variety constitutes an aggregate demand externality in the sense of Schaal and Taschereau-Dumouchel (2019): each new product line raises the welfare-based relative price $\rho(N_t)$, which enters recruiter compensation $w_t^{int} = \rho(N_t)z_t/\mu$ directly and raises the return to vacancy creation, while greater employment expands the resources available for new product-line formation, closing the loop. We segment the product and labor markets so that producers can employ a mass of workers, and thereby do not mechanically link employment and product lines one-for-one as Schaal and Taschereau-Dumouchel (2019). Instead, $u_t$, $v_{t-1}$, and $N_t$ are jointly determined in equilibrium without an accounting restriction.[3] Second, we endogenize firm exit, so that the aggregate destruction rate $\delta_t^e$ responds to aggregate productivity and separation conditions: a recession that reduces profitability or raises separation rates induces higher exit, further depleting the vacancy stock and product variety and further suppressing job creation. Together these extensions sharpen the asymmetry between $\delta$ and $s$ shocks: a firm-destruction shock simultaneously damages all three state variables — raising $u_t$, depleting $v_{t-1}$, and eliminating product lines that would otherwise sustain vacancy creation incentives — while a match-separation shock raises $u_t$ but does not directly affect $v_{t-1}$ or $N_t$. We argue quantitatively that this three-channel amplification helps account for the cross-recession gap in cumulative labor market damage, and we identify the key structural parameters using Bayesian simulated method of moments on unconditional moments, extended with Bayesian impulse response matching to an externally identified firm-destruction shock.

[3]In Schaal and Taschereau-Dumouchel (2019), each matched worker–firm pair produces one variety of an intermediate good, so employment and the range of goods fluctuate one-for-one. The demand externality therefore operates simultaneously through both matching and variety. In our setting, firms employ multiple workers per product line, so employment adds output within existing varieties while the variety externality operates separately through firm entry, governed by the business-formation Euler equation. This separation is what allows $u_t$, $v_{t-1}$, and $N_t$ to be independent state variables.

### 1.1. *Motivating reduced form evidence*

Three empirical facts motivate treating firm destruction as a distinct and quantitatively important margin. *First, establishment closings and openings constitute a large and cyclically sensitive component of gross job flows.* Table 1 extends Table 2 in Jaimovich and Floetotto (2008) through 2024Q4 using Business Economics Database (BED) data. Columns 1–2 of Table 1 report the average fraction of quarterly gross job gains (losses) accounted for by opening (closing) establishments. Columns 3 and 4 report the ratio of the cyclical standard deviation of entry/exit flows to that of total flows, using the Hamilton (2018) regression filter ($h = 8, p = 4$) applied to raw levels. Columns 5 and 6 repeat the calculation with the HP filter ($\lambda = 1600$), which closely reproduces Jaimovich and Floetotto (2008)'s benchmark of approximately 0.33 in the original 1992–2006 sample.

Columns 1 and 2 show that closing establishments account for 19.3 percent of gross job losses on average across all private-sector supersectors — a share that is remarkably stable across industries ranging from construction (20.5 percent) to financial activities (24.6 percent). The relative standard deviation columns show that closing-establishment flows are meaningfully cyclical under both Hamilton and HP filters, with the total-private closing share moving at 0.152 times the cyclical variation of unemployment under the Hamilton filter. Translating these shares into rates, the quarterly establishment exit rate averages 0.93 percent of base employment against a total quarterly separation rate of 9.3 percent: most separations are therefore not firm exits, which makes the $\delta$-vs-$s$ decomposition empirically tractable — the two margins are separately measurable yet distinct enough that conflating them materially distorts model predictions.

| | Mean share | | Rel. std dev (Ham) | | Rel. std dev (HP) | |
|---|---|---|---|---|---|---|
| Industry | $\bar{g}^O$ | $\bar{g}^C$ | $\sigma^O/\sigma$ | $\sigma^C/\sigma$ | $\sigma^O/\sigma$ | $\sigma^C/\sigma$ |
| **Total private** | 0.199 | 0.193 | 0.227 | 0.152 | 0.206 | 0.132 |
| Natural resources & mining | 0.163 | 0.156 | 0.281 | 0.177 | 0.261 | 0.176 |
| Construction | 0.208 | 0.205 | 0.188 | 0.185 | 0.209 | 0.165 |
| Manufacturing | 0.123 | 0.137 | 0.214 | 0.109 | 0.219 | 0.093 |
| Wholesale trade | 0.203 | 0.226 | 0.251 | 0.184 | 0.260 | 0.143 |
| Retail trade | 0.169 | 0.152 | 0.202 | 0.142 | 0.210 | 0.124 |
| Transportation & utilities | 0.163 | 0.169 | 0.236 | 0.118 | 0.194 | 0.097 |
| Information | 0.206 | 0.209 | 0.284 | 0.310 | 0.301 | 0.269 |
| Financial activities | 0.235 | 0.246 | 0.343 | 0.323 | 0.416 | 0.305 |
| Prof. & business services | 0.205 | 0.211 | 0.242 | 0.156 | 0.236 | 0.133 |
| Education & health services | 0.203 | 0.211 | 0.307 | 0.151 | 0.246 | 0.125 |
| Leisure & hospitality | 0.241 | 0.196 | 0.173 | 0.168 | 0.174 | 0.158 |
| Other services | 0.225 | 0.223 | 0.427 | 0.286 | 0.415 | 0.270 |

*Notes.* $\bar{g}^O$ ($\bar{g}^C$): mean share of gross job gains (losses) from opening (closing) establishments, $\sum G_O/\sum G$. Cols 3–4: relative std dev $\sigma^O/\sigma$ ($\sigma^C/\sigma$) via Hamilton (2018) filter ($h$=8, $p$=4), raw levels; robust to GFC/Covid outliers. Cols 5–6: same statistic via HP filter ($\lambda$=1600).

Table 1: Job Gains and Losses from Opening and Closing Establishments (*Extended to 2024Q4 (BED end)*) *Source: BLS Business Employment Dynamics (BED), quarterly, 12 Bartik supersectors. Replication and extension of Jaimovich & Floetotto (2008, JME) Table 2.*

*Second, cumulative establishment exit predicts the severity and persistence of labor market disruption across recessions.* Figure 1 plots the permanence-adjusted establishment exit rate against cumulative unemployment and vacancy gaps across the three post-1992 NBER recessions. The 2008–09 recession generated the largest exit rate (7.8% of base employment) and the largest cumulative unemployment gap (76.6 pp-quarters), reflecting both a deep trough and a protracted recovery. The 2001 recession, despite a more moderate exit rate (5.6%), produced the largest cumulative vacancy shortfall (25.7 pp-quarters) of the three episodes — exceeding even the GFC (18.8 pp-quarters) — driven by an unusually prolonged vacancy formation freeze following the collapse of investment demand after the dot-com bust, a pattern that likely reflects demand-side forces beyond establishment exit alone. The 2020 recession, with the smallest exit rate (2.9%), generated rapid recovery in both labor market margins and negligible cumulative vacancy shortfall (1.7 pp-quarters), consistent with a supply disruption that left the matching technology largely intact. Across these three episodes, larger establishment exit is associated with greater and more persistent labor market damage, particularly in the unemployment dimension.

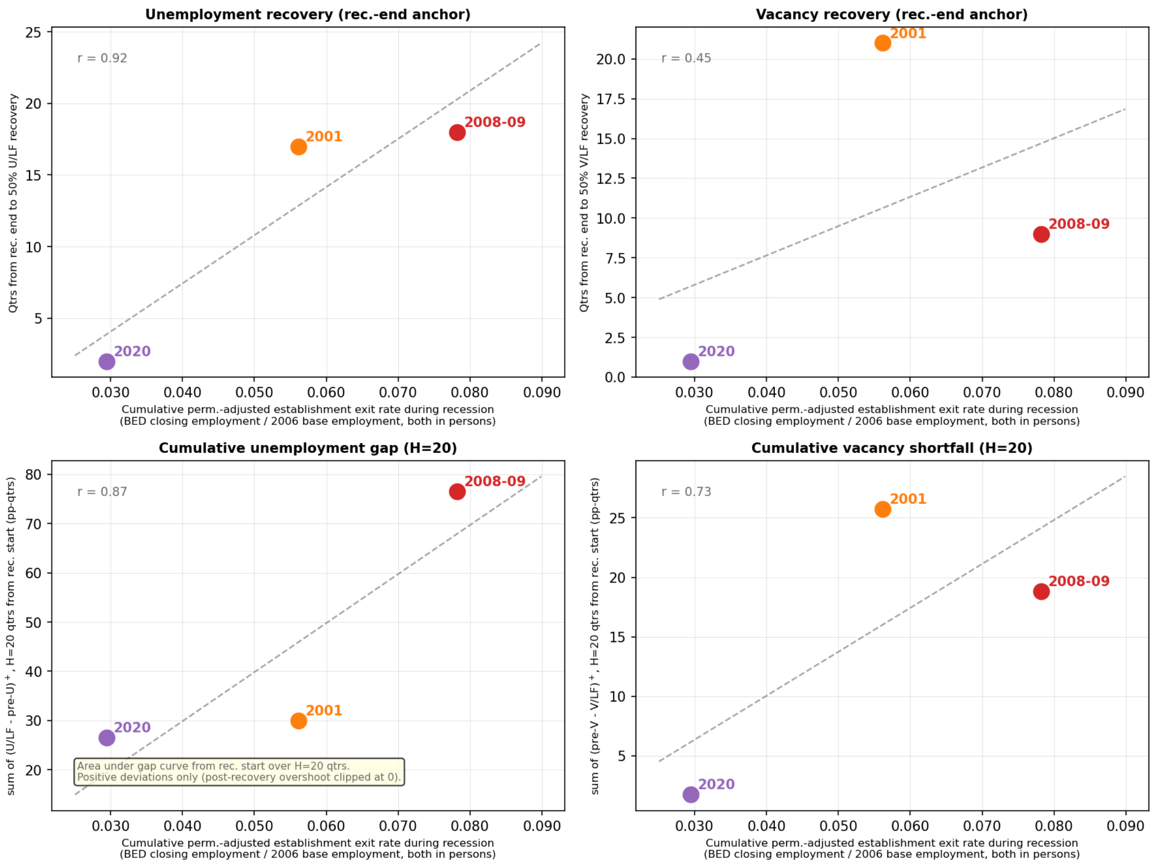


Figure 1: Relationship between cumulative establishment exit rates (BED Deaths) during recessions and labor market outcome measures. The top row shows quarters from the NBER recession end date until unemployment (left) and vacancy (right) rates recover 50% of the trough gap. The bottom row shows the cumulative unemployment gap (CUG, left) and cumulative vacancy shortfall (CVS, right), defined as the sum of positive deviations from the pre-recession baseline over $H = 20$ quarters from recession onset. Exit rates are constructed from BLS Business Employment Dynamics (BED) establishment deaths (establishments absent for four or more consecutive quarters), normalized by 2006 base-year employment.

*Third, states with structurally higher establishment exit rates accumulate larger labor market gaps during recessions.* Figure 2 plots each state's full-sample average Bartik-weighted establishment exit rate against its cumulative unemployment gap (CUG) and cumulative vacancy shortfall (CVS), both summed over 20 quarters from recession onset and averaged across the 2001, 2007–09, and 2020 recessions. The labor-force-weighted cross-sectional correlations are 0.65 (CUG) and 0.28 (CVS): states with structurally higher exit propensity accumulate substantially more total unemployment damage during downturns, with the vacancy pattern weaker but directionally consistent. The stronger CUG correlation is itself informative — it suggests that exit intensity shapes the *duration* of labor market distress more than the *depth* of the vacancy shortfall, consistent with the model's prediction that firm destruction generates persistent Beveridge curve dislocations rather than transitory displacements.[4] However, the cross-sectional pattern conflates structural differences in industrial composition with the causal effect of exit shocks: high-exit states may differ from low-exit states along dimensions — industry mix, institutional environment, average firm age — that independently predict labor market performance. To isolate causal transmission, we construct Bartik shift-share instruments that exploit within-state variation in exit exposure driven by national industry-level closing rates rather than local conditions, building separate instruments for $\delta$ from BED establishment closings and for $s$ from JOLTS layoff and discharge rates. We then estimate local projection impulse response functions to externally identify the dynamic transmission of each shock to state-level unemployment and vacancies, providing a quasi-experimental foundation for the structural parameters prior to Bayesian estimation.

[4]The weaker CVS correlation is also consistent with the model: the vacancy stock partially self-corrects through reposting by surviving firms, attenuating the cross-state vacancy signal relative to the unemployment signal.

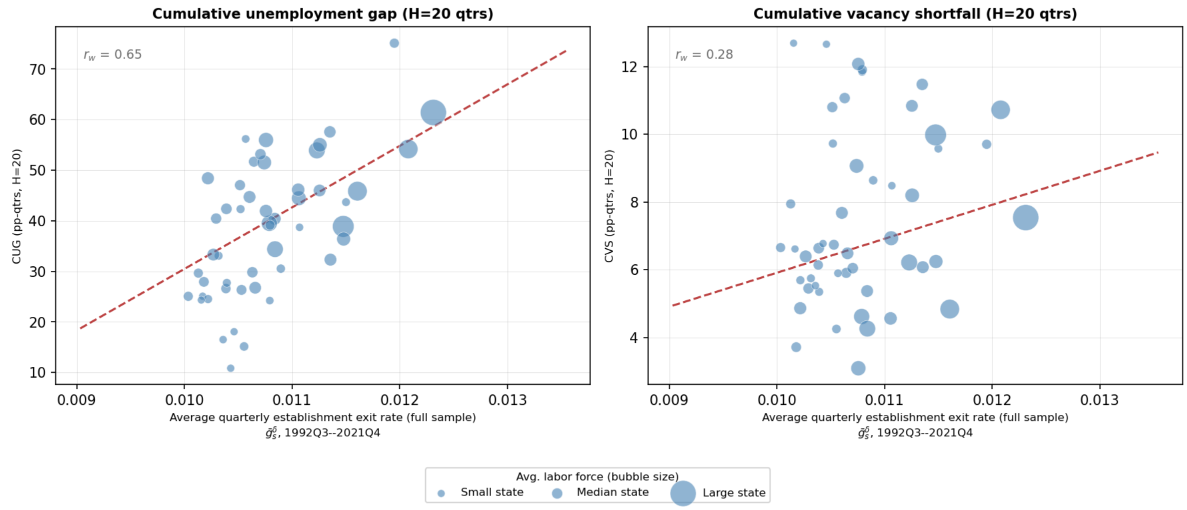


Figure 2: Cross-state relationship between average establishment exit rates and cumulative recession labor market damage. Each point is one of the 50 U.S. states; bubble size is proportional to average labor force. The horizontal axis is the full-sample time average of the state-level Bartik-weighted establishment exit rate ($\bar{g}^{\delta}_{s}$, 1992Q3–2021Q4), constructed from BLS Business Employment Dynamics (BED) data using 2006 base-year employment shares. The left panel shows the cumulative unemployment gap (CUG): the sum of quarterly deviations of the unemployment rate above its pre-recession average over $H = 20$ quarters from recession onset, in percentage-point-quarters. The right panel shows the analogous cumulative vacancy shortfall (CVS) for the vacancy rate. Both outcomes are averaged across the three post-1992 NBER recessions (2001, 2007–2009, 2020), weighted by state labor force at recession onset. The regression line is estimated by WLS with state labor force weights; $r_w$ denotes the labor-force-weighted correlation.

Following Goldsmith-Pinkham, Sorkin, and Swift (2020), we bring quasi-experimental discipline to the $\delta$ transmission channel using Bartik shift-share instruments constructed from BED establishment deaths — the subset of closings absent from the UI payroll base for four or more consecutive quarters, aligning the instrument directly with the model's concept of irreversible product-line destruction. The instrument weights leave-one-out national industry-level death rates by each state's base-year (2006) industry employment shares. The identifying variation is geographic: a state with high pre-recession exposure to industries experiencing national waves of establishment closings — manufacturing in 2008–09, for instance — faces a large exit shock driven by national industry conditions rather than local demand, and can be compared to low-exposure states in the same quarter to isolate the causal labor market effect of firm destruction. The instrument is residualized on aggregate productivity growth to address cross-instrument correlation arising mechanically during aggregate downturns.

We estimate panel local projections (Jordà, 2005) with state and time fixed effects, standard errors clustered at the state level, and outcomes capped at 2019Q4 to exclude the Covid episode. The $\delta$ instrument generates a persistent decline in both unemployment and vacancies: the unemployment response peaks at $+1.57$ percentage points at $h = 19$ quarters and remains significant throughout $h = 0$–$20$; the vacancy response falls monotonically, reaching $-0.426$ log points at $h = 14$ and remaining significant from $h = 1$ onward. Both patterns are consistent with the model's prediction that firm destruction generates a persistent outward shift of the Beveridge curve rather than a transitory displacement along it. The estimated $\delta \to v$ and $\delta \to u$ coefficients approximate the aggregate transmission elasticities and provide external moment conditions for Bayesian impulse response matching, combined with unconditional moments in the Bayesian simulated method of moments estimation described in Section 5.[5]

The model is estimated by Bayesian simulated method of moments, combining two sets of moment conditions. The first is closely related to Coles and Moghaddasi Kelishomi (2018): unconditional moments of unemployment, vacancies, labor productivity, and aggregate job finding and separation rates, augmented with the mean and relative volatility of the permanent establishment exit rate (BED Deaths), which disciplines the level and cyclical variation of the destruction process directly rather than inferring it residually from labor market outcomes. The second set consists of the empirically identified $\delta \to u$ and $\delta \to v$ impulse responses from the Bartik local projections, matched via Bayesian impulse response matching with inverse-variance weights. The two sets are complementary: unconditional moments pin down the destruction process and the steady-state labor market, while the conditional moments discipline the structural transmission parameters governing how costly and slow vacancy replacement is following a firm destruction shock.[6] Quantitative results and the two-model counterfactual — comparing the full model's implied recovery paths to those of a Coles-Kelishomi limiting case that abstracts from variety effects and endogenous exit — are reported in Section 5.

[5]The $\delta$ unemployment response concentrates almost entirely in states experiencing tight financial conditions: at average financial conditions the point estimate is near zero and insignificant at all horizons beyond $h = 0$. This state-dependence in the unemployment channel, absent from the baseline model, is consistent with financial amplification of firm destruction and is left for future work; the vacancy channel, which enters the structural estimation as a moment condition, does not exhibit the same state-dependence.

[6]The unconditional moments identify the level and volatility of the $\delta$ process, while the conditional moments identify the transmission parameters $\xi$ and $x_m$ governing the vacancy creation cost distribution. A model with the same $\delta$ process but different $\xi$ would match the former and fail the latter.

This paper contributes to four strands of literature. The first is search-and-matching models with finitely elastic vacancy creation, where our model builds most directly on Coles and Moghaddasi Kelishomi (2018) and Gabrovski and Silva (2025); Broer, Druedahl, Harmenberg, and Öberg (2025) independently combine finitely elastic vacancy creation with endogenous separation but abstract from product variety and the separate destruction margin. The second is endogenous product variety and firm dynamics in business cycles, where we build on Bilbiie, Ghironi, and Melitz (2012) and depart from the one-firm-one-worker structure of Shao and Silos (2013); Colciago, Fasani, and Rossi (2021) combine endogenous entry with labor market frictions in a related framework but maintain infinitely elastic vacancy creation and abstract from the destruction margin. The third is Bartik shift-share identification in local labor markets (Goldsmith-Pinkham, Sorkin, and Swift, 2020; Nakamura and Steinsson, 2014), discussed alongside the instrument construction in Section 4. The fourth is Bayesian structural estimation of search models, extending the simulated method of moments approach of Coles and Moghaddasi Kelishomi (2018) to incorporate impulse response matching as an additional set of moment conditions (Christiano, Eichenbaum, and Trabandt, 2016). The remainder of the paper is organized as follows. Section 4 presents the motivating evidence and Bartik local projections. Section 2 develops the model environment. Section 3 derives the equilibrium, establishes the $\delta$-vs-$s$ asymmetry formally, and characterizes steady-state properties. Section 5 presents calibration, estimation, and the two-model counterfactual. Section **??** concludes.

## 2. Environment

Time is discrete and infinite. The economy is populated by households, intermediaries (recruiters), and retailers. Households and recruiters each have unit mass; the mass of retailers is endogenous, each supplying one product line. A household consists of a large family of workers who are either employed or unemployed, as in Merz (1995). Employed members supply one unit of labor inelastically. The household consumes and purchases equity. A unit measure of recruiters can each create one vacancy at a time, at a sunk cost drawn from a known distribution, with each vacancy specialized to a single product line. Retailers rent labor services from matched recruiters and produce the specialized variety.

Each period a number $m(u_t, v_t)$ of recruiter–worker pairs form, given unemployment $u_t$ and vacancies $v_t$. We adopt the Cobb-Douglas matching function $m(u, v) = Au^{\eta_L} v^{1-\eta_L}$, where $\eta_L$

is the elasticity of matches with respect to unemployment (Petrongolo and Pissarides, 2001). Constant returns to scale implies that the job-finding rate $f(\theta) = A\theta^{1-\eta_L}$ and the vacancy-filling rate $q(\theta) = A\theta^{-\eta_L}$ depend only on market tightness $\theta \equiv v/u$.

### *2.1. Preferences*

We develop the model with general homothetic preferences as in Bilbiie, Ghironi, and Melitz (2012), then apply specific functional forms in the quantitative analysis. The household consumes a basket $c_t$, a homothetic aggregate over a continuum of goods $\Omega$. The welfare-based price index $P_t$ takes only individual prices as arguments, and demand for variety $j$ satisfies

$$c_{jt} = c_t \partial P_t / \partial p_{jt}. \tag{1}$$

The symmetric price elasticity of demand is in general a function of the measure $N_t$ of goods: $\varepsilon(N_t) = -\frac{\partial \log c_j}{\partial \log p_j}$. The welfare gain from an additional product variety is captured by the relative price and its elasticity:

$$\rho_j = \rho(N) \equiv \frac{p_j}{P}, \quad \zeta(N) = \frac{\rho'(N)}{\rho(N)} N.$$

The relative price rises with variety: with more products available at the same nominal price, the welfare-based cost of living falls, so $\rho$ increases. Equivalently, spreading consumption across more goods benefits the household. The two statistics $\varepsilon(N_t)$ and $\rho(N_t)$ completely characterize symmetric homothetic preferences in the model.

Under standard Dixit–Stiglitz preferences (DS-CES), $\varepsilon$ measures both the taste for diversity and the elasticity of substitution, so both the markup and the variety benefit are independent of the number of producers: $\zeta(N) = \mu(N) - 1 = 1/(\varepsilon - 1)$. The second specification is generalized CES following Benassy (1996), with consumption basket $N^{\zeta - 1/(\varepsilon-1)} \left( \int c_j^{(\varepsilon-1)/\varepsilon} dj \right)^{\varepsilon/(\varepsilon-1)}$; this disentangles the taste for variety $\zeta$ from the elasticity of substitution $\varepsilon$. The third specification uses the translog expenditure function of Feenstra (2003), under which $\varepsilon(N) = 1 + \eta N$, so goods become more substitutable as variety increases. Table 2 summarizes key statistics for all three specifications.

| | DS-CES | General CES | Translog |
|---|---|---|---|
| Consumption index | $\left(\int c_j^{(\varepsilon-1)/\varepsilon} dj\right)^{\varepsilon/(\varepsilon-1)}$ | $N^{\zeta-1/(\varepsilon-1)} \left(\int c_j^{(\varepsilon-1)/\varepsilon} dj\right)^{\varepsilon/(\varepsilon-1)}$ | – |
| Price index | $\left(\int p_j^{1-\varepsilon} dj\right)^{1/(1-\varepsilon)}$ | $N^{-\zeta+1/(\varepsilon-1)} \left(\int p_j^{1-\varepsilon} dj\right)^{1/(1-\varepsilon)}$ | – |
| Markup $\mu(N)$ | $\frac{\varepsilon}{\varepsilon-1}$ | $\frac{\varepsilon}{\varepsilon-1}$ | $1+\frac{1}{\eta N}$ |
| Relative price $\rho(N)$ | $N^{\frac{1}{\varepsilon-1}}$ | $N^{\zeta}$ | $\exp\left\{-\frac{1}{2}\frac{\tilde{N}-N}{\eta\tilde{N}N}\right\}$ |
| Benefit of variety $\zeta(N)$ | $\mu-1$ | $\zeta$ | $\frac{1}{2\eta N}=\frac{\mu(N)-1}{2}$ |

Table 2: Preference specifications: indices, markup, relative price, and benefit of variety

### *2.2. Laws of motion for firms, unemployment, and vacancies*

At the start of period $t$, exogenous destruction $\delta_{t-1}$, realized at the end of $t-1$, and the endogenous continuation-cost draw together determine which product lines are active. Specifically, a product line survives into period $t$ if it was not destroyed exogenously at the end of $t-1$ (probability $1-\delta_{t-1}$) *and* its owner draws a continuation cost below the equilibrium cutoff $\chi_t^c$ at the start of $t$ (probability $F(\chi_t^c)$). Both events occur before production and matching, so $u_t$ is unemployment *after* firm exit but *before* matching. The law of motion for active product lines is

$$N_t = (1-\delta_{t-1})F(x_t^c)(N_{t-1}+N_{t-1}^e) \tag{2}$$

where $N_{t-1}^e$ denotes last-period entrants. The entry lag means a fraction of entrants are destroyed before they produce. The total exit rate combining exogenous destruction and endogenous exit is

$$\delta_{e,t} \equiv 1-(1-\delta_{t-1})F(\chi_t^c), \tag{3}$$

where $\delta_t$ denotes the exogenous AR(1) destruction shock and $\delta_{e,t}$ denotes the total exit rate; we maintain this distinction throughout. The aggregate separation rate flowing into pre-matching unemployment $u_t$ is

$$\tau_t = \delta_{e,t} + s_{t-1}(1-\delta_{e,t}), \tag{4}$$

where $s_{t-1}$ is the match-separation shock that realized at end of $t-1$. The unemployment law of motion is

$$u_t = (1-f(\theta_{t-1}))u_{t-1} + \tau_t\big[(1-u_{t-1}) + f(\theta_{t-1})u_{t-1}\big] \tag{5}$$

The first term is last period's unemployed who did not find a job; the second is the fraction $\tau_t$ of last period's employed (including those who matched in $t-1$) who separate before production in $t$. This has the same form as the textbook model, with the total separation rate $\tau_t$ replacing the constant separation rate of DMP.

Sunk vacancy creation costs render vacancies partially predetermined: current vacancies evolve as

$$v_t = (1-\delta_{e,t})[(1-q(\theta_{t-1}))v_{t-1} + s_{t-1}(1-u_{t-1})] + e_t \tag{6}$$

Current vacancies are the sum of three components: (i) unmatched vacancies from last period that survived product destruction; (ii) filled jobs that experienced a match separation at end of $t-1$ but whose firm survived both margins, reposted at no additional sunk cost; and (iii) vacancies created by new entrants $e_t$, who enter at the start of $t$ before matching. Note that $(1-\delta_{e,t})$ multiplies both (i) and (ii): only firms surviving both exogenous destruction and the continuation-cost draw contribute to either term, so a shock to $\delta_t$ collapses the entire bracket simultaneously. The asymmetric implications of these laws of motion for the transmission of $\delta$ and $s$ shocks are derived formally in Section 3.

*2.3. Stochastic processes*

Technology, product destruction, and idiosyncratic separation each follow an AR(1) process in logs:

$$\log z_t = \rho_z \log z_{t-1} + (1-\rho_z)\log \bar{z} + e_{zt} \tag{7}$$

$$\log \delta_t = \rho_\delta \log \delta_{t-1} + (1-\rho_\delta)\log \bar{\delta} + e_{\delta t} \tag{8}$$

$$\log s_t = \rho_s \log s_{t-1} + (1-\rho_s)\log \bar{s} + e_{st} \tag{9}$$

where $\bar{z}$, $\bar{\delta}$, and $\bar{s}$ are the long-run means of technology, exogenous product destruction, and idiosyncratic separation, respectively, and innovations $e_{zt}$, $e_{\delta t}$, $e_{st}$ are mutually uncorrelated. Shocks $s_t$ and $\delta_t$ realise at the end of period $t$, after production, bargaining, and matching, as illustrated in Figure 3. Agents making vacancy and entry decisions at the start of $t$ therefore act on $s_{t-1}$ and $\delta_{t-1}$ — the shocks that realised at end of $t-1$ — which determines the total exit rate $\delta_{e,t} = 1-(1-\delta_{t-1})F(\chi_t^c)$ and aggregate separation rate $\tau_t = \delta_{e,t} + s_{t-1}(1-\delta_{e,t})$. This lag, combined with sunk vacancy creation costs, is the source of predetermination: vacancies and employment are partially predetermined with respect to current shocks. Here is the complete timing figure code: latex

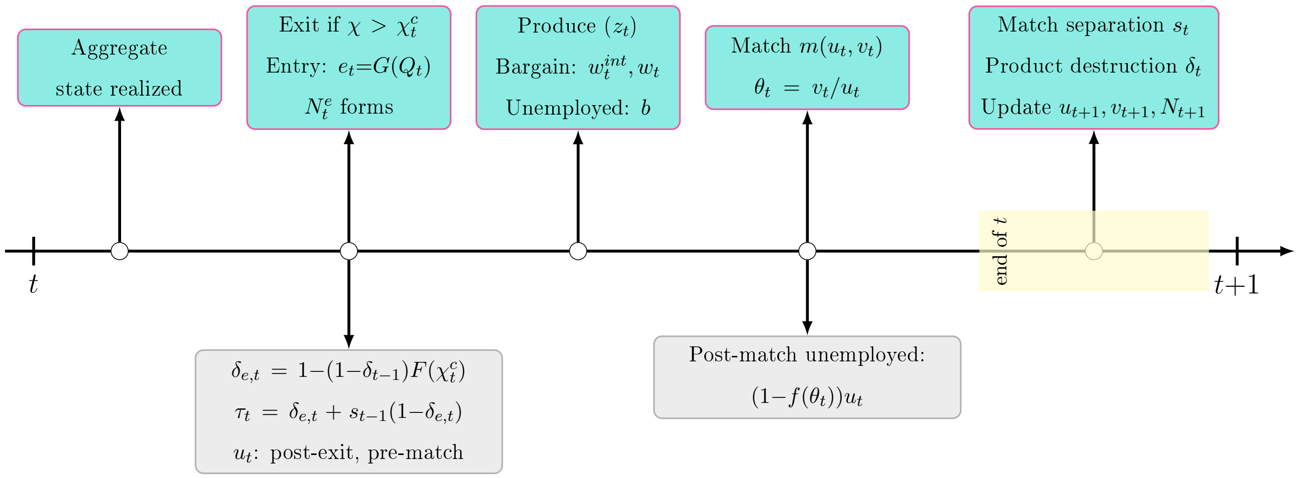


Figure 3: Timing. Turquoise: agent decisions. Grey: implied states. Yellow band: shocks realizing end of $t$, governing $t+1$.

## 3. Equilibrium

This section derives equilibrium given the environment of Section 2. Table A.5 positions the paper relative to the closest antecedents across four model dimensions. Equilibrium comprises three blocks: a business formation block governing the evolution of product lines and firm values, a labor market block governing vacancy creation, matching, and wage determination, and stochastic processes for the three structural shocks. Section 3.9 derives formally the asymmetric transmission of $\delta$ and $s$ shocks that motivates the paper's empirical strategy.

### *3.1. Household problem*

Let $\nu_t$ denote the ex-dividend equity price of the aggregate firm portfolio and $\nu_t^f$ the post-dividend value of an individual firm, following Bilbiie, Ghironi, and Melitz (2012). Given risk-aversion parameter $\sigma$ and discount factor $\beta$, the household chooses consumption $c_t$ and equity shares $a_t$ to maximize expected discounted utility:

$$\max_{c_t,\, a_t} \sum_{t=0}^{\infty} \mathbb{E}_0\, \beta^t c_t^{1-\sigma}/(1-\sigma)$$

subject to

$$c_t + \nu_t a_{t+1} = w_t(1-u_t) + bu_t + (\nu_t + d_t)a_t + D_t^{int} - T_t \tag{10}$$

where $b$ is the value of unemployment, $d_t$ is the per-share dividend, $D_t^{int}$ are intermediary dividends, and $T_t$ are lump-sum taxes financing unemployment benefits. Since households have unit mass, $c_t = C_t$ in equilibrium. The first-order condition for equity shares yields the standard asset-pricing equation:

$$\lambda_t \nu_t = \beta \, \mathbb{E}_t \lambda_{t+1}[\nu_{t+1} + d_{t+1}] \tag{11}$$

where $\lambda_t = c_t^{-\sigma}$ is the marginal utility of consumption and $m_{t+1} \equiv \beta(\lambda_{t+1}/\lambda_t)$ is the stochastic discount factor. To express (11) at the firm level, use the mappings $\nu_t = \nu_t^f(N_t + N_t^e)$ and $\nu_t + d_t = (\nu_t^f + d_t^f)N_t$, together with the firm law of motion (2). This yields the firm-level Euler equation, familiar from BGM and extended here to incorporate endogenous exit:

$$\nu_t^f = \mathbb{E}_t m_{t+1}(1-\delta_t)F(x_{t+1}^c)(\nu_{t+1}^f + d_{t+1}^f) \tag{12}$$

Equation (12) equates the current firm value to the expected discounted sum of next-period firm value and dividends, adjusted for the stochastic discount factor $m_{t+1}$ and the survival probability $(1-\delta_t)F(x_{t+1}^c)$, which combines the probability of surviving exogenous destruction $(1-\delta_t)$ and the probability of drawing a continuation cost below the equilibrium cutoff $F(x_{t+1}^c)$.

### *3.2. Recruiters*

There is a fixed unit measure of recruiters who can create vacancies subject to a sunk cost $x$ drawn from cdf $G(\cdot)$. Additionally, if a vacancy is filled the recruiter pays a fixed real cost $\kappa$ before bargaining with the newly matched worker.[7] After matching, the recruiter rents labor to an incumbent retailer with product line $j$. We introduce recruiters as a separate agent type to avoid the complications of Stole-Zwiebel intra-firm bargaining that arise when workers negotiate directly with retailers facing a downward- sloping demand curve: with a separate recruiter, the bilateral bargaining problem reduces to the standard DMP surplus-splitting equation regardless of the retail market structure.[8] Let $Q_t^j$ and $J_t^j$ denote the values of an unfilled and filled vacancy associated with product line $j$.

[7] Pissarides (2009) first suggested this matching cost to make vacancy posting more responsive; Christiano, Eichenbaum, and Trabandt (2016) estimate it accounts for the dominant share of recruiting costs.

[8] Under Stole-Zwiebel bargaining, retailers would strategically over-hire to improve their bargaining position, generating a fixed-point problem in employment that significantly complicates aggregation. The recruiter segmentation is isomorphic to the retailer– wholesaler structure of Christiano, Eichenbaum, and Trabandt (2016), with recruiters replacing intermediate goods.

The value of an unfilled vacancy for product line $j$ satisfies

$$Q_t^j = \mathbb{E}_t m_{t+1}(1-\delta_t)F(x_{t+1}^c)\left[(1-q(\theta_t))Q_{t+1}^j + q(\theta_t)J_{t+1}^j\right] - q(\theta_t)\kappa$$
$$= \mathbb{E}_t m_{t+1}(1-\delta_t)F(x_{t+1}^c)\left[Q_{t+1}^j + q(\theta_t)(J_{t+1}^j - Q_{t+1}^j)\right] - q(\theta_t)\kappa \tag{13}$$

An unmatched recruiter survives to next period only if the product line survives both exogenous destruction and the continuation-cost draw, with probability $(1-\delta_t)F(x_{t+1}^c)$. In the next period, the vacancy fills with probability $q(\theta_t)$, generating value $J_{t+1}^j$, or remains unfilled with probability $1-q(\theta_t)$. The matching cost $\kappa$ is paid upon filling. The recruiter chooses the best product line: $Q_t \equiv \max_j Q_t^j$. In equilibrium, competition equalizes values across active product lines: $Q_t^j = Q_t$ for all $j$ with $v_t^j > 0$.

Define the flow value of a vacancy as the net benefit of holding a vacancy this period relative to next:

$$K_t \equiv Q_t - \mathbb{E}_t m_{t+1}(1-\delta_t)F(x_{t+1}^c)Q_{t+1}$$

Rearranging (13) places the average hiring cost on the left-hand side:

$$\kappa + \frac{K_t}{q(\theta_t)} = \mathbb{E}_t m_{t+1}(1-\delta_t)F(x_{t+1}^c)(J_{t+1}^j - Q_{t+1}) \tag{14}$$

A recruiter opens a vacancy whenever $x \leq Q_t$, so total newly opened vacancies are $e_t = G(Q_t)$. Following CK, we adopt a power law distribution:

$$G(x) = \left(\frac{x}{x_m}\right)^{\xi}, \quad x \in [0,\, x_m]$$

with density $g(x) = \xi x^{\xi-1}/x_m^{\xi}$.[9] Provided $Q < x_m$, inverting $e = G(Q)$ gives

$$Q_t = e_t^{1/\xi}\, x_m$$

so the elasticity of the vacancy value with respect to entry is $1/\xi$. Free entry of vacancies ($\xi \to \infty$) nests the DMP baseline.

By symmetry of the retail problem (established in Section 3.4), recruiter compensation $w_t^{int,j}$ and the value of a filled vacancy $J_t^j$ are identical across active product lines; we henceforth drop

[9] If $\xi > 1$ the density is rising (relatively more high-cost recruiters); if $\xi < 1$ it is falling; if $\xi = 1$ costs are uniform on $[0, x_m]$.

the superscript $j$. Recruiters supply labor in a competitive market, taking $w_t^{int}$ as given. The value of a matched recruiter is[10]

$$J_t = w_t^{int} - w_t + \mathbb{E}_t m_{t+1}(1-\delta_t)F(x_{t+1}^c)\big(s_t Q_{t+1} + (1-s_t)J_{t+1}\big) \tag{15}$$

The recruiter receives $w_t^{int}$ from the retailer and pays the worker $w_t$. Conditional on product-line survival, the match continues with probability $1-s_t$; otherwise the vacancy is reposted at no additional sunk cost.

*Job creation condition..* Subtracting (13) from (15) and using the definition of $K_t$ yields recruiter surplus:

$$J_t - Q_t = w_t^{int} - w_t - K_t + (1-s_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right) \tag{16}$$

Substituting (16) into the right-hand side of (14) yields the job creation condition:

$$\kappa + \frac{K_t}{q(\theta_t)} = \mathbb{E}_t m_{t+1}(1-\delta_t)F(x_{t+1}^c)\left(w_{t+1}^{int} - w_{t+1} - K_{t+1} + (1-s_{t+1})\left(\kappa + \frac{K_{t+1}}{q(\theta_{t+1})}\right)\right) \tag{17}$$

The left-hand side is the average hiring cost: the fixed matching cost $\kappa$ plus the flow cost $K_t/q(\theta_t)$, equal to the opportunity cost of holding the vacancy this period multiplied by its expected duration $1/q(\theta_t)$. The right-hand side is the present discounted value of the match: future net revenue minus the future flow entry cost, plus the continuation value if the worker is retained, discounted by the stochastic discount factor and the product-line survival probability.

### *3.3. Wage determination*

Nash bargaining between the recruiter and worker yields

$$w_t = \phi\left(w_t^{int} - K_t + f(\theta_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right)\right) + (1-\phi)b \tag{18}$$

where $\phi$ is the worker's bargaining power.[11] Recruiter compensation $w_t^{int} = \rho(N_t)z_t/\mu(N_t)$ — derived in Section 3.4 — enters negatively through the markup. The flow entry cost $K_t$ has

[10] The asset-pricing equations (13) and (15) can equivalently be obtained by having the household purchase shares of recruiter firms directly; the stochastic discount factor ensures equivalence. Writing it via the household problem makes the derivation more compact and facilitates comparison to BGM and Mortensen and Pissarides (1994).

[11] The wage equation (18) has the same general form as Shao and Silos (2013), their equations 21 and 22; their setting includes capital affecting $w_t^{int}$ and writes the wage using the vacancy value $Q_t$ rather than the flow value $K_t$.

opposing effects: it reduces the match surplus (dampening wages by $\phi K_t$) but improves the worker's threat point, raising the tightness slope of the wage curve by $\phi\theta K_t$; the second effect dominates when $\theta > 1$. The fixed cost $\kappa$ raises wages by $\phi f(\theta_t)\kappa$ because a failed negotiation forces the firm to pay $\kappa$ again upon next filling the vacancy. Setting $w_t^{int} = z_t$ and $\kappa = 0$ recovers the CK wage equation.

Substituting (18) into (17) to eliminate $w_t$ yields the wage-substituted job creation condition:

$$\begin{aligned}\kappa + \frac{K_t}{q(\theta_t)} = \mathbb{E}_t m_{t+1}(1-\delta_t)F(x_{t+1}^c)\Bigg[&(1-\phi)\left(w_{t+1}^{int} - K_{t+1} - b\right) \\ &- \phi\theta_{t+1}\left(K_{t+1} + q(\theta_{t+1})\kappa\right) \\ &+ (1-s_{t+1})\left(\kappa + \frac{K_{t+1}}{q(\theta_{t+1})}\right)\Bigg]\end{aligned} \tag{19}$$

*3.4. Retailers*

A retailer selling variety $j$ rents labor from recruiters at wage $w_t^{int,j}$. At the beginning of each period, before production, the firm draws a continuation cost $\chi \sim F_\chi$ in units of the consumption basket, where $F_\chi$ is a mixture between a mass point at zero with probability $1-p_0$ and a bounded power law:

$$F_\chi(\chi) = (1-p_0) + p_0\left(\frac{\chi}{\chi_m}\right)^\psi, \quad \chi \in [0, \chi_m]$$

A firm with draw $\chi > \chi_t^c$ exits; one with $\chi \le \chi_t^c$ pays the cost and continues. Define the survival probability:

$$\Lambda_t \equiv F_\chi(\chi_t^c) = (1-p_0) + p_0\left(\frac{\chi_t^c}{\chi_m}\right)^\psi \tag{20}$$

so that the endogenous exit rate is $1-\Lambda_t = p_0\left(1-(\chi_t^c/\chi_m)^\psi\right)$, consistent with the total exit rate $\delta_{e,t} = 1-(1-\delta_{t-1})\Lambda_t$ from Section 2. The elasticity of the endogenous exit rate with respect to the cutoff is $\psi p_0(\chi_t^c/\chi_m)^\psi/(1-\Lambda_t)$, which increases in $\psi$ and falls as the mass point $1-p_0$ rises.

The value of a firm after its cost realization is:

$$\nu_t^f(\chi) = \max\left\{0,\ \max_{y_t}\left[\rho^d(y_t)y_t - w_t^{int}\frac{y_t}{z_t}\right] - \chi + (1-\delta_t)\mathbb{E}_t m_{t+1}\Lambda_{t+1}\,\nu_{t+1}^{ef}\right\} \tag{21}$$

where $\nu_t^{ef} \equiv (1/\Lambda_t)\int_0^{\chi_t^c}\nu_t^f(\tilde\chi)\,dF_\chi(\tilde\chi)$ is the expected firm value conditional on survival, and $\rho^d(y_t)$ is the inverse demand curve from (1), equal in symmetric equilibrium to $\rho(N_t)$. The profit maximization problem is static: letting $MR \equiv \rho'(y)y + \rho(y)$ denote marginal revenue, the first-order condition is

$$MR_t \cdot z_t = w_t^{int,j}$$

The left-hand side is the marginal revenue product; the right-hand side is recruiter compensation, which exceeds the worker's wage due to matching frictions. Equivalently, $MR_t = MC_t = w_t^{int}/z_t$. The standard markup condition $\rho_t = \mu(N_t)MC_t$ gives recruiter compensation:

$$w_t^{int} = \frac{\rho(N_t)}{\mu(N_t)} z_t \tag{22}$$

Recruiter compensation is increasing in the relative price $\rho(N_t)$ and technology $z_t$ and decreasing in the markup $\mu(N_t)$. Since $\mu'(N_t) \leq 0$, $w_t^{int}$ is increasing in $N_t$ overall.[12] Retail profits gross of the continuation cost are a share $(\mu(N_t) - 1)/\mu(N_t)$ of sales:

$$R_t^f = \frac{\mu(N_t) - 1}{\mu(N_t)} \rho(N_t) y_t$$

*3.5. Firm entry and exit*

By definition $\nu_t^f(\chi_t^c) = 0$. Evaluating the firm Bellman (21) at $\chi = \chi_t^c$ yields the cutoff condition:

$$\chi_t^c = R_t^f + \nu_t^f \tag{23}$$

where $\nu_t^f \equiv \nu_t^{ef} \cdot \Lambda_t$ is the unconditional expected firm value. From the Bellman, the firm value conditional on the cost realization is $\nu_t^f(\chi) = \chi_t^c - \chi$ for all $\chi \leq \chi_t^c$, so that $\nu_t^{ef} = \chi_t^c - E(\chi \,|\, \chi \leq \chi_t^c)$. For the power law component, the conditional mean of a positive draw satisfies $E(\chi \,|\, 0 < \chi \leq \chi_t^c) = \psi_c \chi_t^c$ where $\psi_c \equiv \psi/(\psi + 1)$, so the overall conditional mean (including the mass point at zero) is $E(\chi \,|\, \chi \leq \chi_t^c) = p_0 \psi_c \chi_t^c$. Hence:

$$\nu_t^f = (1 - \delta_t)\mathbb{E}_t m_{t+1} \Lambda_{t+1} \, \nu_{t+1}^{ef} \tag{24}$$

New firms are created using technology $N_t^e = z_t l_t^e / f_e$, where $l_t^e$ is labor hired from recruiters at wage $w_t^{int}$ and $f_e$ is the sunk entry cost in labor units, equivalent to $w_t^{int} f_e / z_t = \rho(N_t) f_e / \mu(N_t)$

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[12] Following BGM, marginal cost is $w_t^{int}/z_t$. Under DS-CES, $\mu = \varepsilon/(\varepsilon - 1)$ is constant, recovering the standard formula.

units of the consumption basket. Firms enter until the post-dividend firm value equals the sunk entry cost:

$$\nu_t^f = \frac{\rho(N_t) f_e}{\mu(N_t)} \tag{25}$$

Combining the free entry condition (25) with the cutoff condition (23) yields:

$$\chi_t^c = R_t^f + \frac{\rho(N_t) f_e}{\mu(N_t)} \tag{26}$$

The cutoff rises with current profitability $R_t^f$ and the value of a new product line $\rho(N_t)f_e/\mu(N_t)$: when conditions improve, marginal firms that would otherwise exit find it worthwhile to continue.

### *3.6. Aggregation*

Labor market clearing implies $L_t = 1 - u_t$. Define gross consumption output $Y_t^c \equiv \rho(N_t) N_t y_t$ and retail employment $L_t^c \equiv N_t l_t^c$. Aggregate continuation costs paid by surviving firms are

$$X_t^c \equiv N_t\, E(\chi \,|\, \chi \leq \chi_t^c) = N_t p_0 \psi_c \chi_t^c \tag{27}$$

where $\psi_c \equiv \psi/(\psi+1)$. Aggregate retail profits gross of continuation costs are $\Pi_t^f = Y_t^c/\varepsilon(N_t) - X_t^c$, and recruiter profits are $\Pi_t^{int} = (w_t^{int} - w_t)L_t$. The average per-firm dividend is $d_t^f = \Pi_t^f/N_t$.

Total vacancy creation costs are the sum of sunk posting costs and fixed matching costs:

$$X_t = X_v + X_f = e_t \frac{\xi}{\xi+1} Q_t + \kappa q(\theta_t) v_t \tag{28}$$

where $X_v = x_m^{-\xi} \frac{\xi}{\xi+1} Q_t^{\xi+1}$ follows from integrating $xg(x)$ over $[0, Q_t]$. Given $e_t$ and $Q_t$, the parameter $x_m$ has no additional effect on entry costs provided $Q_t < x_m$.

Aggregating (10) with asset market clearing $a_{t+1} = a_t$ and $T_t = bu_t$, and using $Y_t^{Gross} = Y_t^c + \nu_t^f N_t^e$, real GDP net of intermediate inputs is:

$$Y_t \equiv Y_t^{Gross} - X_t - X_t^c = C_t + \nu_t^f N_t^e = w_t L_t + \Pi_t^f + \Pi_t^{int} \tag{29}$$

GDP comprises consumption and investment in new product lines. Moreover, decomposing output into income confirms the national accounts identity.

Using free entry (25) and $d_t^f = \Pi_t^f/N_t$, the firm-level Euler equation (12) becomes the business formation equation:

$$f_e \rho(N_t) = (1-\delta_t)\mathbb{E}_t m_{t+1} \Lambda_{t+1} \frac{\mu(N_t)}{\mu(N_{t+1})} \left\{ f_e \rho(N_{t+1}) + \frac{Y_{t+1}^c}{N_{t+1}} \left( \mu(N_{t+1}) - 1 - \mu(N_{t+1}) \frac{X_{t+1}^c}{Y_{t+1}^c} \right) \right\}$$

The left-hand side is the current cost of creating a product line. The right-hand side is the expected discounted return, adjusted for survival probability $(1-\delta_t)\Lambda_{t+1}$ and the current-to-future markup ratio $\mu(N_t)/\mu(N_{t+1})$. The term in braces is the next-period firm value plus the private marginal product of a new product line net of continuation costs; the net markup $\mu(N_{t+1})-1$ governs the return to entry.

### 3.7. Definition of equilibrium

**Definition 1.** *Given $\rho(N)$, $\mu(N)$, $F_\chi$, initial exogenous states $\{z_0, \delta_0, s_0\}$, and initial endogenous states $\{v_{-1}, u_{-1}, u_0, N_0\}$, an equilibrium is a sequence $\{N_t, C_t, \chi_t^c, Y_t^c, \theta_t, K_t, Q_t, e_t, v_t, u_t\}_{t=0}^{\infty}$ satisfying the business formation block:*

$$N_t = (1-\delta_{t-1})\Lambda_t \left( N_{t-1} + \frac{z_{t-1}(1-u_{t-1})}{f_e} - \frac{Y_{t-1}^c}{f_e \rho(N_{t-1})} \right) \tag{30}$$

$$f_e\rho(N_t) = (1-\delta_t)\mathbb{E}_t m_{t+1}\Lambda_{t+1}\frac{\mu(N_t)}{\mu(N_{t+1})} \left\{ f_e\rho(N_{t+1}) + \frac{Y_{t+1}^c}{N_{t+1}} \left( \mu(N_{t+1}) - 1 - \mu(N_{t+1})\frac{X_{t+1}^c}{Y_{t+1}^c} \right) \right\} \tag{31}$$

$$Y_t^c = C_t + X_t + X_t^c \tag{32}$$

$$\chi_t^c = \frac{Y_t^c(\mu(N_t)-1)}{\mu(N_t)N_t} + \frac{f_e\rho(N_t)}{\mu(N_t)} \tag{33}$$

*and the labor market block:*

$$\kappa + \frac{K_t}{q(\theta_t)} = \mathbb{E}_t m_{t+1}(1-\delta_t)\Lambda_{t+1} \Bigg( (1-\phi)\left( \frac{\rho(N_{t+1})z_{t+1}}{\mu(N_{t+1})} - K_{t+1} - b \right) - \phi\theta_{t+1}\left(K_{t+1} + q(\theta_{t+1})\kappa\right) + (1-s_{t+1})\left( \kappa + \frac{K_{t+1}}{q(\theta_{t+1})} \right) \Bigg) \tag{34}$$

$$K_t = Q_t - (1-\delta_t)\mathbb{E}_t m_{t+1}\Lambda_{t+1}Q_{t+1} \tag{35}$$

$$e_t = \left( \frac{Q_t}{x_m} \right)^{\xi} \tag{36}$$

*together with laws of motion (6)–(5), auxiliary definitions $\delta_{e,t}$ (3), $\tau_t$ (4), $\Lambda_t$ (20), $X_t$ (28), $X_t^c$ (27), $m_t$ (11), and stochastic processes (7)–(9).*

The business formation block jointly determines $N_t$, $C_t$, $Y_t^c$, and $\chi_t^c$: the firm LOM propagates the product-line stock, the business formation Euler governs entry, the resource constraint clears goods markets, and the cutoff equates the exit threshold to current profitability plus the option value of a new product line. The labor market block jointly determines $u_t, v_t, \theta_t$, $K_t$, and $Q_t$: the job creation condition equates average hiring cost to the present discounted value of a filled vacancy, and free entry pins down the vacancy creation margin. The two blocks interact through $\rho(N_t)$ and $\mu(N_t)$ in the job creation condition and through $\delta_t$ in the survival-adjusted discount factor $(1-\delta_t)\Lambda_{t+1}$, which appears in both.

### *3.8. Limiting cases*

The next several propositions characterize the relationship between the business formation block and labor market block, and establish the connection to benchmark frameworks in the literature. Proofs are in Appendix G.

**Proposition 1** (Independence of labor market block)**.** *Consider general CES preferences with elasticity of substitution $\varepsilon$. The labor market block $\{u_{t+1}, v_t, e_t, Q_t, K_t, w_t\}_{t=0}^{\infty}$ can be solved independently of the business formation block if and only if $\sigma = 0$, $\zeta = 0$, and $p_0 = 0$.*

When $\sigma = 0$, risk neutrality fixes $m_{t+1} = \beta$, eliminating the consumption dependence of the stochastic discount factor. When $\zeta = 0$, the taste for variety vanishes and $\rho(N_t) = 1$, removing $N_t$ from recruiter compensation $w_t^{int} = \rho(N_t)z_t/\mu(N_t)$ and from the job creation condition while leaving $\mu$ and hence the business formation block otherwise intact. When $p_0 = 0$, endogenous exit is absent so $\Lambda_t = 1$ and $\delta_{e,t} = \delta_{t-1}$ depends only on the exogenous shock. Under all three conditions, the labor market block closes on its own; the business formation block then determines $C_t$ and $N_t$ taking $u_t$ and $X_t$ as given from the labor market block.

The model nests GS and CK as the taste for variety vanishes, which under DS-CES corresponds to $\varepsilon \to \infty$. This limit is not directly workable, however: with $f_e$ fixed, vanishing profits ($\mu \to 1$) make entry unprofitable and $N \to 0$ from the resource constraint, so that $Y_t^c = \rho(N_t)N_t y_t \to 0$ with positive employment, rendering aggregate production degenerate. One might restore a non-degenerate limit by setting $\delta \to 0$, but this would eliminate the destruction margin that is central to the vacancy law of motion in GS and CK — precisely the channel we wish to preserve. We instead seek a joint limit in which entry costs shrink proportionally with profits, keeping the product-line stock well-defined and positive while making business formation irrelevant to aggregate labor market allocations. Proposition 2 shows that scaling $f_e = \bar{f}_e/\varepsilon$ jointly with $\varepsilon \to \infty$ achieves exactly this.

**Proposition 2** (Double limit)**.** *Consider DS-CES preferences. Set $p_0 = 0$, and let $\varepsilon \to \infty$ jointly with $f_e = \bar{f}_e/\varepsilon$ for fixed $\bar{f}_e > 0$. Then:* (i) *the product-line mass has a unique positive steady-state value $\bar{N} > 0$ satisfying $\bar{\delta}\bar{N} = \bar{z}\bar{L}\ln \bar{N}/\bar{f}_e$; local stability of the $N_t$ dynamics around $\bar{N}$ holds provided $\bar{N} > e^{1-\bar{\delta}}$, a condition satisfied for sufficiently small $\bar{f}_e$ given other parameters.*

(ii) *firm value* $\nu_t^f$, *investment* $\nu_t^f N_t^e$, *and labor devoted to entry* $l_t^e$ *all vanish at rate* $1/\varepsilon$; (iii) *recruiter compensation* $w_t^{int} \to z_t$; *and* (iv) *aggregate allocations* $\{\theta_t, u_t, v_t, C_t, Y_t^c\}$ *are independent of* $\bar{N}$, *so the normalization* $\bar{N} = 1$ *is without loss of generality for aggregate allocations. In the limit, the business formation block is irrelevant to real allocations.*

As $\varepsilon \to \infty$ under DS-CES, $\zeta = 1/(\varepsilon - 1) \to 0$, satisfying one of the separability conditions of Proposition 1. Proposition 2 is stronger, however: it establishes not merely that the labor market block separates but that business formation becomes irrelevant to aggregate allocations entirely — a consequence of profits and entry costs vanishing jointly at the same rate. Proposition 3 below implements the limiting economy directly via clone replacement, maintaining $N_t = 1$ by construction rather than through the limit passage.

**Proposition 3** (Clone-replacement equilibrium)**.** *Under clone replacement with* $N_t = 1$ *and* $\mu = 1$, *the business formation block drops out and equilibrium labor market variables* $\{u_{t+1}, v_t, e_t, Q_t, K_t\}_{t=0}^{\infty}$ *satisfy:*

$$\kappa + \frac{K_t}{q(\theta_t)} = \mathbb{E}_t m_{t+1}(1-\delta_t)\left((1-\phi)(z_{t+1} - K_{t+1} - b)\right.$$
$$\left. -\phi\theta_{t+1}(K_{t+1} + q(\theta_{t+1})\kappa) + (1 - s_{t+1})\left(\kappa + \frac{K_{t+1}}{q(\theta_{t+1})}\right)\right) \tag{37}$$

$$K_t = Q_t - (1-\delta_t)\mathbb{E}_t m_{t+1} Q_{t+1}$$
$$v_t = (1-\delta_{t-1})\left[(1-q(\theta_{t-1}))v_{t-1} + s_{t-1}(1-u_{t-1})\right] + e_t$$
$$u_t = (1 - f(\theta_{t-1}))u_{t-1} + \tau_t\left[(1-u_{t-1}) + f(\theta_{t-1})u_{t-1}\right] \tag{38}$$

*together with* $e_t = G(Q_t)$, $m_t = \beta(C_t/C_{t-1})^{-\sigma}$,

$$C_t = z_t(1-u_t) - X_t, \tag{39}$$

*and stochastic processes* (7)–(9), *with* $\delta_{e,t} = \delta_{t-1}$ *since endogenous exit is absent by construction.*

We dub this the *augmented Gabrovski-Silva* (AGS) model. Relative to AGS, the baseline model adds two channels affecting labor market dynamics: variety effects in the job creation condition through $\rho(N_t)/\mu(N_t)$, and endogenous exit through variation in the survival cutoff $\chi_t^c$, as developed formally in Section 3.9 and evaluated quantitatively in Section 5.

**Remark 1** (Nested special cases of AGS)**.** *The AGS model nests two benchmark frameworks as limiting cases. Setting* $\sigma = \kappa = 0$ *renders the AGS system isomorphic to Gabrovski and*

*Silva (2025), extended here with aggregate AR(1) shocks to $\delta_t$ and $s_t$. Additionally setting $s_t = 0$ recovers Coles and Moghaddasi Kelishomi (2018) exactly, with $w_t^{int} = z_t$ and consumption determined residually from $z_t(1-u_t) = C_t + \frac{\xi}{\xi+1}Q_t^{\xi+1}$.*

Strictly speaking, the baseline model does not nest Bilbiie, Ghironi, and Melitz (2012) since there is no intensive labor margin. However, it does nest a restricted version. Define *labor-restricted BGM* (LR-BGM) as the Bilbiie, Ghironi, and Melitz (2012) model with labor dropped from preferences: the household maximizes $\sum_t \beta^t c_t^{1-\sigma}/(1-\sigma)$ subject to $c_t = w_t L_t + \pi_t - X_t$, where $\{L_t\}$ and $\{X_t\}$ are exogenous sequences and $X_t$ is the vacancy-creation cost absorbed as an intermediate input. The resource constraint of LR-BGM is therefore $Y_t = C_t + \nu_t^f N_t^e$, matching the GDP identity (29), with $X_t$ netted out on both sides.

**Proposition 4** (Nesting LR-BGM)**.** *Set $p_0 = 0$, so $X_t^c = 0$, $\Lambda_t = 1$, and $\delta_{e,t} = \delta_{t-1}$. For any exogenous sequences $\{L_t\}$ and $\{X_t\}$, the business formation block of the baseline model—equations (30), (31), and (29)—is isomorphic to LR-BGM evaluated at the same sequences.*

The isomorphism holds at the level of GDP (29) rather than gross output (32): vacancy costs $X_t$ are intermediate inputs in both models and cancel from the resource constraint, leaving $Y_t = C_t + \nu_t^f N_t^e$ in each case. The sole departure of the baseline from LR-BGM is that $L_t$ and $X_t$ are endogenous, determined jointly by the DMP block (6)–(5). Restoring $p_0 > 0$ further adds endogenous exit to the LR-BGM core.

### *3.9. Shock transmission and the $\delta$–$s$ asymmetry*

The equilibrium laws of motion imply qualitatively distinct labor-market dynamics under product destruction versus match separation shocks. This section establishes the asymmetry analytically, motivating the paper's focus on $\delta$ shocks throughout the quantitative analysis.

**Proposition 5** ($\delta$–$s$ asymmetry in unemployment-vacancy comovement)**.**

1. ($s$ shock, exogenous exit.) *Suppose $p_0 = 0$. Then for any $\rho_s \in [0, 1)$, a positive $s_t$ shock raises unemployment and vacancies simultaneously at horizon $h = 1$, generating positive $u$–$v$ comovement.*
2. ($s$ shock, endogenous exit.) *Under DS-CES with*

$$f_e \bar{N} < \frac{\bar{z}(1-\bar{u})(\varepsilon - 2)}{\varepsilon - 1}, \tag{40}$$

*the preceding result extends to the full model ($p_0 > 0$) if the reposting inflow* $(1-\delta_t)\Lambda_{t+1}(1-u_t)$ *weakly dominates the endogenous exit drain* $(1-\delta_t)F'(\chi^c_{t+1})|\partial\chi^c_{t+1}/\partial s_t|\cdot[(1-q(\theta_t))v_t + s_t(1-u_t)]$ *on the pre-committed vacancy stock. The drain is increasing in $\rho_s$, so the condition holds for all $\rho_s \in [0, \bar{\rho}_s)$ for some $\bar{\rho}_s > 0$.*

3. ($\delta$ shock.) *A positive $\delta_t$ shock raises unemployment and reduces total vacancies at horizon $h = 1$, generating negative $u$–$v$ comovement, provided entry does not fully offset pre-committed vacancy destruction: $\partial e_{t+1}/\partial\delta_t < |\partial v_{pre,t+1}/\partial\delta_t|$. A sufficient condition is that $\bar{\delta}_e/\bar{\tau}$ is small: when only a fraction of total separations flows through the destruction channel, duration shortening generates a weak entry cushion and the direct vacancy destruction dominates.*

See Appendix H for the proof.

Define $g(N, u) \equiv d^f(N, u) + f_e\rho(N)/\mu$ as the continuation profit threshold: a firm survives whenever its fixed-cost draw satisfies $\chi^c \leq g$. Under DS-CES, $g_N$ balances two opposing forces: flow profit $d^f \propto N^{(2-\varepsilon)/(\varepsilon-1)}$ combines a market-share dilution effect $(1/N)$ and a love-of-variety demand expansion ($\rho(N) \propto N^{1/(\varepsilon-1)}$), with net exponent $(2-\varepsilon)/(\varepsilon-1)$ negative for $\varepsilon > 2$, so dilution dominates and $d^f$ falls in $N$; and an option value of entry $f_e\rho(N)/\mu \propto N^{1/(\varepsilon-1)}$, which always rises in $N$. Condition (40) ensures the profit decline dominates the option-value rise ($g_N < 0$, $\det J > 1$): $\varepsilon > 2$ is necessary (it makes the right-hand side positive), while the bound on $f_e\bar{N}$ ensures the option-value channel does not overturn it. Both hold at standard calibrations.

From the vacancy law of motion (6), a positive $\delta_t$ shock raises $\delta_{e,t+1}$ and thereby destroys a fraction $\hat{\delta}_{e,t+1}$ of the pre-committed vacancy stock $\bar{v}^{pre} = \bar{v} - \bar{e}$; see Lemma 1 in Appendix H. Whether *total* vacancies fall depends on the entry response. Duration shortening raises $K$ (E.11) and hence the vacancy value $Q$ (E.12), stimulating entry $e_{t+1} = G(Q_{t+1})$ that partially backfills destroyed vacancies. Under free entry ($\xi \to \infty$, $Q \approx x_m$ pinned), $K$ rises with $\delta_e$, the job-creation condition (34) implies $\theta$ falls, and total vacancies $v = \theta u$ have an ambiguous sign: positive $u$–$v$ comovement requires the $u$ rise to exceed the $\theta$ fall.

A sufficient condition for the $\theta$ fall to dominate is that $\bar{\delta}_e/\bar{\tau}$ is small: when only a fraction of total separations flow through the destruction channel, the proportional $u$ rise from a $\delta$ shock is small while the duration-shortening effect on $K$ is also correspondingly weak, so the $\theta$ decline dominates. At the empirically calibrated $\bar{\delta}_e/\bar{\tau} \approx 0.21$, negative $u$–$v$ comovement holds even under free entry, as confirmed Appendix F.1; convex entry costs govern the *degree* of cushioning, not the *sign* of comovement.

### 3.10. Steady state

A steady state is a fixed point of the equilibrium system in which all variables are constant and shocks are at their long-run means $(\bar{z}, \bar{\delta}, \bar{s})$. We characterize it in two blocks. The *exit block* in $(\chi^c, \delta_e)$ space determines the exit threshold and total destruction rate given aggregate profitability. The *labor market and entry block* in $(\theta, N)$ space determines market tightness and product variety given $\delta_e$. The two blocks interact through $\delta_e$ and $\rho(N)$, and must be solved jointly. We focus on the $(\theta, N)$ block for the diagrammatic analysis, treating $\delta_e$ as a parameter; the full steady-state system and key derivations are in Appendix D.

*Key steady-state ratios.* Setting $\delta_t = \bar{\delta}$ and $s_t = \bar{s}$, the Euler equations yield $N^e/N = \delta_e/(1-\delta_e)$. Retail profits as a fraction of consumption output are

$$\pi_s \equiv \frac{Nd^f}{Y^c} = \frac{1}{\varepsilon(N)} - \frac{X^c}{Y^c} \tag{41}$$

The profit share $\pi_s$ captures the net return to operating a product line after continuation costs: it falls with the elasticity of substitution (competition erodes markups) and with the continuation cost burden $X^c/Y^c$. The equilibrium relationship between total employment and product lines is:

$$N = \frac{\pi_s z L(1-\delta_e)}{f_e\left(\frac{r+\delta_e}{\mu(N)} + \delta_e \pi_s\right)} \tag{42}$$

At a given profit share, a higher destruction rate $\delta_e$ reduces product lines relative to employment by raising the effective discount rate on future profits and increasing the entry required to maintain the product-line stock. The full derivation of steady-state output shares and labor allocation is in Appendix D.

*Four steady-state curves.* The *exit threshold curve* gives the locus $(\chi^c, \delta_e)$ consistent with the cutoff condition (33) at steady state:

$$\chi^c = \frac{\rho(N) f_e}{\mu(N)} \left( \frac{r+\delta_e}{1-\delta_e} \cdot \frac{r+\delta_e+\psi_c(1-\delta_e)}{(1-\psi_c)(r+\delta_e)} + 1 \right) \tag{43}$$

A higher destruction rate requires higher per-period profits to compensate, raising $\chi^c$; higher variety $N$ shifts the curve upward through $\rho(N)$. The *product destruction curve* follows from $\Lambda(\chi^c)(1-\delta) = (1-\delta_e)$:

$$\chi^c = \left(\frac{1-\delta_e}{1-\delta}\right)^{1/\psi} \chi_m \tag{44}$$

It is negatively sloped: a higher cutoff $\chi^c$ raises the endogenous survival probability and therefore lowers $\delta_e$.

For the labor market block, express $e$ and $K$ as functions of tightness $\theta$ using the steady-state unemployment and vacancy laws of motion:

$$e(\theta) = \delta_e \, \frac{\tau\theta + (1-\delta_e)f(\theta)}{\tau + (1-\delta_e)f(\theta)}, \qquad K(\theta) = \frac{r+\delta_e}{1+r}\, e(\theta)^{1/\xi}\, x_m$$

The *job creation curve* is the steady-state JCC (34):

$$\left(\kappa + \frac{K(\theta)}{q(\theta)}\right)(r + \tau + (1-\delta_e)\phi\theta q(\theta)) = (1-\delta_e)(1-\phi)\left(\frac{\rho(N)z}{\mu(N)} - K(\theta) - b\right) \tag{45}$$

Its right-hand side is increasing in $N$ through $\rho(N)$: more product lines raise recruiter compensation, stimulating vacancy creation — the employment–variety feedback emphasized by Schaal and Taschereau-Dumouchel (2019) and derived here in a more flexible setting. Under general CES with $\zeta = 0$, $\rho = 1$ and (45) determines $\theta$ independently of $N$, yielding a vertical line in $(\theta, N)$ space. The *resource constraint curve* combines the aggregate resource constraint, free entry (25), and the $N$-$L$ relationship (42):

$$N = \frac{(\mu(N)-1)zL(\theta)(1-\delta_e)}{f_e(\delta_e\mu(N)+r)} \tag{46}$$

Employment $L(\theta)$ is increasing and concave in $\theta$, so (46) is increasing, concave, and bounded above by $\bar{N} = (\mu-1)z(1-\delta_e)/(f_e(\delta_e\mu+r))$, the product variety consistent with full employment.

**Proposition 6** (Variety effects and elasticity of substitution)**.** *The exit threshold curve* (43)*, job creation curve* (45)*, and resource constraint curve* (46) *each depend on the elasticity of substitution* $\varepsilon(N)$*. Only the exit threshold and job creation curves depend on the benefit of variety* $\zeta(N)$*. The product destruction curve* (44) *is independent of both.*

The result follows directly from the expressions above: $\varepsilon(N)$ enters $\mu(N)$ and $\pi_s$ in all three non-destruction curves, while $\zeta(N)$ enters only through $\rho(N)$, which is absent from the resource constraint curve (46). The product destruction curve depends only on $\psi$, $\chi_m$, and $\bar{\delta}$. The implication is that empirical variation in exit rates across recessions identifies the destruction curve independently of preference parameters.

**Proposition 7** (Steady-state equilibria under DS-CES)**.** *Restrict to DS-CES preferences ($\rho(N) = N^{1/(\varepsilon-1)}$, $\mu = \varepsilon/(\varepsilon-1)$ constant) and let $p_0 \to 0$ so that $\delta_e$ is a parameter. If $b > 0$ or $\kappa > 0$, the job creation curve (45) satisfies*

$$N^0_{JC} \equiv \lim_{\theta \to 0} N_{JC}(\theta) = \left( \frac{\mu}{z} \left( \frac{\kappa(r+\tau)}{(1-\delta_e)(1-\phi)} + b \right) \right)^{\varepsilon-1} > 0, \tag{47}$$

*while the resource constraint curve (46) satisfies $N_{RC}(0) = 0$ and $N_{RC}(\theta) \to \bar{N} < \infty$. As $\theta \to \infty$, $N_{JC}(\theta) \to \infty$. The two curves therefore cross an even number of times; generically exactly two equilibria exist in $(\theta, N)$ space, one with strictly higher tightness and product variety than the other.*

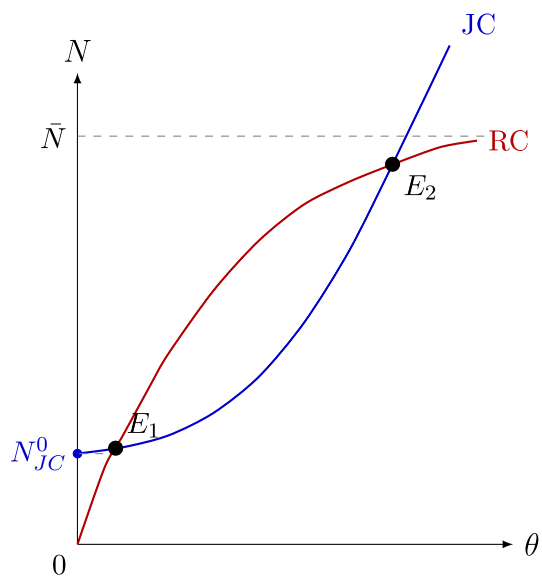


Figure 4: Steady-state equilibria in $(\theta, N)$ space. The job creation curve (JC) is convex and intersects the vertical axis at $N^0_{JC} > 0$ whenever $b > 0$ or $\kappa > 0$; see (47). The resource constraint curve (RC) is concave, starts at the origin, and is bounded above by $\bar{N} = (\mu - 1)\bar{z}(1 - \bar{\delta}_e)/(f_e(\bar{\delta}_e\mu + r))$. Two equilibria exist generically; $E_1$ is the low and $E_2$ the high tightness–variety equilibrium.

Figure 4 illustrates the two equilibria. A positive productivity shock $z$ shifts both curves outward: higher $z$ raises recruiter compensation $w^{int} = \rho(N)z/\mu$, increasing the surplus from posting vacancies and shifting (45) rightward; simultaneously higher aggregate income supports more product lines for a given employment level, shifting (46) upward. The high equilibrium moves from $E_2$ to $E_2'$ with both higher tightness and more product lines, as illustrated in Figure 5.

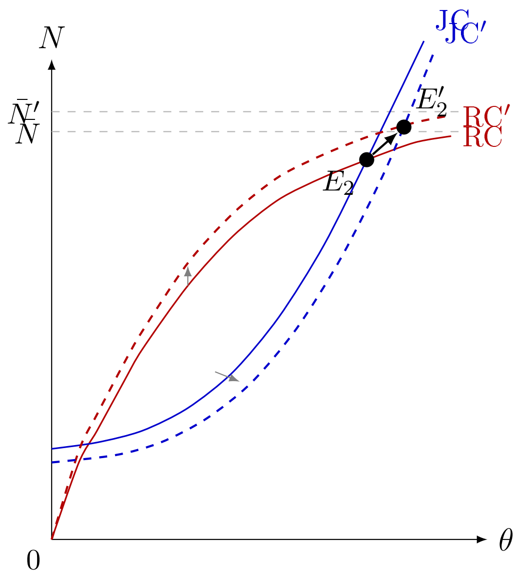


Figure 5: Effect of a positive productivity shock $z\uparrow$ on the steady-state equilibrium. Higher $z$ raises recruiter compensation $w^{int} = \rho(N)z/\mu$, shifting the job creation curve rightward (JC to JC′); simultaneously higher income supports more product lines for given employment, shifting the resource constraint curve upward (RC to RC′, with $\bar{N}' > \bar{N}$). The high equilibrium moves from $E_2$ to $E_2'$, with strictly higher market tightness and product variety.

## 4. Empirical Evidence and Bartik Local Projections

### *4.1. Overview and identification strategy*

The cross-state evidence in Section 1 establishes that states with structurally higher establishment exit rates accumulate larger unemployment gaps during recessions, but the correlation conflates the causal effect of exit shocks with persistent differences in industrial composition, institutional environment, and average firm age across states. A state panel regression of labor market outcomes on local exit rates would confound the structural $\delta$ shock with the local demand conditions that drive both exits and unemployment simultaneously — the classic omitted variable in any cross-sectional or within-state identification strategy. The standard panel remedy of adding state and time fixed effects absorbs permanent cross-state heterogeneity and aggregate time variation but leaves untouched the within-state, within-time comovement between local exit activity and local business conditions.

We address this using a Bartik shift-share design (Bartik, 1991; Goldsmith-Pinkham, Sorkin, and Swift, 2020). The instrument for state $s$ at quarter $t$ is a weighted average of *national* industry-level establishment exit rates, where the weights are the state's pre-determined industry employment shares. To see why this breaks the confound, consider Michigan in 2008: its instrument is large because manufacturing nationally — not just in Michigan — was experiencing a wave of permanent closings, and Michigan had a high pre-recession exposure to manufacturing. The national exit rate is computed leaving out Michigan's own establishments (leave-one-out), so the instrument cannot be mechanically driven by Michigan-specific conditions. The instrument thus isolates variation in exit exposure driven by a state's inherited industry structure interacting with national industry cycles, rather than by local demand or policy.

The identifying assumption is that, conditional on state and time fixed effects and lagged controls, the Bartik instrument is uncorrelated with state-specific shocks to unemployment and vacancy creation beyond the exit channel. Section 4.2 describes instrument construction in detail; Section 4.3 states the local projection specification; Section 4.4 reports results. Section 4.4 provides two validation exercises: a severity placebo that tests whether the vacancy IRF is driven by recession depth rather than exit exposure, and a joint local projection that checks robustness to controlling simultaneously for the layoff-and-discharge instrument.

### *4.2. Instrument construction*

*BED data and the permanence problem..* The underlying data source is the BLS Business Employment Dynamics (BED) survey, which tracks establishment-level employment via state unemployment insurance payroll records at quarterly frequency from 1992Q3. An establishment is classified as a "closing" in a given quarter if it exits the UI payroll base between consecutive March reference dates. Coverage is all private-sector establishments across 12 supersectors and all 50 states.

Raw BED closings conflate two economically distinct events: *permanent exits* — establishments that dissolve irreversibly, destroying the associated product line and vacancy slot — and *temporary shutdowns*, establishments that suspend operations for one to three quarters before reopening. The model's $\delta$ shock corresponds exclusively to permanent exit: only an irreversible dissolution removes a product line from the variety space and eliminates the recruiter's vacancy. Using raw closings as the shock proxy therefore overstates the true destruction margin and introduces attenuation bias when the temporary-shutdown share varies over time. The contamination is most severe during the 2020Q2 lockdown, where roughly 400,000 of the recorded closings reflected government-mandated suspensions that subsequently reopened, but it is present at lower intensity throughout the sample: seasonal businesses, firms temporarily suspending operations during downturns, and establishments undergoing ownership transitions all appear as closings in BED without constituting permanent product-line destruction.

*BED Deaths as the measure of permanent exits..* We address this using a second BED data series that resolves the permanence problem directly. In addition to closings, BED separately reports *deaths* — establishments absent from the UI payroll base for four or more consecutive quarters — which by construction excludes temporary shutdowns.[13] Since a temporarily closed establishment re-enters the payroll base within one to three quarters, it never accumulates four consecutive absences and therefore never appears in BED Deaths. Our $\delta$ instrument uses BED Deaths — not raw closings — as the numerator of the national exit rate.

Figure 6 shows employment-weighted mean BED Closings and Deaths rates across 12 supersectors from 1992Q3 to 2019Q4, together with 10th–90th industry percentile bands. The Deaths/Closings ratio (lower panel) averages 0.63 across the full sample and is remarkably stable over the business cycle, dipping modestly during the GFC but remaining well below unity throughout. This stability implies that the fraction of closings that are temporary does not vary sharply over time, and that using Deaths directly gives a cleaner series than either raw closings or a time-varying permanence correction.

The Deaths/Closings ratio exhibits substantial cross-sector heterogeneity. Mining has the lowest mean ratio (0.37), and Construction is similarly low (0.52), consistent with the prevalence of project-based and seasonal establishments that suspend operations temporarily without constituting permanent product-line destruction. Leisure and hospitality (0.58) and Other services (0.59) are also below the sample average. At the other end, Manufacturing (0.71), Retail trade (0.70), and Information (0.70) have the highest ratios, reflecting structurally lower rates of temporary closure in those sectors. No supersector reaches the theoretical upper bound of unity in any quarter of the sample. The employment-weighted mean across all 12 supersectors

[13]BED data classes: 06 (Closings) records establishments absent from the next quarterly payroll file; 08 (Deaths) records establishments absent for four or more consecutive quarters. Deaths $\subseteq$ Closings in every industry$\times$quarter cell, so ratio is bounded above by unity by construction.

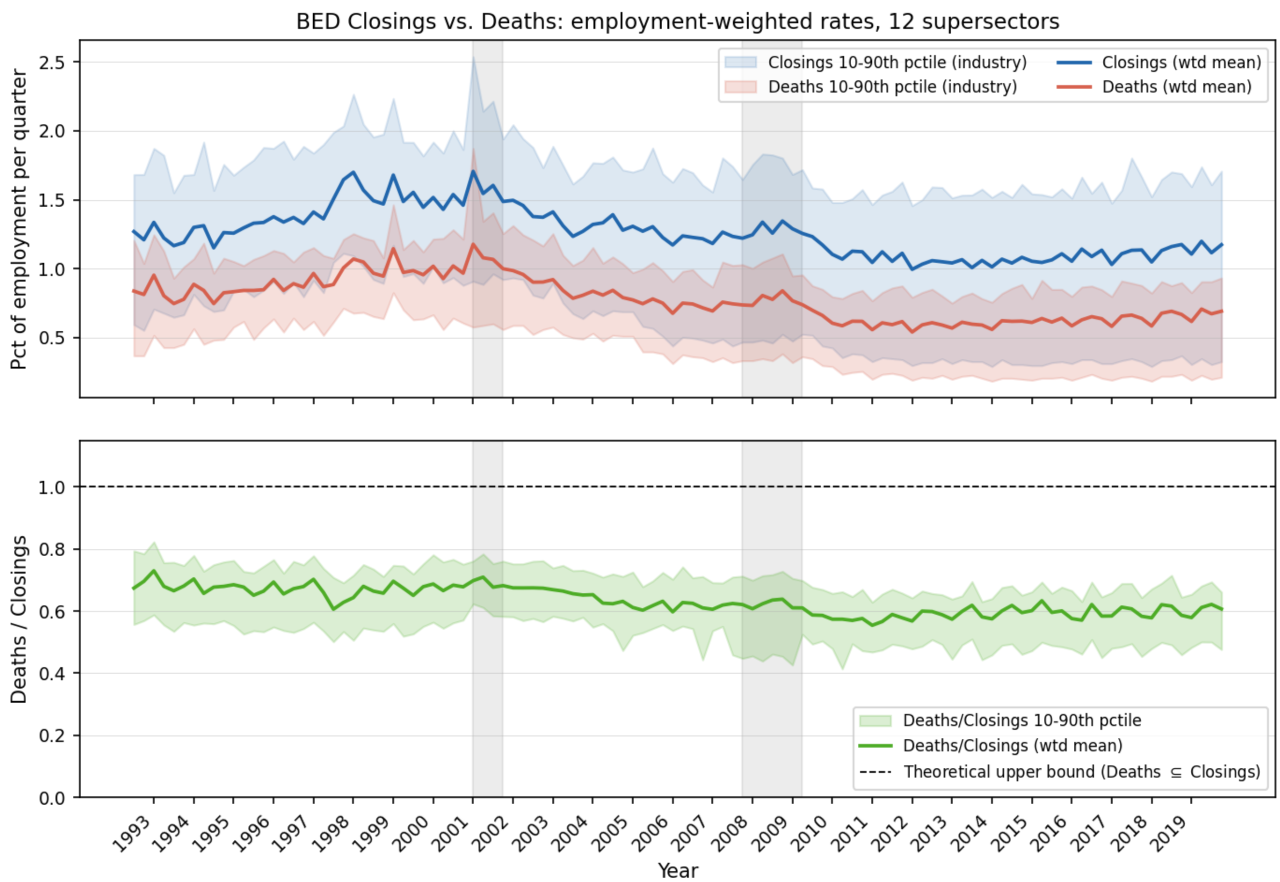


Figure 6: BED Closings and Deaths: employment-weighted rates across 12 supersectors, 1992Q3–2019Q4. *Top panel:* mean closing rate (blue) and death rate (red) as percent of employment per quarter, with 10th–90th industry percentile bands. Grey shading denotes NBER recessions. *Bottom panel:* Deaths/Closings ratio (dashed line at 1.0 marks the theoretical upper bound; ratio is bounded above by unity since Deaths $\subseteq$ Closings in every industry$\times$quarter cell).

is 0.63 (2001–2019); supersector time series are shown in Appendix C.1.

Most papers using BED data for shock identification (e.g., Elsby and Michaels, 2013) implicitly treat all closings as permanent. We are not aware of a prior paper using the BED Deaths series to align the instrument with the structural concept of irreversible product-line destruction.

*Leave-one-out construction and base-year shares..* The leave-one-out (LOO) national permanent exit rate for state $s$, supersector $j$, quarter $t$ is

$$g^{\delta}_{-s,j,t} = \frac{\text{deaths}_{j,t}}{E^{\text{nat}}_{-s,j,t_0}},$$

where $E^{\text{nat}}_{-s,j,t_0} = E^{\text{nat}}_{j,t_0} - E_{s,j,t_0}$ is base-year national employment in supersector $j$ minus state $s$'s own base-year employment. The LOO correction in the denominator ensures that state $s$'s own exit activity cannot mechanically inflate its instrument value. Numerator LOO — subtracting state $s$'s own deaths from the national count — is not implemented because BED state-level deaths by supersector are not publicly available. Employment shares $\omega_{s,j}$ are drawn from the 2006 QCEW (agglvl=54, private sector), a stable pre-GFC year that predates the bulk of the identifying variation. The Bartik instrument is:

$$B^{\delta}_{s,t} \;=\; \sum_{j} \omega_{s,j} \cdot g^{\delta}_{-s,j,t}. \tag{48}$$

*Residualization..* Raw BED exit rates co-move with industry-specific demand conditions: when a sector contracts nationally, establishments exit at higher rates *and* surviving firms simultaneously reduce vacancy posting and raise layoffs. An instrument built on raw $g^{\delta}_{j,t}$ would therefore partly identify the effect of industry demand contractions rather than the structural $\delta$ shock. We address this by projecting each log shock rate on predetermined demand controls before Bartik aggregation, retaining the residual as the instrument input.

The baseline residualization (v2) runs a within-industry OLS regression:

$$\log g^{\delta}_{j,t} \;=\; \alpha_j + \gamma\,\Delta\log p_t + \lambda\,\Delta\log VA_{j,t-1} + \nu^{\delta}_{j,t}, \tag{49}$$

where $\Delta\log p_t$ is aggregate nonfarm business productivity growth (FRED: OPHNFB) and $\Delta\log VA_{j,t-1}$ is real value-added growth in supersector $j$ lagged one quarter (BEA via FRED series RVAM, RVAC, etc.). Both regressors are predetermined relative to the current-period shock draw: aggregate productivity is contemporaneous but common to all industries and thus absorbed by the time fixed effects in the second stage; industry VA is lagged one quarter to ensure strict predetermination. The residual $\nu^{\delta}_{j,t}$ is then Bartik-aggregated in place of the raw rate, yielding $\tilde{B}^{\delta}_{s,t}$.

The FRED BEA real value-added series begin in 2005Q1, which would restrict the $\delta$ residualization sample to 2005Q2. To recover the full 1997Q1+ sample, we extend the quarterly VA series back to 1992Q1 using a constrained Chow-Lin (1971) temporal disaggregation. An indicator regression estimated on the 2005Q1+ post-sample relates quarterly sectoral VA growth to real GDP growth: $\Delta\log VA_{j,t} = \alpha_j + \beta_j\,\Delta\log \text{GDP}_t + \varepsilon_{j,t}$. The pre-2005 quarterly path is recovered by applying the implied growth rates $\hat{\alpha}_j + \hat{\beta}_j\,\Delta\log \text{GDP}_t$ recursively from the 2005Q1 anchor level. The backcast is then constrained to match published BEA annual benchmarks for 1997–2004 via a proportional Denton adjustment: for each year $A$ with annual benchmark $Y_{j,A}$, quarterly values are scaled by $Y_{j,A}/\overline{VA}^{\text{unconstrained}}_{j,A}$ so that the within-year average equals the published figure exactly. Industries with low GDP–VA correlation ($\mathrm{R}^2$ below threshold) receive $\hat{\beta}_j = 0$, using only the sector-specific mean growth rate to avoid injecting noisy cyclical variation into the pre-period.

After v2 residualization the cross-state correlation between the $\delta$ and layoffs-and-discharges (LD) Bartik instruments falls from approximately 0.54 on raw rates to 0.33.[14] The persistence of this residual correlation after controlling for industry demand cycles and aggregate productivity is informative: the shared variation is not a demand artifact but most likely reflects industry-specific financial conditions that move both exit rates and layoff rates simultaneously. This motivates the joint local projection robustness check in Section 4.4.

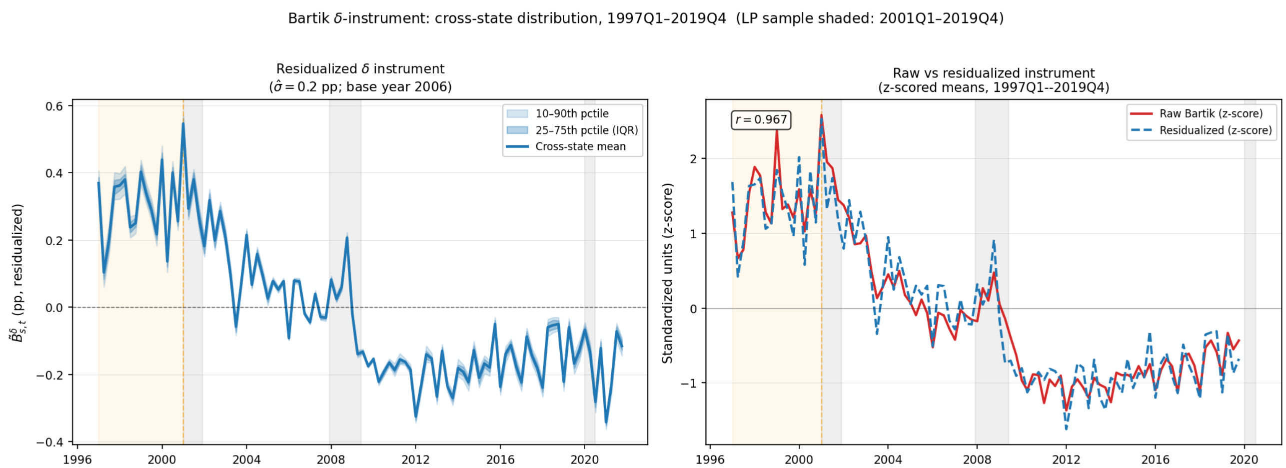


Figure 7: Cross-state $\delta$ Bartik instrument distribution, 1997Q1–2019Q4. *Left:* residualized instrument $\tilde{B}^{\delta}_{s,t}$ (pp of base-year employment); shading shows 25th–75th (dark) and 10th–90th (light) cross-state percentile bands. *Right:* z-scored cross-state means of raw (red) and residualized (blue dashed) instruments; $r = 0.905$. Orange shading (1997Q1–2000Q4): pre-LP period available for $\delta \to u$ but not vacancy or LD/QU. Grey shading: NBER recessions.

Figure 7 documents the cross-state distribution of the instrument and the effect of residualization. Both panels extend to 1997Q1, the start of the Chow–Lin-extended value-added series;

[14] As a robustness check we further augment equation (49) with a monetary policy sensitivity term $\psi(\phi_j \times \Delta MP_t)$, where $\phi_j$ is log average establishment size in supersector $j$ (Census CBP, 2000) and $\Delta MP_t$ is the quarterly change in the Wu–Xia (2016) shadow federal funds rate (v3 specification). The v3 results are reported in Appendix ??; v2 is the baseline throughout.

the orange-shaded region (1997Q1–2000Q4) marks the pre-JOLTS period that is available for the $\delta \to u$ specification but not for the vacancy LP. The left panel shows substantial cross-sectional dispersion throughout, with the interquartile range widening sharply during the GFC. Negative values in some states and quarters reflect lower-than-predicted closing rates given prevailing aggregate conditions and industry composition; since the local projection absorbs the cross-state mean via time fixed effects, identification comes from cross-sectional variation in residualized exposure, which remains well-defined regardless of sign.

The right panel compares the z-scored cross-state means of the raw and residualized instruments ($r = 0.905$). The two series track each other closely throughout the sample, indicating that residualization does not materially alter identifying variation in normal times. Two episodes diverge. During the GFC, the residualized mean rises somewhat more than the raw instrument at 2008Q4 ($z = 1.51$ vs. 0.95), consistent with residualization removing attenuating aggregate and industry-level variation. The more striking divergence would appear in 2020Q2 (not shown, excluded from LP sample): the raw Bartik spikes to $z = 2.33$ while the residualized instrument falls to $z = -0.45$, because 2020Q2 BED closings were dominated by government-mandated temporary shutdowns that correlate strongly with industry composition and aggregate conditions; residualization correctly removes this variation, leaving below-average permanent destruction. This episode illustrates precisely why the LP sample is capped at 2019Q4.

The cross-state mean of the instrument exhibits a pronounced secular decline over the sample, falling roughly one-third from the late 1990s to the late 2010s. This mirrors the well-documented decline in aggregate labor market fluidity: the Shimer (2005) quarterly separation rate fell proportionally over the same period, and the ratio of the two series is nearly constant throughout (coefficient of variation 14%), indicating that permanent establishment destruction has declined in tandem with broader separation margins rather than becoming relatively more or less important over time.

Two features of the left panel deserve note. First, the cross-sectional percentile bands are narrow relative to the time-series movement of the mean: the within-quarter p90–p10 spread averages 0.045 pp, a coefficient of variation of about 15%. The pooled standard deviation across all state-quarters ($\hat{\sigma} = 17.4$ pp) is therefore predominantly a time-series number, reflecting the common component of the instrument that moves all states together. The time fixed effects in the local projection absorb this common component; identification comes from the within-period cross-sectional deviations, which have a typical magnitude of roughly 0.02 pp. Second, some state-quarter cells take negative values, reflecting lower-than-predicted closing rates given industry composition and aggregate conditions; these are well-defined as deviations from the cross-state mean and cause no identification problem.

### 4.3. Local projection specification

For each horizon $h = 0, 1, \ldots, 20$ quarters, the panel local projection estimating equation is

$$y_{s,t+h} - y_{s,t-1} \;=\; \alpha_s + \alpha_t + \beta_h\, \tilde{B}^{\delta}_{s,t} + \gamma_1\, y_{s,t-1} + \gamma_2\, \log L_{s,t-1} + \varepsilon_{s,t+h},$$

where $y_{s,t}$ is the labor market outcome of interest for state $s$ in quarter $t$, $\tilde{B}^{\delta}_{s,t}$ is the residualized $\delta$ Bartik instrument constructed in Section 4.2, $\alpha_s$ and $\alpha_t$ are state and time fixed effects, $y_{s,t-1}$ is the lagged outcome, and $L_{s,t-1}$ is the lagged labor force. The left-hand side measures the cumulative $h$-period change relative to the quarter before the shock, so the sequence $\{\hat{\beta}_h\}$ traces the impulse response directly. Standard errors are clustered at the state level ($n = 50$).

The two outcomes are the state unemployment rate $u_{s,t}$ and vacancy rate $v_{s,t}$, both in percentage points (unemployed persons or vacancies as a share of the labor force). Concretely, $v_{s,t} = 100 \times V_{s,t}/L_{s,t}$, where $V_{s,t}$ is the quarterly average of monthly JOLTS job-openings stocks (seasonally adjusted) and $L_{s,t}$ is the LAUS labor force. The two specifications differ only in the autoregressive control $y_{s,t-1}$.

The estimation sample runs from 2001Q1, when JOLTS state vacancy data become available, through 2019Q4; observations with $t + h >$ 2019Q4 are dropped to exclude COVID-19.[15] The instrument $\tilde{B}^{\delta}_{s,t}$ has cross-sectional standard deviation $\hat{\sigma}_{\tilde{B}^{\delta}} = 17.4$ pp; all reported $\hat{\beta}_h$ are scaled by $\hat{\sigma}^{-1}_{\tilde{B}^{\delta}}$ so that the impulse responses correspond to a one-standard-deviation shock.

We also construct Bartik instruments for layoffs-and-discharges $\tilde{B}^{\mathrm{LD}}$ (JOLTS LD) and quits $\tilde{B}^{\mathrm{QU}}$ (JOLTS quits), using the same 2006 QCEW base-year shares and v2 residualization. Neither is treated as a well-identified separate estimation target. $\tilde{B}^{\mathrm{LD}}$ conflates genuine match dissolution at surviving establishments with vacancy withdrawals at partially closing firms, mixing $s$- and $\delta$-type events in proportions that vary across industries and cycles; moreover,

[15] The $\delta$ unemployment equation can be estimated back to 1997Q1 using the Barnichon (2010) Composite Help-Wanted Index as a pre-JOLTS vacancy proxy; extending the sample attenuates the IRF modestly (peak +1.07 pp vs. +1.50 pp) because the 1997–2000 expansion years exhibit faster reabsorption of displaced workers. We use the 2001Q1 baseline throughout to maintain a common sample across both outcomes.

$\tilde{B}^{\delta}$ Granger-causes future LD through $h = 12$ while the reverse predictability fades by $h = 8$, indicating that LD is partly endogenous to destruction. $\tilde{B}^{\mathrm{QU}}$ is procyclical and driven largely by job-to-job transitions, which represent a worker gain for one employer offset by a hire at another and have no clear mapping to the model's match separation shock $s$; we include it as a diagnostic instrument only.[16] Both instruments and their local projections are reported in Appendix C.

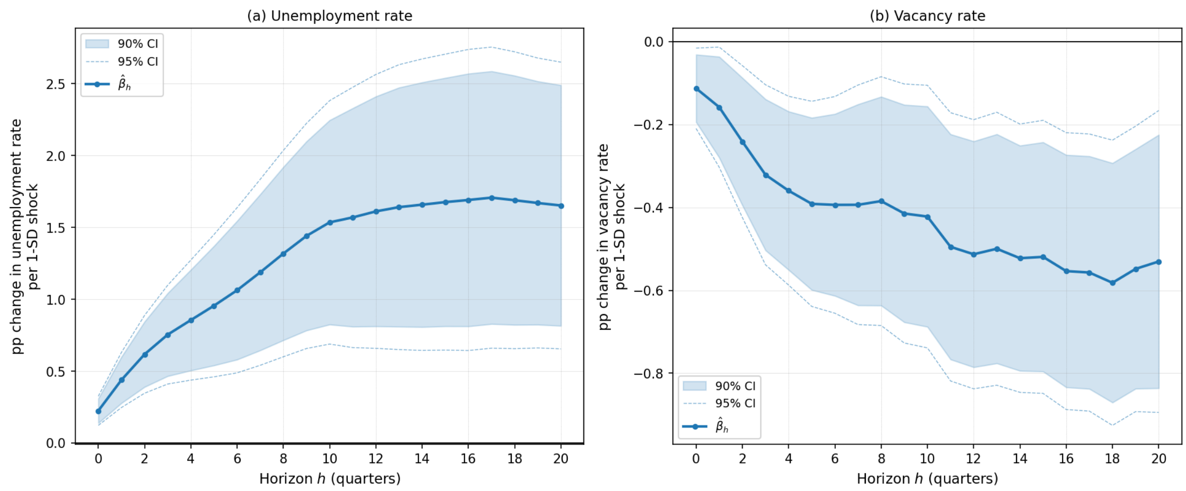


Figure 8: Impulse responses of the state unemployment rate (left) and vacancy rate (right) to a one-standard-deviation increase in the residualised $\delta$ Bartik instrument ($\hat{\sigma}_{\tilde{B}^{\delta}} = 17.4$ pp, 2001Q1–2019Q4). Both outcomes are estimated from panel local projections with state and time fixed effects; standard errors are clustered at the state level ($n = 50$). Shaded band: 90% confidence interval; dashed lines: 95% confidence interval. The instrument is BED establishment deaths (equation (48)), residualized on aggregate productivity growth and lagged industry value-added growth (equation (49))

## *4.4. Results*

Figure 8 reports the impulse responses of the state unemployment rate and vacancy rate to a one-standard-deviation increase in the residualized $\delta$ Bartik instrument ($\hat{\sigma}_{\tilde{B}^{\delta}} = 17.4$ pp, 2001Q1–2019Q4 sample). The unemployment response rises steadily from 0.22 pp at $h = 0$ to a plateau of +1.71 pp by $h = 17$–20, remaining highly significant throughout ($p < 0.001$ from $h = 0$). The vacancy response falls persistently from $-0.11$ pp at $h = 0$ to a trough of $-0.58$

[16] In particular, a high quit rate in an industry does not imply that vacancies go unfilled — the quitting worker is typically replaced by a job-to-job mover from elsewhere, so the net effect on aggregate vacancies is near zero. This is conceptually distinct from the $s$ shock in the model, which destroys a match and triggers a vacancy repost.

pp at $h = 18$, significant at $p < 0.01$ from $h = 2$ onward and remaining so through $h = 20$. The simultaneous and persistent rise in unemployment alongside the fall in vacancies traces the outward leg of a Beveridge loop, consistent with the model's central prediction that permanent product-line destruction depletes the vacancy stock rather than merely suspending it.

A notable feature of the two IRFs is their asymmetric dynamics at medium horizons: the vacancy response stabilizes around $h = 12$–$14$ while the unemployment response continues to rise through $h = 20$. This pattern is consistent with a recovery path in which new vacancy creation resumes before the unemployment pool fully clears. Whether the calibrated model replicates this relative timing is testable and will be assessed in Section 5.

A natural concern is that the $\delta$ Bartik instrument may inadvertently capture recession severity: states with structurally high establishment turnover may also experience worse aggregate downturns, so the IRF could reflect demand-driven labor-market deterioration rather than product-line destruction per se. We address this with a severity placebo that augments the baseline local projection with two interactions: $\tilde{B}_t^{\delta} \times \bar{u}_t^{\text{nat}}$, a level proxy for underlying economic slack, and $\tilde{B}_t^{\delta} \times \Delta\bar{u}_t^{\text{nat}}$, a flow proxy for cyclical momentum. If the instrument were a severity proxy, these interactions should absorb the baseline IRF. Instead, both interaction coefficients are insignificant at every horizon $h = 0, \ldots, 20$ for the vacancy outcome, and the baseline $\delta \to v$ coefficients are unchanged in magnitude and significance. This is the cleanest external validity check in the paper: the vacancy response to establishment destruction is not explained by where the aggregate economy happens to stand in the cycle. The severity placebo also passes for the unemployment outcome, though the interpretation is less sharp given the natural correlation between any adverse state-level shock and local unemployment. Robustness results for the alternative instruments and the joint local projection are collected in Appendix C.

## 5. Quantitative analysis

### *5.1. Relationship to Coles and Kelishomi*

Our framework nests Coles and Moghaddasi Kelishomi (2018) as a special case, as formalized in Proposition 4. Relative to their calibration, we make two departures. First, we set $\delta$ to the BED Deaths employment-weighted exit rate ($\bar{\delta} \approx 1.1\%$ per quarter) rather than equating it to the full aggregate separation rate; Gabrovski and Silva (2025) show this correction materially improves the model's fit on the unemployment–productivity correlation. Second, we expand the estimation moment set to include the permanent exit rate $\delta$ and the entry flow $N_t^e$ (proxied by BFS high-propensity business applications), and supplement unconditional second moments with identified IRFs from the Bartik local projections of Section 4 to separately discipline the propagation dynamics of $\delta$ and $s$ shocks.

### *5.2. Calibration and estimation*

This section sets out our estimation and calibration strategy, the taxonomy of parameters, and the explicit mapping from empirical targets to model objects.

**Estimation framework.** We estimate the model by Bayesian simulated method of moments, combining two complementary sets of moment conditions that together over-identify the parameter vector.

The first set consists of unconditional second moments of HP-filtered aggregate time series, extending Coles and Moghaddasi Kelishomi (2018). We target three groups: (i) standard deviations ($m_1$), (ii) contemporaneous pairwise correlations ($m_2$), and (iii) first-order autocorrelations ($m_3$) for unemployment, vacancies, separations, the job-finding rate, the permanent exit rate, business applications, and labor productivity. The full empirical moment matrix is reported in Table 3. All series are log-transformed and HP-filtered with smoothing parameter $\lambda = 1{,}600$ before computing moments; the identical filter is applied to simulated model data when evaluating $m(\theta)$. This symmetric treatment renders HP-filtered second moments a valid indirect inference target: Canova and Ferroni (2011) establish that applying the same filter to data and simulated output controls for any filter-induced distortion, even when the filter introduces spurious dynamics in isolation. We prefer $\lambda = 1{,}600$ over the $\lambda = 100{,}000$ convention of Coles and Moghaddasi Kelishomi (2018) because the higher smoothing parameter strips out the business-cycle frequency variation in $\delta$ at which the destruction mechanism operates, reversing the sign of $\mathrm{cor}(\delta, v)$ and driving $\mathrm{cor}(\delta, u)$ to near zero — an artifact of over-smoothing rather than a feature of the data. Appendix Appendix C.6 documents this explicitly: across the Hamilton (2018) regression filter, first differences, and HP with $\lambda = 1{,}600$, the correlations $\mathrm{cor}(\delta, u) > 0$ and $\mathrm{cor}(\delta, v) < 0$ are robust; $\lambda = 100{,}000$ is the anomalous specification.

Table 3: Business Cycle Moments

| | $\sigma(x)$ | Contemporaneous correlations | | | | | | | $\rho_1(x)$ |
|---|---|---|---|---|---|---|---|---|---|
| | | $u$ | $v$ | $\tau$ | $f$ | $\delta$ | $N^e$ | $z$ | |
| Unemployment | 0.1496 | 1 | | | | | | | 0.792 |
| Vacancies | 0.1419 | -0.804 | 1 | | | | | | 0.902 |
| Separations | 0.1019 | 0.577 | -0.422 | 1 | | | | | 0.250 |
| Job-finding rate | 0.0977 | -0.791 | 0.620 | 0.020 | 1 | | | | 0.808 |
| Exit rate | 0.0793 | 0.310 | -0.246 | 0.681 | 0.213 | 1 | | | 0.114 |
| Entry flow ($N^e$) | 0.0813 | -0.105 | 0.482 | -0.345 | -0.226 | -0.496 | 1 | | 0.545 |
| Labor productivity | 0.0128 | -0.009 | 0.306 | -0.228 | -0.212 | -0.250 | 0.381 | 1 | 0.766 |
| *Derived statistics* | | | | | | | | | |
| $\sigma(\theta)/\sigma(z)$ | 11.70 | | | | | | | | |

*Note:* HP filter with $\lambda$ = 1,600 applied to log-levels of each series. Correlations use pairwise-complete observations; the shortest series ($N^e$, entry flow) begins 2004Q3. Sources — $u$: BLS LAUS; $v$: Barnichon (2010) composite HWI spliced with BLS JOLTS at 2001Q1; $s$, $f$: Shimer (2012) continuous-time flows from CPS; $\delta$: BLS BED Deaths (dataclass 08), employment-weighted; $N^e$: BFS high-propensity business applications (BAWBATOTALSAUS) per capita, proxy for model entry flow; $z$: BLS PRS85006163 (output per hour, nonfarm business).

Table 3 highlights several salient features of the data. Unemployment and vacancies are highly volatile relative to labor productivity ($\sigma(u)/\sigma(z) \approx 11.7$, $\sigma(v)/\sigma(z) \approx 11.1$) and display a pronounced Beveridge curve ($\text{cor}(u, v) = -0.80$). The unemployment-labor productivity correlation is near zero ($\text{cor}(u, z) = -0.01$), consistent with weak technology-shock transmission in models with endogenous separation. The exit rate co-moves strongly with the match separation rate ($\text{cor}(\delta, \tau) = 0.68$), reflecting a common cyclical component of firm destruction and worker-job dissolution. Importantly, the exit rate is positively correlated with unemployment ($\text{cor}(\delta, u) = 0.31$) and negatively correlated with vacancies ($\text{cor}(\delta, v) = -0.25$), consistent with the simultaneous vacancy-destruction mechanism central to our model.[17] Labor market variables and entry display strong persistence ($\rho_1 \in [0.55, 0.90]$), while separations and the exit

[17] These signs reverse under $\lambda = 10^5$, an artifact of over-smoothing documented inAppendix C.6.

rate exhibit substantially less ($\rho_1 \approx 0.11$–$0.25$), consistent with their role as shock-propagation mechanisms rather than persistent states.

The second set consists of empirically identified impulse responses from the Bartik local projections of Section 4: the $h = 0, \ldots, H$ quarter paths of unemployment and vacancies following a one-standard-deviation $\delta$ shock. Following Christiano, Eichenbaum, and Trabandt (2016), we form a Gaussian quasi-likelihood based on the sampling distribution of the LP estimator:

$$\log \mathcal{L}_{\text{IRF}}(\theta) = -\tfrac{1}{2}(\hat{\beta} - \beta(\theta))' \Omega_\beta^{-1} (\hat{\beta} - \beta(\theta)),$$

where $\hat{\beta}$ stacks the two estimated IRF paths, $\beta(\theta)$ is the corresponding model-implied path computed from the linearized state-space representation, and $\Omega_\beta$ is the sampling covariance estimated by wild cluster bootstrap with $n = 50$ state clusters.

The quasi-likelihood is a limited-information Bayesian update: rather than conditioning on the raw panel data, it conditions on $\hat{\beta}$ as a sufficient statistic for the model-implied path $\beta(\theta)$. Writing the full likelihood $\mathcal{L}(\text{data}|\theta)$ for the panel would require the model to generate 50 state-level employment series, cross-state variation in industry composition, and the full Bartik instrument—objects entirely outside a representative-agent aggregate economy. Even setting this aside, any reduced-form likelihood over $\hat{\beta}$ would require distributional assumptions on the LP residuals, the instrument's conditional distribution, and the finite-sample behavior of a clustered two-stage estimator—auxiliary assumptions with no grounding in the structural model. The quasi-likelihood avoids both problems by relying only on the asymptotic normality of $\hat{\beta}$, which follows from the LP moment conditions alone (**?**).

What is the value of using the IRF block instead of relying on the structural likelihood on aggregate time series? The answer is identification. The full likelihood recovers structural parameters from the time-series covariance structure of observables, but in a three-shock model the unconditional second moments cannot separately discipline the propagation of $\delta$ and $s$ shocks: both raise unemployment, and their vacancy responses—the qualitative distinction between them—are collinear in second moments because they depend jointly on shock persistence, sunk-cost dynamics, and the entry margin. The Bartik design provides the only clean exogenous variation in $\delta$ and therefore the only moment condition that separately identifies the parameters governing the model's central mechanism. Omitting the IRF block would leave the asymmetric vacancy-destruction channel weakly identified from time-series covariances alone.

A filtering asymmetry between the two blocks is present by design: unconditional moments are statistics of HP-filtered data, while IRF coefficients are estimated from level regressions. This is resolved on the model side. Model moments $m(\theta)$ are computed from simulated-then-filtered data, mirroring the data construction. Model IRFs $\beta(\theta)$ are read directly from the state-space impulse response without filtering, mirroring the LP estimation. The two objects therefore target genuinely distinct features of the model—cyclical second-moment structure and shock-specific propagation dynamics—in a way that is internally consistent on both the data and model sides.

The two blocks provide complementary identification. Unconditional moments discipline the level and cyclical volatility of the destruction and separation processes, the amplification and comovement structure, and the persistence of key aggregates. The IRF block separately identifies the dynamic transmission of $\delta$ shocks—in particular, the simultaneous elimination of vacancies that distinguishes $\delta$ from $s$ and generates persistent Beveridge-curve dislocations—identification unavailable from unconditional moments alone, which conflate the contributions of all three shocks.

To prevent the IRF block from mechanically dominating the joint objective by virtue of moment count, we normalize each block's quadratic form by its dimension. Letting $K_\beta$ denote the number of IRF moment conditions and $K_m$ the number of unconditional moments, the log-posterior kernel is

$$\log p(\theta \mid \text{data}) \propto \log \pi(\theta) - \frac{1}{2K_\beta}(\hat{\beta} - \beta(\theta))'\Omega_\beta^{-1}(\hat{\beta} - \beta(\theta)) - \frac{1}{2K_m}(\hat{m} - m(\theta))'\Omega_m^{-1}(\hat{m} - m(\theta)),$$

where $\hat{m}$ collects the empirical moments from Table 3, $m(\theta)$ the corresponding model-implied moments from simulated HP-filtered data, and $\Omega_m$ a HAC covariance estimated by block bootstrap with block length eight quarters. The HP filter induces serial correlation in cyclical residuals at all lags, so HAC correction is required regardless of the choice of $\lambda$; the block length of eight quarters is sufficient to cover the effective filter width at $\lambda = 1{,}600$. The degrees-of-freedom normalization ensures each block contributes equally in expectation when the model is correctly specified, since each quadratic form is approximately $\chi^2(K)$ with expectation $K$ under the null (Dridi, Guay, and Renault, 2007). Appendix **??** reports robustness to rescaling the unconditional moment block by factors of one-half and two.

**Parameter taxonomy.** We partition the full parameter space $\Theta$ into three blocks.

*Estimated block* $\Theta_e$: structural parameters and shock processes whose identification comes from business-cycle moments. This block contains the replacement value $b$; the sunk-cost share $x_v$ of total recruiting expenditure (which maps to the fixed matching cost $\kappa$ in Stage 4 below); the inverse elasticity of entry with respect to vacancy value $\xi^{-1}$; the endogenous exit share $\omega_\delta \equiv (1 - F(\chi^c))/\delta_e$; the mass weight $p_0$ on the continuous part of the continuation-cost distribution $F$; the inverse intertemporal elasticity of substitution $\sigma$; and the persistence $\rho_x$ and conditional standard deviation $\sigma_x$ of each shock $x \in \{z, \delta, s\}$.

*External block*: parameters set directly from microeconomic evidence or long-run data. We fix the annual discount rate $r = 4\%$ and the matching-function elasticity $\eta_L = 0.6$. Following Jaimovich and Floetotto (2008), 21% of job destruction is attributable to establishment exit; combined with an aggregate monthly separation rate of $\tau = 3.1\%$, this implies a monthly destruction rate $\delta_e = 0.651\%$ (annual: 7.54%). We set the elasticity of substitution $\varepsilon = 4.3$, implying a gross markup $\mu = \varepsilon/(\varepsilon - 1) \approx 1.30$.

*Dependent block* $\Theta_d$: parameters implied jointly by the normalizations, the external block, and a draw from $\Theta_e$ via the four-stage calibration procedure described below. We normalize the steady-state mass of firms $N = 1$ and the wage $w = 1$. The remaining parameters—$z$, $\phi$, $f_e$, $A$, $s$, $\psi$, $\chi_m$, $x_m$, $\kappa$, $\delta$—are recovered from the following targets:

**Calibration algorithm.** Given a draw from $\Theta_e$ and the external block, we recover $\Theta_d$ in four sequential stages.

*Stage 1: direct conversions (targets → fundamentals).* The destruction rate target $\delta_e^{ann}$ and endogenous share $\omega_\delta$ pin

$$\delta_e = 1 - (1 - \delta_e^{ann})^{1/12}, \qquad \delta = (1 - \omega_\delta)\, \delta_e.$$

The match separation rate follows from the aggregate target:

$$s = \frac{\tau - \delta_e}{1 - \delta_e}.$$

The empirical job-finding and vacancy-filling rates are corrected for destruction before use: $f = f^{emp}/(1 - \delta_e)$, $q = q^{emp}/(1 - \delta_e)$. Market tightness $\theta = f/q$, the unemployment rate $u = \tau/(\tau + (1 - \delta_e)f)$, and the matching-function level parameter $A = f/\theta^{1-\eta_L}$ are then computed analytically.

Table 4: Calibration targets for the dependent block $\Theta_d$. Job-finding and vacancy-filling rates are gross JOLTS rates, corrected by $(1-\delta_e)$ in Stage 1. $X/Y$ and $X^c/Y$ are external targets; all other $\Theta_e$ parameters listed in the estimated block are drawn by the SMM sampler.

| Target | Value | Source | Parameters identified |
|---|---|---|---|
| Ann. destruction rate $\delta_e^{ann}$ | 7.54% | BED Deaths | Monthly $\delta_e$; with $\omega_\delta \in \Theta_e$: exogenous $\delta$ |
| Job-finding rate $f^{emp}$ | 41% | JOLTS | Corrected rate $f$, tightness $\theta$, level $A$ |
| Vacancy-filling rate $q^{emp}$ | 80% | JOLTS | Corrected rate $q$, tightness $\theta$ |
| Separation rate $\tau$ | 3.1% | Shimer (2005) | Match separation $s = (\tau - \delta_e)/(1-\delta_e)$ |
| Recruiting share $X/Y$ | 1.5% | CET 2016 | Vacancy value $Q$; then $\kappa$, $z$, $f_e$, $x_m$, $\phi$ |
| Fixed cost share $X^c/Y$ | 10% | See text | Retail ratio $X^c/Y^c$, profit share $\pi_s$ |
| Wage $w = 1$ | — | — | Productivity $z$ |
| Firm mass $N = 1$ | — | — | Entry cost $f_e$ |

*Stage 2: profit share (root-find $X^c/Y^c$).* The fixed-cost share of retail output $X^c/Y^c$ is not directly observed; we recover it by solving

$$\frac{X^c/Y}{(Y^c/Y^{Gross})(Y^{Gross}/Y)} = \frac{X^c}{Y^c},$$

where $Y^c/Y^{Gross} = (r+\delta_e)/(r+\delta_e+\delta_e\pi_s)$ and $Y^{Gross}/Y = 1 + X/Y + X^c/Y$. The profit share $\pi_s = 1/\varepsilon - X^c/Y^c$ is a residual: retail value added equals the Lerner markup $1/\varepsilon$ after paying stochastic fixed costs.

*Stage 3: cost-distribution parameters (root-find $\psi$).* Given $\pi_s$, we recover the mass fraction of continuing cost draws, *cons*, from

$$\pi_s = \frac{\mu - 1}{\mu}\,(1 - cons)\,\frac{r+\delta_e}{r+\delta_e+cons\,(1-\delta_e)},$$

then map *cons* to the power-law shape parameter $\psi = (cons/p_0)\,/\,(1 - cons/p_0)$ and location parameter $\chi_m = \chi^c/\varsigma^{1/\psi}$, where $\varsigma = ([(1-\delta_e)/(1-\delta)] - (1-p_0))/p_0$ is the quantile of the exit threshold within the continuous part of the cost distribution. The mass weight $p_0 \in \Theta_e$ is estimated: given $\pi_s$, only the product $cons = (\psi/(1+\psi))p_0$ is pinned by Stage 3, so $p_0$ and $\psi$ are not separately identified from steady-state targets alone and require business-cycle moment restrictions.

*Stage 4: vacancy value (root-find $Q$).* We solve for the steady-state vacancy value $Q$ by requiring the recruiting-cost share to equal its target: $X/Y = 1.5\%$. Given a trial value $Q$, the contact-rate value $K = (r+\delta_e)Q/(1+r)$ and the fixed matching cost $\kappa = (1-x_v)/x_v \cdot K/q$ follow from the sunk-cost split $x_v$. Total recruiting expenditure $X = e/(1+\xi^{-1}) \cdot Q + \kappa q v$, the interior wage $w^{int}$, and firm-level output $Y^c = \rho z L^c$ are computed sequentially. The value of $Q$ is chosen so that output implied by the recruiting-cost target, $Y = X/(X/Y)$, coincides with expenditure-side output $Y = C + \nu^f N^e$. The remaining dependent parameters are then recovered analytically:

$$z = (\mu/\rho)\, w^{int}, \tag{50}$$

$$f_e = \pi_s\, z\, L^c\, (1-\delta_e)\, \mu \,/\, (N\,(r+\delta_e)), \tag{51}$$

$$\phi = \frac{w-b}{w^{int} - K + \theta(K + q\kappa) - b}, \tag{52}$$

$$x_m = Q \,/\, e^{\xi^{-1}}.$$

The normalizations provide clean intuition: $w = 1$ pins $z$ via (50) because the interior wage $w^{int} = \rho z/\mu$ is proportional to productivity; $N = 1$ pins $f_e$ via (51) because free entry ties the mass of product lines to the entry cost; and the recruiting-cost share target $X/Y = 1.5\%$, once $Q$ is found, pins bargaining power $\phi$ via (52).

**Discussion of $b$ and $\phi$.** In our labor market framework, the labor share and the search wedge $w/w^{int}$ both depend on Nash bargaining power $\phi$ and the worker outside option $b$. Shimer (2005) interprets $b$ as unemployment insurance and calibrates it to 0.4, finding that technology shocks generate an order-of-magnitude too little unemployment and vacancy volatility, attributing this to excessively wage-elastic labor markets. Hagedorn and Manovskii (2008) show that even adjusting $\phi$ to match the mild empirical wage cyclicality leaves tightness volatility unchanged; what matters is the proportional response of profits to productivity. We resolve this debate by estimating $b$ via simulated method of moments and, given the draw of $b$, selecting $\phi$ internally to satisfy (52)—ensuring that the wage normalization and labor-share implications are mutually consistent. Our model links market power with search frictions, making the labor share a natural discipline on both $\phi$ and $b$. Matching the consumption share and labor share jointly also implies a realistic stock-market capitalization-to-output ratio without sacrificing fit on business-cycle moments.

### *5.3. Inspecting the mechanism*

Here we quantify novel features of the model mechanism. Across specifications, we set replacement-income ratio $b/w^{int} = 0.9$, a sunk-cost share of recruiting expenditure $x_v = 0.5$, and an endogenous exit share $\omega_\delta = 0.5$, so that half of steady-state destruction operates through the continuation-cost margin. The exit-elasticity target is $\psi\varsigma/(1-\varsigma) = 5$—the percent increase in the endogenous exit rate for a one-percent rise in the continuation-cost threshold—which pins the shape parameter $\psi \approx 0.033$ analytically.[18] The resulting steady state has a monthly destruction rate $\delta_e \approx 0.653\%$, exogenous component $\bar{\delta} \approx 0.326\%$, total separation rate $\tau = 3.1\%$, and unemployment $u \approx 6.8\%$.

**Shock calibration.** The stochastic processes for technology and firm destruction are disciplined by a bivariate structural VAR(1) estimated at quarterly frequency on the HP($\lambda = 1{,}600$)-filtered cyclical components of log labor productivity $z_t$ (BLS nonfarm business output per hour, PRS85006163) and log aggregate firm destruction $\delta_t$ (BLS BED Deaths, national employment-weighted exit rate) over 1992Q3–2019Q4 ($T = 110$).[19] The reduced-form system is

$$\begin{pmatrix} \log z_t \\ \log \delta_t \end{pmatrix} = A_1 \begin{pmatrix} \log z_{t-1} \\ \log \delta_{t-1} \end{pmatrix} + \mathbf{u}_t, \qquad \mathbf{u}_t \sim \mathcal{N}(\mathbf{0}, \Sigma),$$

with estimated companion matrix and pre-identification innovation correlation:

$$\hat{A}_1 = \begin{pmatrix} 0.735 & -0.002 \\ -1.050 & 0.207 \end{pmatrix}, \qquad \widehat{\text{corr}}(u_t^z, u_t^\delta) = -0.284.$$

Structural shocks are identified by a Cholesky decomposition ordering technology first, consistent with the within-period timing in which aggregate productivity is realized before firm-level

[18] With the survival quantile $\varsigma \approx 0.993$ at the calibrated exit rate, feasibility limits the destruction elasticity to roughly 10 before implied fixed-cost shares exceed 18 % of output.

[19] The $\delta$ series is the national employment-weighted BED Deaths exit rate—the same underlying series that enters the Bartik instrument's national rates—aggregated directly rather than Bartik-weighted. Using the Bartik aggregate would conflate the shock process with the instrument construction.

continuation decisions. With $P$ the lower-triangular Cholesky factor of $\hat{\Sigma}$, the structural innovations are $\varepsilon_t^z = u_t^z / P_{11}$ and $\varepsilon_t^\delta = (u_t^\delta - P_{21}\varepsilon_t^z)/P_{22}$, so $\mathrm{Cov}(\varepsilon^z, \varepsilon^\delta) = 0$ by construction.

The two off-diagonal entries of $\hat{A}_1$ are excluded from the model's exogenous shock processes, for distinct reasons. The entry $\hat{a}_{21} = -1.050$ (HC3 SE $= 0.830$) reflects the tendency of exit to rise following productivity declines; this channel is generated endogenously by the model through the JCC and the continuation-cost threshold $\chi_t^c$, so including it in the exogenous process would double-count the mechanism under study. The entry $\hat{a}_{12} = -0.002$ is two orders of magnitude smaller in absolute value and is consistent with $z$ being an autonomous technology process unaffected by market-structure shocks. Treating both off-diagonal terms as equilibrium outcomes rather than primitive forcing processes leaves the diagonal entries as the persistence parameters of the exogenous shocks.

The quarterly diagonal persistence parameters are converted to monthly via $\rho^m = (\rho^Q)^{1/3}$, reflecting the cube-root relationship between quarterly and monthly AR(1) coefficients. Innovation standard deviations are rescaled to match the unconditional variance of the quarterly process: $\sigma^m = \sigma^Q \sqrt{(1-(\rho^Q)^2)/(1-(\rho^m)^2)}$. This yields $\rho_z = 0.902$, $\sigma_z = 0.0092$ for technology and $\rho_\delta = 0.592$, $\sigma_\delta = 0.0669$ for destruction, entering equations (7) and (8) directly. The destruction shock has a monthly half-life of roughly two months, so prolonged unemployment responses to $\delta$ shocks reflect propagation through the firm and vacancy stocks rather than the persistence of the driving process itself.

**Choice of shocks and comparisons.** We report impulse responses to the $z$ and $\delta$ shocks throughout and abstract from the $s$ shock. Proposition 5 establishes that a positive $s$ shock generates weakly positive $u$–$v$ comovement — the opposite of the Beveridge curve — so the $s$ shock does not engage the vacancy-destruction and variety channels that are central to the model's amplification. The $s$ shock therefore provides a useful theoretical benchmark rather than a quantitative focus. The $z$ and $\delta$ shocks, by contrast, interact with the firm-stock state $N_t$ through the variety channel in equation (22) and through the firm and vacancy laws of motion in equations (30) and (6), which is precisely where the model departs from the augmented Gabrovski-Silva benchmark.

We organize the comparisons around three counterfactuals. *Comparison A* examines the role of the level of $\delta_e$ at steady state. To facilitate comparison, it abstracts from endogenous entry entirely: $p_0 = 0, \delta_e = \delta$. The baseline sets ($\delta_e = \bar{\delta}$ and replaces the endogenous component by

$s$: $s = (\tau - \delta_e)/(1 - \delta_e)$. The high-$\delta$ counterfactual sets $\delta_e = \tau$, and thus $s = 0$, which replicates the Coles and Moghaddasi Kelishomi (2018) setup in which all separations are from destruction events. fixed. This comparison extends Gabrovski and Silva (2025), who document the LOM-multiplier mechanism for technology shocks, to the $\delta$-shock dimension. *Comparison B* isolates the endogenous exit margin: we set $p_0 = 0$ and equate the exogenous destruction parameter $\bar{\delta}$. We keep the aggregate separation rate $\tau$ the same, which requires $s$ in the counterfactual to compensate for the loss of endogenous destruction. *Comparison C* sets $\zeta = 0$ so that $\rho_t \equiv 1$ regardless of $N_t$, removing the $N_t \to \rho_t \to w_t^{int} \to$ JCC channel and quantifying what the baseline offers beyond the augmented Gabrovski-Silva framework without product variety. The markup $\mu = \varepsilon/(\varepsilon - 1)$ is preserved in both arms; only the love-of-variety externality is shut off.

**Comparison A: the level of destruction.** Figures 9 and 10 report impulse responses to one-standard-deviation shocks. To place both on a common scale in which unemployment rises, the $z$ figure shows a *negative* technology shock and the $\delta$ figure shows a positive destruction shock. Each panel traces the transmission chain from labor market outcomes (rows 1–2) through the extensive margin (row 3) to the vacancy-creation chain (row 4). Appendix Appendix E.5 derives the log-linearized propagation equations that underlie the discussion below.

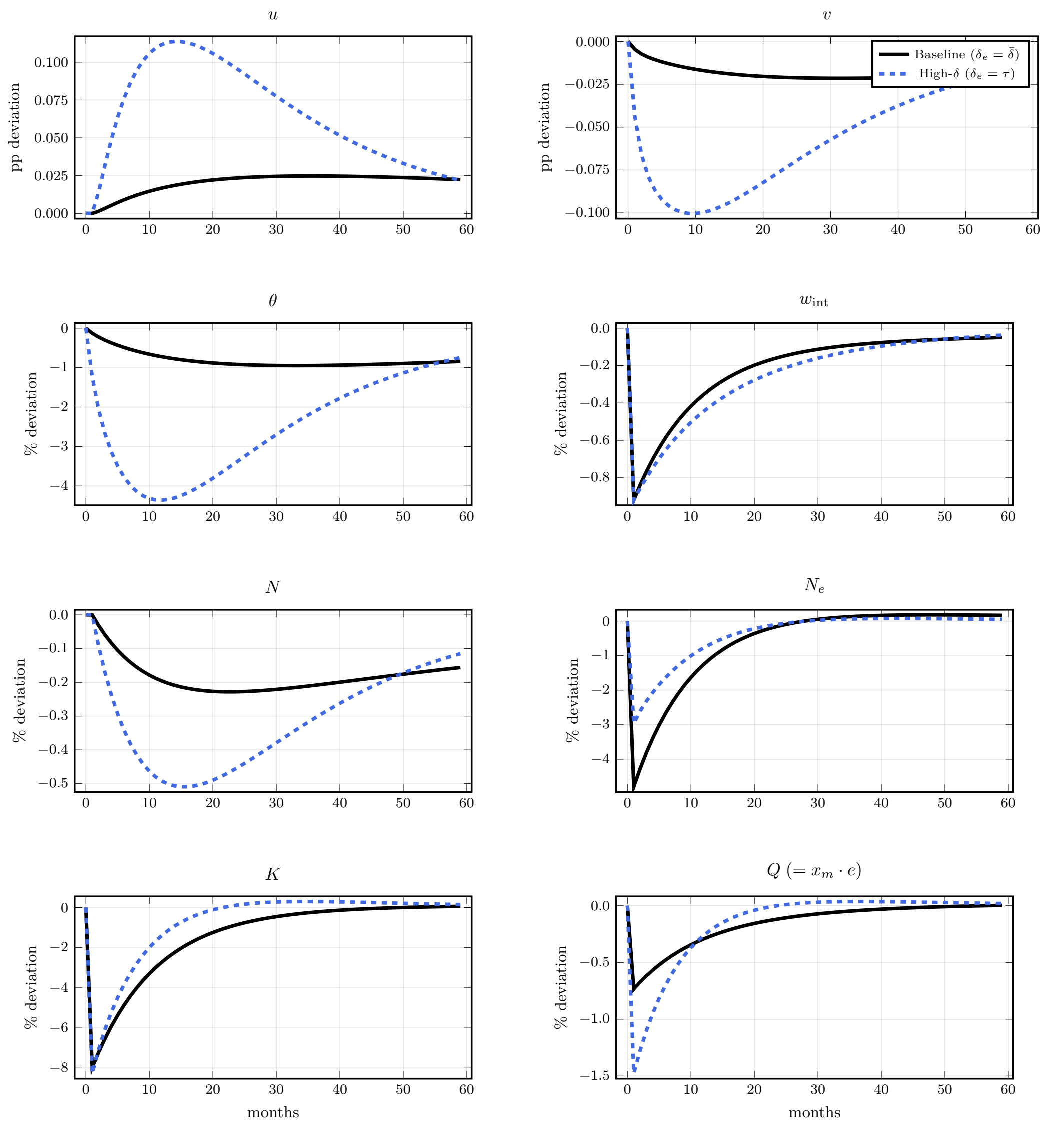


Figure 9: Comparison A — impulse responses to a one-standard-deviation *negative* technology shock. Solid black: baseline ($\delta_e = \bar{\delta} \approx 0.653\%$/month). Dashed blue: high-$\delta$ variant ($\delta_e = \tau = 3.1\%$/month; CK timing, $p_0 = 0$). Row 1 in percentage-point level deviation from steady state; rows 2–4 in $100 \times$ log deviation.

Under the *negative technology shock* (Figure 9), the high-$\delta$ variant generates a substantially larger response of $N_t$ despite a nearly identical response of the flow vacancy value $K_t$. Log linearize the firm law of motion (30)

$$\hat{N}_{t+1} = (1 - \delta_e)\,\hat{N}_t \;+\; \delta_e\,\hat{N}^e_t, \tag{53}$$

where hats denote log-deviations from steady state and we use the steady-state relation $N^e/N = \delta_e/(1 - \delta_e)$. The loading on the entry flow equals $\delta_e$ itself, so the four-fold increase in $\delta_e$ from baseline to the high-$\delta$ variant translates a given entry response into a commensurately larger accumulation of product lines. A complementary channel operates through the vacancy value. Log-linearising (35) and using $\mathbb{E}_t\hat{Q}_{t+1} = \rho\hat{Q}_t$, which holds exactly under a one-shock approximation by undetermined coefficients,[20] yields

$$\hat{Q}_t = \underbrace{\frac{r+\delta_e}{1+r-\rho(1-\delta_e)}}_{\mathcal{D}(\delta_e,\rho)}\hat{K}_t,$$

where $\beta = 1/(1+r)$ and $\rho \in \{\rho_z, \rho_\delta\}$ is the persistence of the driving shock. The multiplier $\mathcal{D}$ is strictly *increasing* in $\delta_e$, since $\partial\mathcal{D}/\partial\delta_e = (1+r)(1-\rho)/\text{den}^2 > 0$. The mechanism is that a higher survival discount $\beta(1-\delta_e)$ in the baseline causes the continuation term $\beta(1-\delta_e)\mathbb{E}_t\hat{Q}_{t+1}$ in (35) to nearly offset $\hat{Q}_t$, dampening the transmission from $\hat{K}_t$ to $\hat{Q}_t$. In the high-$\delta$ variant the continuation term is smaller, so $\hat{Q}_t$ — and through $\hat{e}_t = \xi\hat{Q}_t$ in (36), new postings — respond more strongly on impact, as visible in row 4 of Figure 9. Both $\mathcal{D}$ and the LOM entry-loading $\delta_e$ in (53) therefore favour the high-$\delta$ variant on impact. However, the baseline generates more persistence via the survival term $(1-\delta_e) \approx 0.993$ in (53), which prolongs the $\rho_t \downarrow \to w_t^{int} \downarrow \to \theta_t \downarrow \to u_t \uparrow$ chain documented above.

Under the *destruction shock* (Figure 10), the same two forces operate as under the technology shock, but now both scale with $\delta_e$ because the shock itself directly destroys product lines through (53). The high-$\delta$ variant therefore suffers a deeper $N$ trough on impact — the destruction term is larger — but also recovers faster, since the replacement rate $N^e/N = \delta_e/(1-\delta_e)$ and the entry-loading $\mathcal{D}$ are both larger. The net effect on persistence is thus a race between a deeper initial hit and faster repair, and the figures show that recovery speed dominates: the high-$\delta$ variant converges markedly faster. This is consistent with the limiting case: as $\delta_e \to 0$ the $\delta$ shock becomes irrelevant regardless of persistence, so the baseline necessarily generates a smaller but longer-lived response. In the baseline, the monthly replacement rate is only 0.66% of the firm stock, so after a destruction impulse the variety depression is long-lived: $\rho_t$ stays depressed, $w_t^{int} = \rho_t z_t/\mu$ remains low, and the job creation condition (34) stays tight, prolonging the recovery of $\theta_t$ and $u_t$. The central quantitative message of Comparison A is therefore the same across both shocks: a model calibrated to the empirical ratio $\delta_e/\tau \approx 10\%$ necessarily operates with a low-churning firm stock, generating persistent responses through the slow $N$-

[20] Guess $\hat{Q}_t = \phi_Q\hat{\delta}_t$; then $\mathbb{E}_t\hat{Q}_{t+1} = \phi_Q\rho_\delta\hat{\delta}_t = \rho_\delta\hat{Q}_t$. For a $z$ shock replace $\rho_\delta$ with $\rho_z$. The full perturbation solution additionally accounts for the $N_t$ state but the qualitative result is unchanged.

recovery channel independently of endogenous exit or variety effects. For the destruction shock specifically, the low level of $\delta_e$ also limits the initial amplitude of the $N$ disruption, so the model produces a modest but highly persistent unemployment response — precisely the shape visible in the empirical IRF of Section 4.

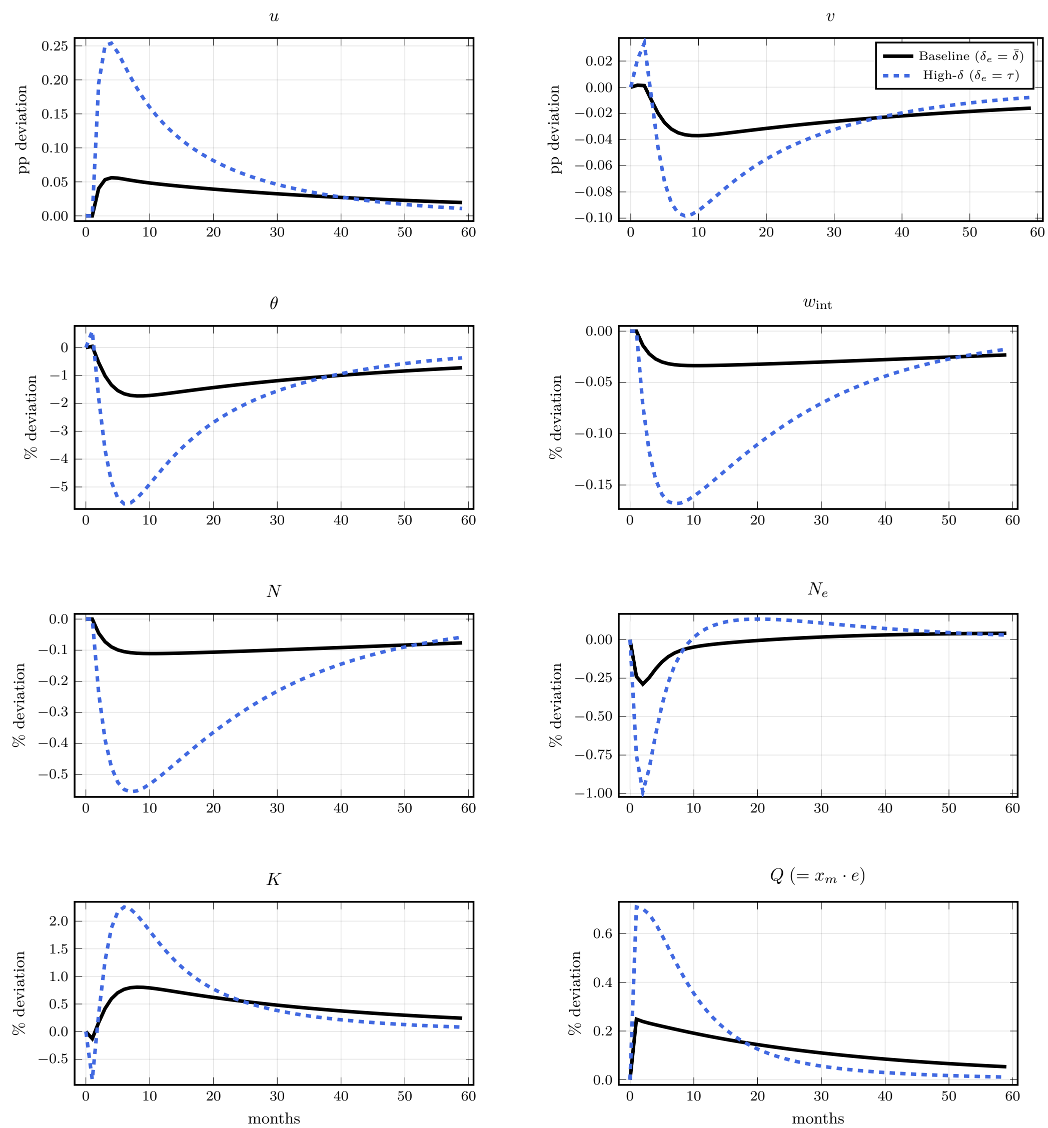


Figure 10: Comparison A — impulse responses to a one-standard-deviation destruction shock. Lines and units as in Figure 9.

**Comparison B: endogenous versus exogenous exit.** Figures 11 and 12 compare the full baseline with an exogenous-exit model in which $p_0 = 0$ and $\omega_\delta = 0$. The exogenous exit variant keeps the same rate of exogenous destruction $\bar{\delta}$ and aggregate separation $\tau$, which requires raising $s$ accordingly. By equations (30), (6), and (5), this keeps steady-state $u$, $v$, and $\theta$ equal so that percentage-point IRF differences are directly comparable.

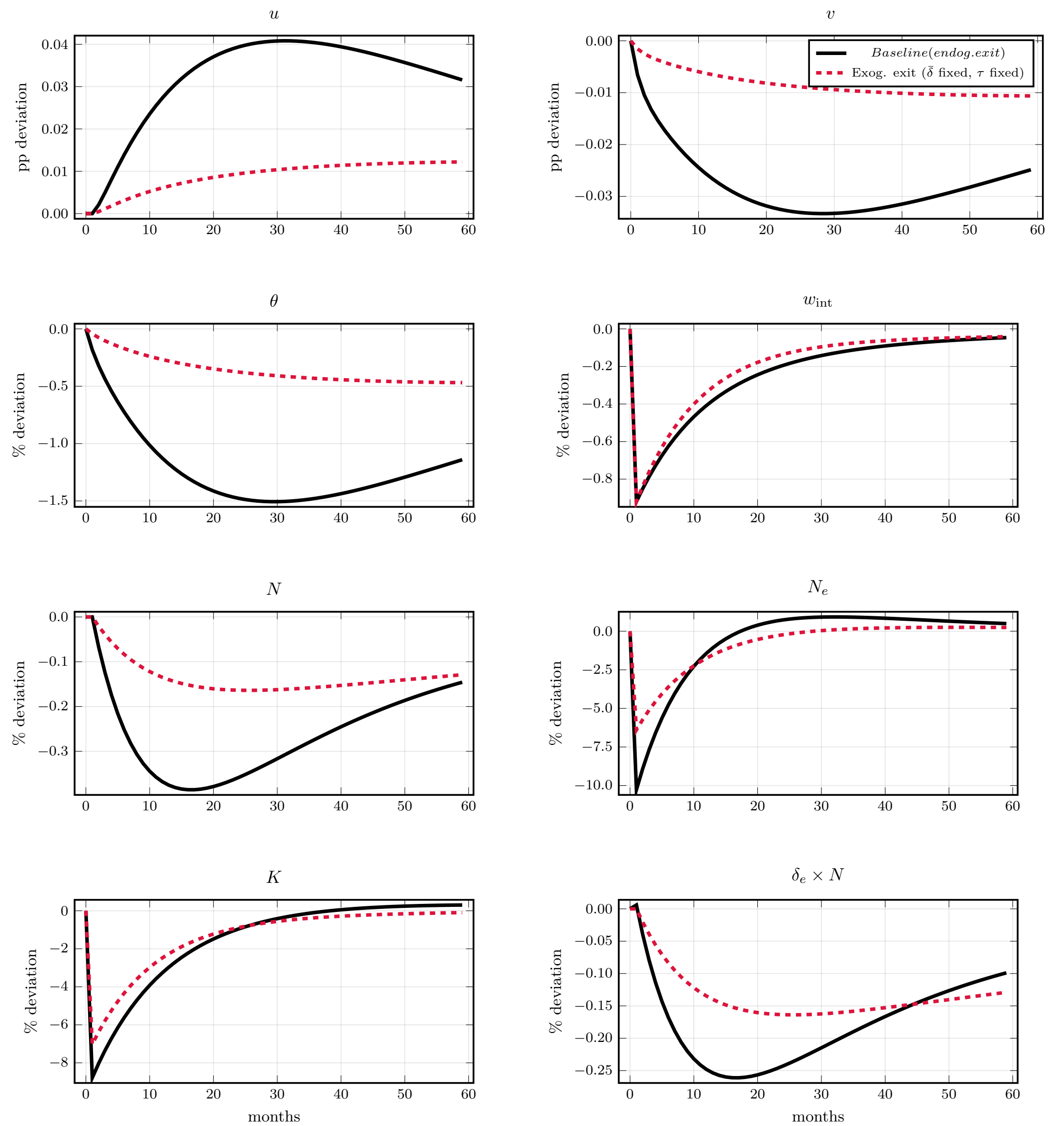


Figure 11: Comparison B — impulse responses to a one-standard-deviation *negative* technology shock. Solid black: full baseline (endogenous exit, $\omega_\delta = 0.5$, $\varepsilon_x = 5$). Dashed crimson: exogenous exit ($\bar{\delta}$ equated, $\tau$ fixed, $s$ rises to compensate). Units as in Figure 9.

For the *negative technology shock* (Figure 11), the endogenous-exit model generates a larger response of $N_t$ that builds over time. The continuation threshold satisfies

$$\chi_t^c = \underbrace{\frac{1}{\varepsilon}\frac{Y_t^c}{N_t}}_{\text{per-firm profit}} + \underbrace{\nu_{f,t}}_{\text{cont. value}}, \tag{54}$$

where $\nu_{f,t} = \rho_t f_e/\mu$ from free entry (25). A negative $z$ shock lowers $Y_t^c = \rho_t z_t \bar{z} L_{c,t}$ immediately, compressing per-firm profit and lowering $\chi_t^c$ on impact, which induces exit. Two forces govern subsequent dynamics. First, $N_t$ itself has an ambiguous effect on $\chi_t^c$: a fall in $N$ raises per-firm profit (market-share effect, elasticity $(2-\varepsilon)/(\varepsilon-1)$) but simultaneously lowers the continuation value $\nu_f = \rho_t f_e/\mu$ through $\rho_t = N_t^{1/(\varepsilon-1)}$ (elasticity $1/(\varepsilon-1)$). At $\varepsilon \approx 4.3$, the profit-share elasticity ($\approx -0.70$) dominates the continuation-value elasticity ($\approx 0.30$), so falling $N$ tends to *raise* $\chi^c$ — a stabilizing feedback that partly offsets the initial $z$-driven decline. Second, the endogenous model calibrates a lower sunk entry cost $f_e$: with endogenous exit, steady-state profits are lower (some continuation costs are paid), so free entry requires a smaller $f_e$, amplifying entry responses at any given $\nu_f$. The exit-flow panel shows that the two models' exit flows are similar on impact but diverge at the peak, reflecting the net of these competing forces as they cumulate over the transition path; the elasticity $\varepsilon_x = \psi\varsigma/(1-\varsigma) = 5$ governs the local sensitivity of exit to the threshold throughout.

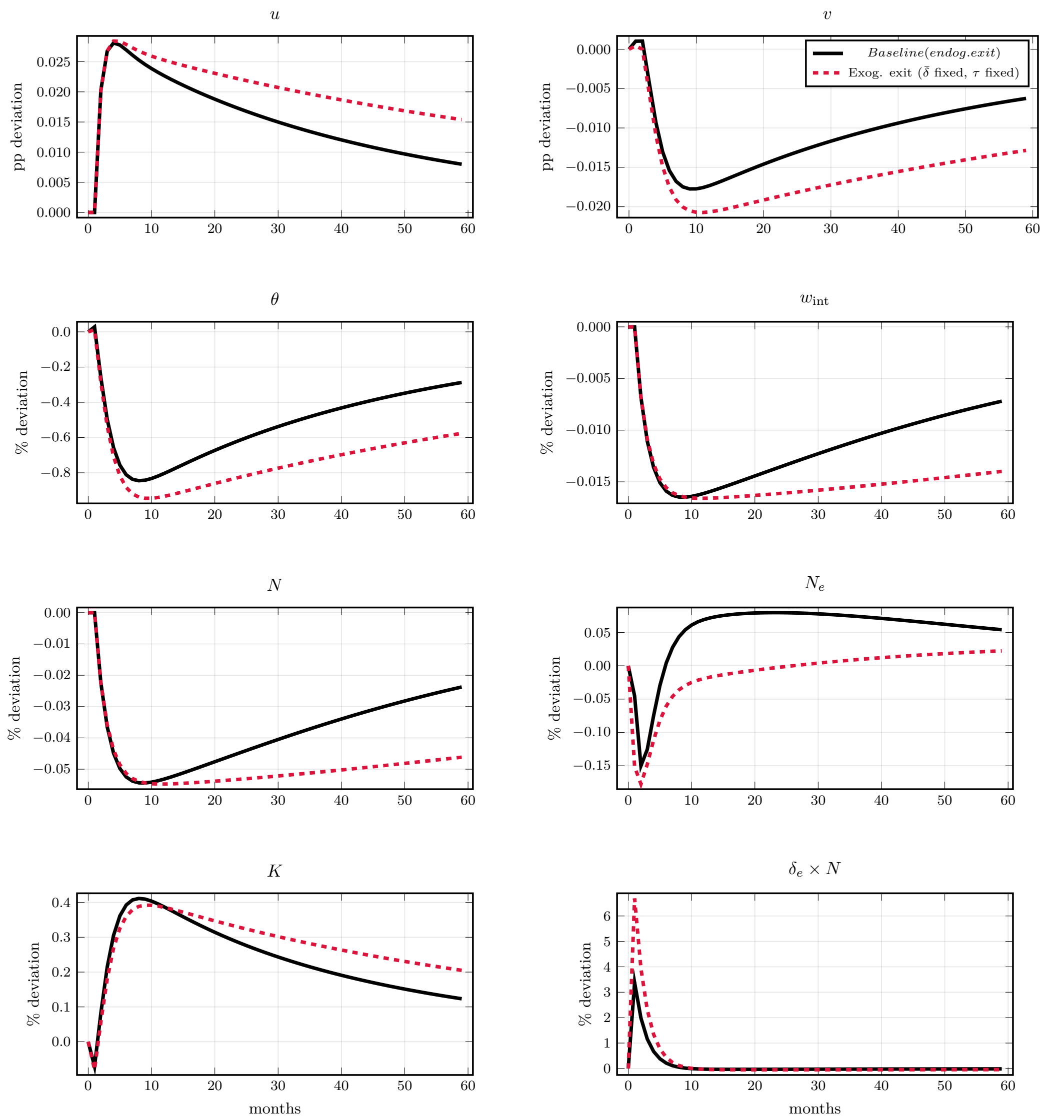


Figure 12: Comparison B — impulse responses to a one-standard-deviation destruction shock. Lines and units as in Figure 11.

For the *destruction shock* (Figure 12), the design equates $\bar{\delta}$ across both models so that a proportional $\delta$ shock delivers the same first-period impulse to $\delta_{e,t}$ in both. Impact responses in $u$ and $N$ are therefore similar. Recovery, however, is markedly faster in the endogenous-exit model, for the same reason identified in Comparison A: the steady-state level of $\delta_e$ governs mean reversion. To equate $\bar{\delta}$ while eliminating endogenous exit, the exogenous model must set $\delta_{e,\text{exog}} = \bar{\delta}_{\text{endog}} \approx 0.326\%/\text{month}$, roughly half of $\delta_{e,\text{endog}} \approx 0.653\%$. From the N law of motion (53), the entry coefficient equals $\delta_e$ and the survival coefficient equals $(1-\delta_e)$: higher $\delta_e$ in the endogenous model means both faster replenishment of the firm stock and stronger mean reversion of any $N$ deviation. The exogenous model's low-churning firm stock therefore keeps

$N_t$ depressed for longer, sustaining the $\rho_t \downarrow \rightarrow w_t^{int} \downarrow \rightarrow \theta_t \downarrow$ chain and prolonging unemployment. Comparison B thus shows, as a mirror image of Comparison A, that equating $\bar{\delta}$ while reducing $\delta_e$ produces greater persistence — the endogenous-exit specification recovers faster precisely because its higher steady-state destruction rate implies a higher replacement rate.

The $\chi^c$ margin contributes a secondary channel for the destruction shock as well, though the direction is ambiguous. A $\delta \uparrow$ impulse directly destroys firms, lowering $N_t$; by (54) this raises per-firm profit (market-share effect) but lowers $\nu_f$ through $\rho_t$. The net effect on $\chi^c$ — and hence on additional endogenous exit — depends on which elasticity dominates along the transition, a quantitative question the IRFs resolve. The dominant effect of endogenous exit in Comparison B remains the structural difference in the replacement rate: the $\tau$-fixing constraint halves $\delta_e$ in the exogenous model, slowing $N$-recovery and prolonging unemployment.

**Comparison C: variety effects.** The third comparison sets $\zeta = 0$ so that $\rho_t \equiv 1$ regardless of $N_t$, removing the link between product variety and the interior wage in equation (22). The markup $\mu = \varepsilon/(\varepsilon - 1)$ is unchanged; only the love-of-variety externality is shut off. Figures 13 and 14 report the results.

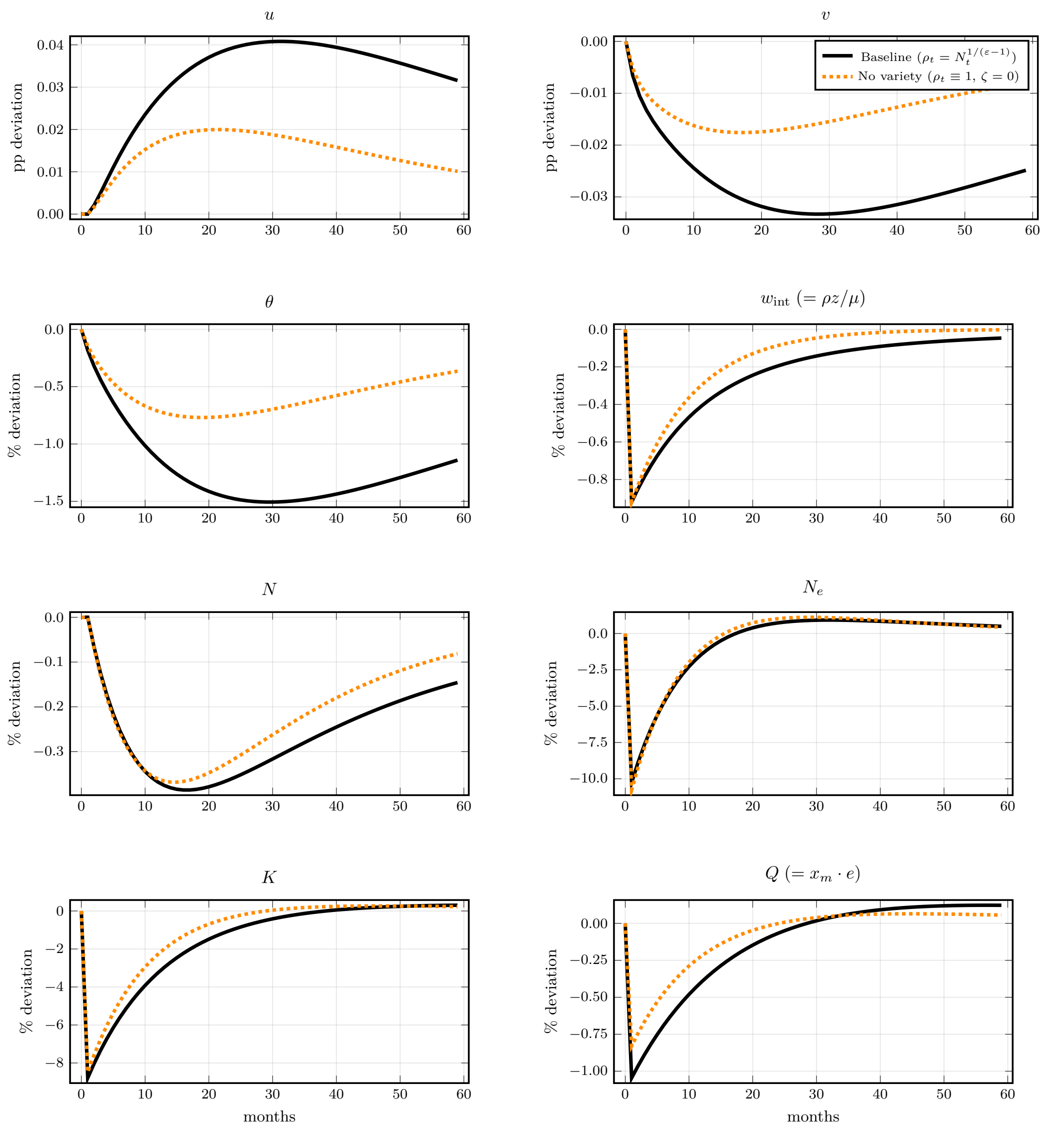


Figure 13: Comparison C — impulse responses to a one-standard-deviation *negative* technology shock. Solid black: baseline ($\rho_t = N_t^{1/(\varepsilon-1)}$, full variety). Dotted orange: no-variety counterfactual ($\rho_t \equiv 1$, $\zeta = 0$; markup $\mu$ unchanged). Row 1 in percentage-point level deviation from steady state; rows 2–4 in $100 \times$ log deviation.

Under the *negative technology shock* (Figure 13), the two models share the same direct $z \to w_t^{int} = \rho_t z_t/\mu$ channel, but diverge through the indirect variety channel. In the baseline, $z \downarrow$ reduces entry, $N_t$ falls, $\rho_t = N_t^{1/(\varepsilon-1)}$ falls, and $w_t^{int}$ declines by more than the direct $z$ impulse alone. With $\rho_t \equiv 1$, the second-round compression of $w_t^{int}$ is absent: the interior-wage panel shows a smaller response in the no-variety arm at all horizons, and the gap is the pure $N \to \rho$ contribution. Because $w_t^{int}$ pins the job-creation surplus (34), a smaller $w_t^{int}$ response implies a smaller $\theta_t$ response, which attenuates both the unemployment rise and the vacancy fall. The $w_t^{int}$ gap is visible from impact—$z$ directly enters $w_t^{int}$ even before $N$ adjusts, so the two lines diverge as the $N$ transition cumulates over the first few months.

The $N$ and $N_e$ panels show a further distinction. With $\rho_t \equiv 1$, the firm value $\nu_f = f_e/\mu$ is a constant: it does not rise when $N$ recovers, so the entry Euler (24) provides no forward-looking amplification of the entry margin. The no-variety $N$ response is consequently more compressed and recovers faster—entry is purely backward-looking in the counterfactual, governed only by current profits rather than the anticipated $\rho(N)$ appreciation.

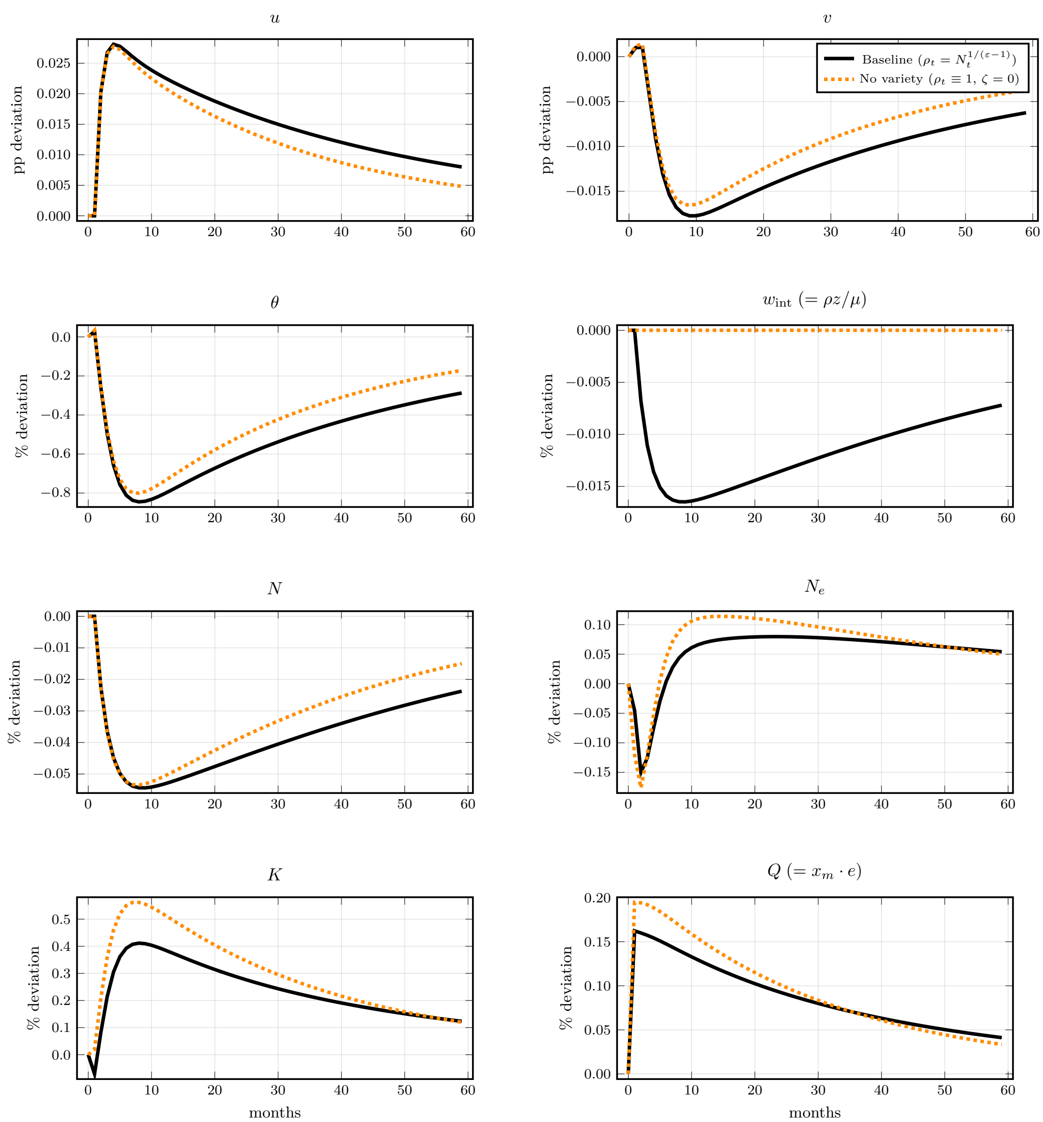


Figure 14: Comparison C — impulse responses to a one-standard-deviation destruction shock. Lines and units as in Figure 13.

Under the *destruction shock* (Figure 14), the variety channel works in the amplifying direction. A $\delta \uparrow$ impulse destroys firms, $N_t$ falls, $\rho_t$ falls, and $w_t^{int}$ falls beyond what the exogenous destruction alone would imply. The lower $w_t^{int}$ tightens the JCC—reducing the surplus firms extract from a filled job—and simultaneously depresses entry through the free-entry condition $\nu_f = \rho_t f_e/\mu$. Unemployment rises more and vacancies fall more in the baseline than in the no-variety arm because the same $N$ depression feeds back into wages and entry values in the full model. Absent taste for diversity, $\nu_f = f_e/\mu$ is fixed, so the entry Euler is unresponsive to the $N$ depression: recovery is faster but the initial impact is also attenuated, since the $\rho \to \nu_f$ mechanism that amplifies destruction shocks into a prolonged Beveridge-curve response is absent. In particular, $w_t^{int}$ does not respond at all.

Taken together, Comparisons A–C decompose the model's amplification into three orthogonal channels: the level of $\delta_e$ governing asset duration and the LOM multiplier (A), the $\chi^c$ margin governing endogenous exit amplification (B), and the $N \to \rho$ variety externality governing the wage and entry-value response (C). Each channel is active in both the technology and destruction directions, but with different relative magnitudes that the IRF comparisons quantify.

### *5.4. Posterior estimates*

## 6. Conclusion

## Appendix A. Relation to literature

| | Finitely elastic vacancies | Endogenous variety | Endogenous separation | Separate destruction margin |
|---|---|---|---|---|
| Bilbiie, Ghironi, and Melitz (2012) | NA | ✓ | NA | NA |
| Coles and Kelishomi (2018) | ✓ | × | × | × |
| Broer, Druedahl, Harmenberg, and Öberg (2025) | ✓ | × | ✓ | × |
| Gabrovski and Silva (2025) | ✓ | × | × | ✓ |
| Shao and Silos (2013) | ✓ | ✓ | × | × |
| Christiano, Eichenbaum, and Trabandt (2016) | × | × | × | × |
| Colciago, Fasani, and Rossi (2021) | × | ✓ | × | × |
| Cacciatore and Fiori (2016) | × | ✓ | ✓ | × |
| **This paper** | ✓ | ✓ | ✓ | ✓ |

Table A.5: Comparison of model ingredients. *Finitely elastic vacancies*: sunk vacancy creation costs generating a positive vacancy value. *Endogenous variety*: equilibrium number of product lines responds to economic conditions. *Endogenous separation*: firm exit driven by idiosyncratic profitability draws. *Separate destruction margin*: exogenous product-line destruction distinguished from match separation.

## Appendix B. Data sources

Table B.6: Data sources

| Series | Source | Frequency & coverage | Unit |
|---|---|---|---|
| *Panel A: Instrument construction* | | | |
| BED establishment deaths (dc 08) | BLS BED | Quarterly, 1992Q3–2024Q2 | National × supersector |
| QCEW employment shares | BLS QCEW | Annual, base year 2006 | State × supersector |
| Layoffs & discharges (LD) | BLS JOLTS | Monthly, 2001Q1–2023Q1 | National × supersector |
| Quits (QU) | BLS JOLTS | Monthly, 2001Q1–2023Q1 | National × supersector |
| *Panel B: Residualization controls* | | | |
| Labor productivity (OPHNFB) | BLS, FRED | Quarterly, 1947Q1–present | National |
| Value added by industry | BEA, FRED | Quarterly, 2005Q1–present | National × supersector |
| *Panel C: LP outcome variables* | | | |
| Unemployment rate | BLS LAUS | Monthly, 1990M1–present | State |
| Civilian labor force | BLS LAUS | Monthly, 1990M1–present | State, thousands |
| Job openings | BLS JOLTS | Monthly, 2001M1–present | State, thousands |
| *Panel D: Bayesian estimation moments* | | | |
| Job-finding rate ($f_t$) | BLS CPS | Monthly, 1948M1–present | National |
| Separation rate ($s_t$) | BLS CPS | Monthly, 1967M1–present | National |
| Business formations (BF8) | Census BFS | Quarterly, 2004Q3–present | National |

Panel A instruments are constructed as Bartik shift-share aggregates. BED deaths (dataclass 08) counts employment at establishments absent for four or more consecutive quarters, excluding temporary shutdowns by construction. QCEW shares are fixed at the 2006 base year and leave-one-out corrected. LD and QU are summed from monthly to quarterly; QU serves as a placebo instrument (Appendix Appendix C.5). Panel B controls enter the v2 residualization as first log-differences: aggregate productivity growth (OPHNFB) and lagged industry value-added growth (12 BEA supersector series via FRED: VAPGDPIP, VAMFNIP, VACONIP, VATRDIP, VATRIIP, VAINIP, VAFNSIP, VAPBSIP, VAEDUHIP, VALAHIP, VAOSIP, VAGOSIP); the value-added series are extended to 2001Q1 via Chow–Lin temporal disaggregation using BLS employment as the indicator. Panel C outcomes are averaged from monthly to quarterly; the vacancy rate is defined as job openings divided by civilian labor force, expressed in percentage points. Panel D moments are national aggregates used as targets in Bayesian estimation. Job-finding and separation rates are constructed from CPS unemployment flows with the continuous-time correction for time-aggregation bias (Shimer, 2005). BF8 counts business applications with planned wages that appear in QCEW within eight quarters, and proxies for the model entry flow $N_t^e$.

## Appendix C. Instrument diagnostics and robustness

### *Appendix C.1. Deaths/Closings ratio by supersector*

Figure C.15 plots the Deaths/Closings ratio for each of the 12 BLS supersectors separately, divided into the six with the highest mean ratio (left panel) and the six with the lowest (right panel). Across all sectors and quarters the ratio remains strictly below unity, confirming that BED Deaths are always a subset of BED Closings. The cross-sector range is wide: Manufacturing, Retail, and Information cluster near 0.70, while Mining (0.37) and Construction (0.52) are substantially lower, consistent with the prevalence of project-based and cyclically suspended establishments in those sectors. The ratio is broadly stable within each supersector over time, with no systematic cyclical pattern, supporting the use of a fixed Deaths/Closings fraction as the implicit permanence assumption in our instrument.

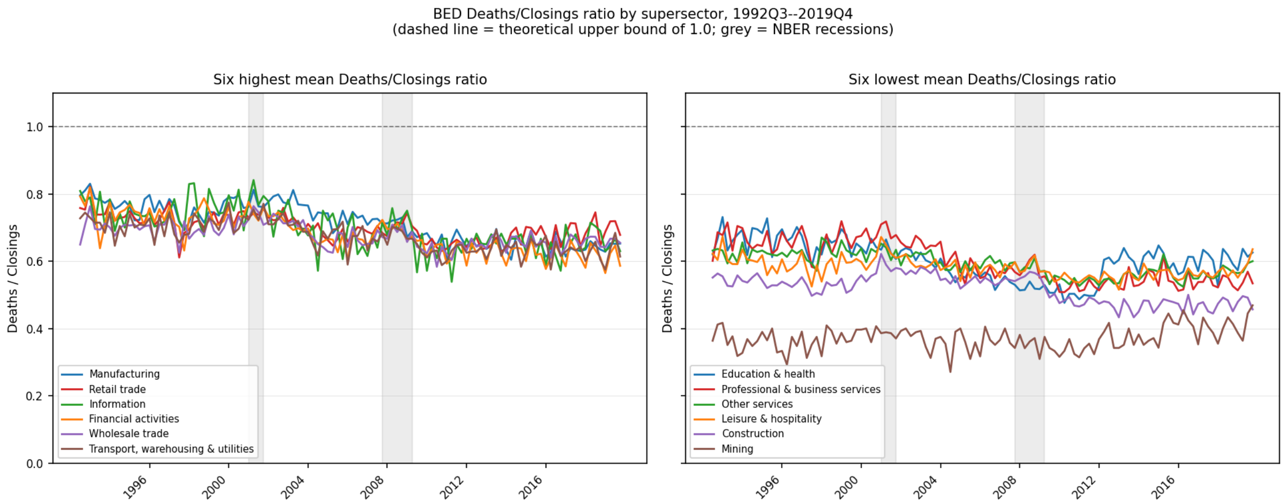


Figure C.15: BED Deaths/Closings ratio by supersector, 1992Q3–2019Q4. *Left panel:* six supersectors with highest mean ratio. *Right panel:* six supersectors with lowest mean ratio. Dashed horizontal line at 1.0 is the theoretical upper bound (Deaths $\subseteq$ Closings by construction). Grey shading denotes NBER recessions.

### *Appendix C.2. Residual δ–LD correlation*

After v2 residualization the cross-state correlation between the $\delta$ and layoffs-and-discharges instruments is $r(\tilde{B}^{\delta}, \tilde{B}^{\text{LD}}) = 0.334$, down from 0.54 on raw rates. The residualization controls for aggregate productivity growth, lagged industry value-added, and lagged labor-market tightness, ruling out the most obvious industry-demand and aggregate-cycle explanations for the shared variation. The leading interpretation is industry-specific financial conditions: when credit tightens in a sector, both product-line exit ($\uparrow \delta$) and layoffs at surviving establishments ($\uparrow$ LD) rise simultaneously, generating positive covariance without a structural link between the two shocks. Consistent with this, the Granger-causality evidence shows that $\delta$ predicts future LD at horizons $h = 4$–$12$ while LD predicts future $\delta$ only at $h = 0$–$4$; the long-horizon dependence runs from destruction to layoffs and not the reverse. A natural further step would be to residualize on a measure of sector-level credit supply, such as SLOOS net C&I tightening interacted with industry leverage exposure; we leave this for future work.

### *Appendix C.3. LD local projections*

The separate LD local projections are subsumed by the joint LP figure (Figure C.16 below), which overlays the LD separate-specification IRF alongside the joint-LP coefficient for the same instrument. In the separate specification the LD unemployment response rises monotonically to a peak of +1.68 pp at $h = 20$, significant throughout, while the vacancy response is persistently negative (trough $-0.45$ pp at $h = 20$). Both patterns survive the severity placebo. The monotonically rising unemployment IRF is anomalous given the estimated LD persistence $\hat{\rho}_{\mathrm{LD}} \approx 0.3$: a transitory shock should produce a hump-shaped response, not a path that continues to rise five years out. The most likely explanation is GFC contamination — the 2008–09 recession produced a severe and prolonged layoff episode whose industry-composition variation is not fully absorbed by the v2 residualization — rather than a structural property of the $s$ shock. We regard the LD IRFs as indicative rather than structural, and do not use them as estimation targets.

*Appendix C.4. Joint local projection*

Figure C.16 reports the joint versus separate local projections for the vacancy outcome. Each panel overlays the joint-LP coefficient (red, solid) against the corresponding separate-LP coefficient (blue, solid), with 90% confidence bands.

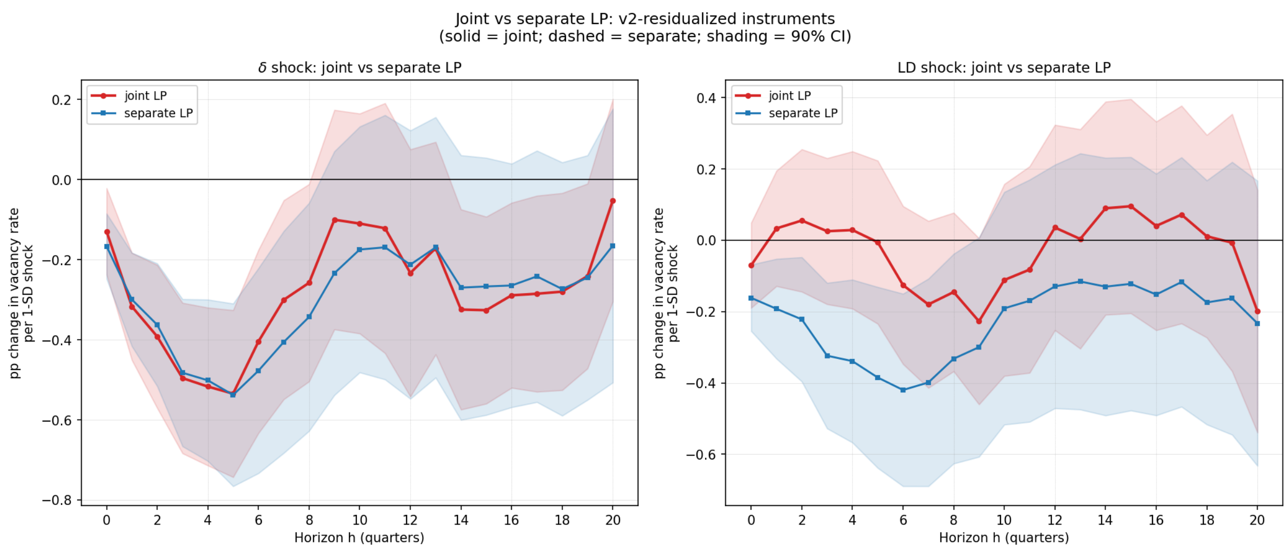


Figure C.16: Joint versus separate local projections: vacancy rate outcome. *Left panel:* $\delta$ instrument — joint LP (red) vs. separate LP (blue). *Right panel:* LD instrument — joint LP (red) vs. separate LP (blue). Shading denotes 90% confidence intervals; standard errors clustered at the state level ($n = 50$). Instruments are v2-residualized; sample 2001Q1–2019Q4.

Three findings emerge from the figure. First, the $\delta$ vacancy IRF is qualitatively stable across specifications: the joint and separate coefficients share the same sign, timing, and shape throughout, and the confidence bands overlap at most horizons. The trough in the joint specification ($-0.535$ pp at $h = 5$) is somewhat shallower and earlier than in the separate specification

($-0.582$ pp at $h = 18$), reflecting partial absorption of the shared variation with LD, but the persistent negative response to product-line destruction is not an artifact of LD exposure.

Second, the $\delta$ unemployment IRF shrinks by roughly 0.6–0.8 pp in the joint specification relative to the separate LP, reflecting absorption of the variation that $\tilde{B}^{\delta}$ and $\tilde{B}^{\mathrm{LD}}$ share. The joint peak of +1.15 pp remains positive and significant, so the unemployment result is robust, if attenuated.

Third, the LD vacancy coefficient collapses from its separate-LP value to approximately zero and is insignificant at all horizons once $\delta$ is controlled for (right panel). This collapse should *not* be read as evidence that $\delta$ subsumes a genuine LD effect on vacancies. Rather, it reflects that $r(\tilde{B}^{\delta}, \tilde{B}^{\mathrm{LD}}) = 0.334$ is large enough that the two instruments do not span orthogonal directions in the shock space: the variation in $\tilde{B}^{\mathrm{LD}}$ orthogonal to $\tilde{B}^{\delta}$ is insufficient to separately identify the LD vacancy response. The appropriate conclusion is that LD is not separately identified as a structural shock in this setting, not that separation shocks have zero causal effect on vacancies.

We therefore treat the separate $\delta$ LP as the primary specification. The joint LP establishes that the headline result — a persistent vacancy decline in response to product-line destruction — is robust to controlling for LD, and that the failure to separately identify LD is an instrument-space problem rather than a finding about vacancy dynamics.

*Appendix C.5. QU placebo instrument*

$\tilde{B}^{\mathrm{QU}}$ is constructed from JOLTS quit rates using the same shares and residualization as the primary instruments. Quits represent voluntary job-to-job transitions: the quitting worker moves directly to another employer, leaving the origin firm with an unfilled position but supplying the destination firm with a worker. The net effect on aggregate unemployment and vacancies is therefore ambiguous and governed by matching frictions at the destination, not by the separation event at the origin. This is conceptually distinct from the model's $s$ shock, which dissolves a match and triggers an unambiguous vacancy repost. Furthermore, the quit rate is procyclical — it rises when labor markets tighten — so $\tilde{B}^{\mathrm{QU}}$ partly reflects the same industry-level demand cycle that v2 residualization attempts to remove.

Empirically, the QU instrument does not behave as a flat-zero placebo. Its unemployment and vacancy IRFs are nearly identical in magnitude to those of $\tilde{B}^{\delta}$, and the within-instrument correlation $\mathrm{corr}(\hat{\beta}^{u}_{h}, \hat{\beta}^{v}_{h}) = -0.971$ across $h = 0, \ldots, 20$ — the same mechanical proportionality observed for $\tilde{B}^{\delta}$ ($-0.948$). This signature indicates that QU identifies industry-level demand contractions (which move $u$ and $v$ symmetrically along the Beveridge curve) rather than a structurally distinct separation shock. The contamination is likely structural: high-quit industries tend to be procyclical, so the industry component of the quit rate is correlated with demand conditions even after residualization on aggregate productivity and lagged VA. The QU failure confirms that a well-identified $s$-type instrument remains an open challenge and motivates treating the LD results as indicative rather than structural.

*Appendix C.6. Filter robustness for $\delta$-related moments*

Table 3 is computed with HP filter $\lambda = 1{,}600$, departing from the $\lambda = 10^5$ convention of (Shimer, 2005; Coles and Moghaddasi Kelishomi, 2018) and other papers. Here we show that smoothening at $\lambda = 10^5$ generates anomalous $\delta$ dynamic correlations with respect to other filters.

Table C.7 reports dynamic cross-correlations $\text{cor}(\delta_t, x_{t+h})$ for $x \in \{u, v\}$ and $h \in \{-4, \ldots, +4\}$ under four conventions: HP with $\lambda = 1{,}600$ and $\lambda = 10^5$; the Hamilton (2018) regression filter (projection of $y_t$ on $y_{t-h}, \ldots, y_{t-h-3}$ with $h = 8$ quarters); and first differences. All series are log-transformed before filtering.

First, $\text{cor}(\delta_t, u_{t+h})$ is positive at $h = 0$ under every specification except $\lambda = 100^5$ (HP-100k: +0.02; HP-1600: +0.39; Hamilton: +0.19; FD: +0.60). Second, $\text{cor}(\delta_t, v_{t+h})$ is negative at $h = 0$ and $h = +1$ under HP-1600, Hamilton, and FD, while HP-100k delivers the opposite sign. HP-100k is flagged as an outlier at five of the nine horizons for the vacancy correlation.

Table C.7: Dynamic correlations cor$(\delta_t, x_{t+h})$ by detrending filter

| | $x$ = Unemployment ($u$) | | | | $x$ = Vacancies ($v$) | | | |
|---|---|---|---|---|---|---|---|---|
| $h$ | HP-1600 | HP-100k | Hamilton | FD | HP-1600 | HP-100k | Hamilton | FD |
| −4 | −0.265 | −0.411 | −0.332 | −0.013 | 0.296 | 0.557 | 0.448 | 0.175 |
| −3 | −0.343 | −0.435 | −0.334 | −0.115 | 0.211 | 0.490 | 0.357 | 0.068 |
| −2 | −0.271 | −0.365 | −0.257 | −0.018 | 0.089 | 0.372 | 0.229 | 0.032 |
| −1 | −0.197 | −0.309 | −0.176 | −0.414 | −0.029 | 0.281 | 0.085 | 0.137 |
| 0 | +0.386 | +0.018 | +0.185 | +0.601 | −0.225 | +0.148 | −0.057 | −0.143 |
| +1 | +0.216 | −0.048 | +0.172 | −0.129 | −0.312 | +0.065 | −0.143 | 0.003 |
| +2 | +0.198 | −0.031 | +0.158 | −0.026 | −0.384 | −0.005 | −0.205 | −0.197 |
| +3 | +0.201 | −0.003 | +0.133 | −0.016 | −0.292 | +0.021 | −0.144 | −0.169 |
| +4 | +0.215 | +0.031 | +0.122 | +0.127 | −0.058 | +0.133 | −0.028 | 0.174 |

*Note:* All series log-transformed before filtering. Sample: 1992Q3–2021Q4 ($N$ = 118); FD loses one observation. HP-1600: Hodrick-Prescott filter with $\lambda = 1{,}600$. HP-100k: HP filter with $\lambda = 10^5$. Hamilton: regression filter of Hamilton (2018), $h$ = 8 quarters, $p$ = 4 lags. FD: quarter-on-quarter first differences. Entries in **bold** indicate horizons where HP-100k has the opposite sign from all three other filters.

## Appendix D. Steady state

We summarize the steady state of the system below:

$$
\begin{aligned}
\nu^f &= \rho(N) f_e / \mu(N) \\
w^{int} &= \rho(N) z / \mu(N) \\
\frac{d^f}{\nu^f} &= \frac{r+\delta_e}{1-\delta_e} \\
\kappa + \frac{K}{q(\theta)} &= \frac{1-\delta_e}{r+\tau}(w^{int} - w - K) \\
w &= \phi\left[w^{int} - K + \theta(K + q(\theta)\kappa)\right] + (1-\phi)b \\
Q &= e^{1/\xi} x_m \\
1-\delta_e &= (1-\delta)F(\chi^c), \quad \tau = 1-(1-\delta_e)(1-s) \\
K &= \frac{r+\delta_e}{1+r} Q \\
u &= \frac{\tau}{\tau + (1-\delta_e) f(\theta)}
\end{aligned}
\tag{D.1}
$$

$$e = \delta_e(v + 1 - u) \tag{D.2}$$

$$N^e = \delta_e N/(1 - \delta_e)$$

$$L^c = \frac{r + \delta_e}{r + \delta_e + \delta_e \pi_s \mu} L, \quad L^e = \frac{\delta_e \pi_s \mu}{r + \delta_e + \delta_e \pi_s L \mu} \tag{D.3}$$

$$\chi^c = \frac{\rho f_e}{\mu} \left( \frac{r + \delta_e}{1 - \delta_e} \frac{\mu - 1}{\mu \pi_s} + 1 \right)$$

$$\pi_s = \frac{\mu - 1}{\mu} \frac{(1 - \psi_c)(r + \delta_e)}{r + \delta_e + \psi_c(1 - \delta_e)}$$

$$Y^{Gross} = Y^C + \nu^f N^e = w^{int} L + N d^f$$

$$Y = C + \nu^f N^e$$

$$X = e \frac{\xi}{\xi + 1} Q + \kappa q v$$

$$X^c = N \psi_c \chi^c$$

$$Y^c = \rho z L^c = C + X + X^c$$

Most of these equations are self-evident from the equilibrium system. For instance, evaluating (5) yields (D.1). To find $e$, start with the steady-state vacancy law of motion:

$$[1 - (1 - \delta_e)(1 - q)]v = (1 - \delta_e)s(1 - u) + e$$

Expanding the left-hand side as $\delta_e v + (1 - \delta_e)qv$ and rewriting $(1 - \delta_e)qv = \tau(1 - u)$, we have

$$\delta_e v + \tau(1 - u) = (1 - \delta_e)s(1 - u) + e$$

$$\delta_e v + [\tau - (1 - \delta_e)s](1 - u) = e$$

so $e = \delta_e(v + 1 - u)$. Equation (D.2) makes clear how the contribution of new entrants to the vacancy pool is governed by the total exit rate $\delta_e$.

To find labor in entry $L^e$ in terms of aggregate labor, (D.3), we plug the resource constraint curve into the expression for $L$:

$$\begin{aligned} L^e &= \frac{\delta_e}{1 - \delta_e} N f_e / z \\ &= \frac{\delta_e}{1 - \delta_e} \left[ \frac{(\mu - 1) z L (1 - \delta_e)}{f_e(\delta_e \mu + r)} \right] f_e / z \\ &= \frac{\delta_e(\mu - 1)}{\delta_e \mu + r} L \end{aligned}$$

The steady-state job creation and wage equations can be arranged as

$$\kappa + \frac{K}{q} = \frac{1-\delta_e}{r+\tau}\left[(1-\phi)(w^{int} - K - b) - \phi\theta q\left(\kappa + \frac{K}{q}\right)\right] \tag{D.4}$$

Equation (D.4) equates the average recruiting cost to the present-discounted value of the match, adjusted for the wage bargain. The right-hand side incorporates a fraction $1-\phi$ of the flow surplus $(w^{int} - K - b)$ and an adjustment for the outside option of the worker. Whereas in the standard DMP model, the analogous expression would characterize tightness, here $K$ and $w^{int}$ are additional endogenous variables. Moreover, the average recruiting cost has the compound form $\kappa + K/q$, and discounting is adjusted by the overall firm survival rate $1-\delta_e$.

## Appendix E. Derivations

*Appendix E.1. Relationship between profit share and Power law shape parameter $\psi$*

Here, we can establish the relationship between $\pi_s$ and $\psi_c = \psi/(1+\psi)$. and the probability of a positive fixed cost $p_0$. Recall that the cutoff satisfies $\chi^c = Y^c[(\mu-1)/(\mu N)] + \rho f_e/\mu$. So, aggregate fixed costs are $X^c = \psi_c p_0(Y^c(\mu-1)/\mu + N\rho f_e/\mu)$. Divide by $Y^c$ and use $Y^c = \rho z L^c$ to show that

$$\frac{X^c}{Y^c} = \frac{\psi_c p_0}{\mu}\left(\mu - 1 + \frac{f_e N}{z L^c}\right)$$

Now, plug in (**??**) to show

$$\frac{X^c}{Y^c} = \frac{\psi_c p_0}{\mu}\left(\mu - 1 + \frac{1-\delta_e}{r+\delta_e}\mu\pi_s\right)$$

Hence,

$$\pi_s = \frac{\mu-1}{\mu}(1-\psi_c p_0) - \frac{\psi_c p_0(1-\delta_e)\pi_s}{r+\delta_e}$$

We solve for $\pi_s$ as

$$\pi_s = \frac{\mu-1}{\mu}\frac{(1-\psi_c p_0)(r+\delta_e)}{r+\delta_e+\psi_c p_0(1-\delta_e)} \tag{E.1}$$

It is easy to verify that, holding $\delta_e$ fixed, $\pi_s$ is increasing with $\psi_c$. In particular, for $\psi_c = 0, \pi_s = (\mu-1)/\mu = 1/\varepsilon$, as Bilbiie, Ghironi, and Melitz (2012). We can rewrite the cutoff $\chi^c$ in terms of both the profit share and $\psi_c p_0$:

$$\chi^c = \frac{\rho}{\mu}\left(\frac{zL^c(\mu-1)}{N} + f_e\right)$$

$$= \frac{\rho f_e}{\mu}\left(\frac{r+\delta_e}{1-\delta_e}\frac{\mu-1}{\mu\pi_s}+1\right)$$
$$= \frac{\rho f_e}{\mu}\left(\frac{r+\delta_e}{1-\delta_e}\frac{r+\delta_e+\psi_c p_0(1-\delta_e)}{(1-\psi_c p_0)(r+\delta_e)}+1\right)$$

*Appendix E.2. Resource constraint curve*

Expressing the $N$–$L$ relationship in terms of the distributional parameter $\psi_c$, rearrange (42) as

$$N = \frac{zL(1-\delta_e)}{f_e\left(\frac{r+\delta_e}{\mu\pi_s}+\delta_e\right)}$$

Substitute the value $\pi_s$ from (E.1) to obtain

$$N = \frac{zL(\mu-1)(1-\psi_c p_0)}{f_e[r+\delta_e+\psi_c(1-\delta_e)+\delta_e(\mu-1)(1-\psi_c)]}$$

Hence, labor in entry is

$$L^e = \frac{\delta_e}{1-\delta_e}\frac{Nf_e}{z}$$
$$= \frac{\delta_e(\mu-1-\psi_c p_0)L}{\delta_e\mu+r+\psi_c p_0(1-\mu\delta_e)}$$

Thus, labor in the consumption sector is

$$L^c = \frac{r+\delta_e+\psi_c p_0(1-(\mu-1)\delta_e)}{\delta_e\mu+r+\psi_c p_0(1-\mu\delta_e)}L$$

We can substitute to solve for $X^c$ as

$$X^c = \frac{\rho z L}{\mu}\psi_c p_0\left(\mu-1+\frac{(\mu-1-\psi_c p_0)(1-\delta_e\mu)}{\delta_e\mu+r+\psi_c p_0(1-\mu\delta_e)}\right)$$

Hence, the ratio $X^c/Y^c$ satisfies

$$\frac{X^c}{Y^c} = \frac{L}{L^c}\frac{\psi_c p_0}{\mu}\left(\mu-1+\frac{(\mu-1-\psi_c p_0)(1-\delta_e\mu)}{\delta_e\mu+r+\psi_c p_0(1-\mu\delta_e)}\right)$$

*Appendix E.3. Labor share of GDP*

The aggregate resource constraint can be rearranged to pin down the income of recruiters:

$$w^{int}L = \frac{Y^c}{\mu}+\nu^f N^e$$

Thus, the income share of recruiters relative to gross output is

$$\begin{aligned}\frac{w^{int}L}{Y^{Gross}} &= \left(\frac{r+\delta_e}{r+\delta_e+\delta_e\pi_s}\right)\frac{1}{\mu} + \frac{\delta_e\pi_s}{r+\delta_e+\pi_s} \\ &= \frac{r+\delta_e+\delta_e\mu\pi_s}{\mu(r+\delta_e+\delta_e\pi_s)}\end{aligned}$$

The labor share of GDP is

$$\frac{wL}{Y} = (1+\frac{X+X^c}{Y})\frac{w}{w^{int}}\frac{r+\delta_e+\delta_e\mu\pi_s}{\mu(r+\delta_e+\delta_e\pi_s)}$$

The steady-state labor share is composed of three factors. The first factor just reflects the wedge between gross output and GDP, which depends on the recruiting cost and fixed cost shares. The second and third comprise a search wedge $w/w^{int}$ and a market power wedge $w^{int}L/Y^{Gross}$. Through the latter, the labor share depends positively on $\delta_e$ and $\rho$, and negatively on $\pi_s$. Intuitively, a higher product destruction rate requires more labor resources, and thus income accruing to labor. A higher price markup raises profits and thus lowers the labor share. Indeed, in standard DSGE models with monopolistic competition, the labor share just the inverse of the price markup. Here that arises as a limiting case in which search frictions vanish ($w \to w^{int}$) and $\delta \to 0$. The latter entails eliminating the entry sector, which affects the labor share since only retailers charge markups. Note that the search wedge, in turn, depends on $\phi$ and $b$. Moreover, variety effects affect the search wedge through $w^{int}$ but otherwise do not affect the steady-state shares.

The special case studied by Bilbiie, Ghironi, and Melitz (2012) arises if we shut down labor market frictions and fixed costs in production. In this limit $\delta_e = \delta$ (no endogenous exit), $w^{int} = w$, $X = X^c = 0$, and $\pi_s = 1/\varepsilon$, giving

$$\frac{wL}{Y} = \frac{(r+\delta)(\varepsilon-1)+\delta}{\varepsilon(r+\delta)+\delta} = \frac{C}{Y}\frac{1}{\mu} + \frac{\nu^f N^e}{C}$$

which equals the consumption share divided by the markup plus the entry share.

*Appendix E.4. Wage determination*

We characterize the wage $w_t$. To characterize these value functions, we revisit the value of the large household:

$$\begin{aligned}W_t = \max_{c_t,a_t} \frac{c_t^{1-\sigma}}{1-\sigma} + \beta\mathbb{E}_t W_{t+1} \quad \text{s.t.} \\ c_t + \nu_t a_{t+1} = w_t L_t + b(1-L_t) + (\nu_t+d_t)a_t + D_t^{int} - T_t\end{aligned}$$

Note that, given the continuum of members in a household, the surplus of moving an unemployed household into employment is $\frac{\partial W_t}{\partial L_t}$. Nash bargaining solves

$$w_t = \arg\max[\frac{\partial W_t(w_t)}{\partial L_t}]^{\phi}[J_t(w_t) - Q_t]^{1-\phi} \tag{E.2}$$

For convenience, we define the match surpluses $S_T^H = \frac{\partial W_t}{\partial L_t}$. and $S_t^J = J_t - Q_t$.

From the Bellman equation,

$$\frac{\partial W_t}{\partial L_t} = c_t^{-\sigma}\frac{\partial c_t}{\partial L_t} + \beta\mathbb{E}_t[\frac{\partial W_{t+1}}{\partial L_{t+1}}\frac{\partial L_{t+1}}{\partial L_t}] \tag{E.3}$$

From the budget constraint, $\frac{\partial c_t}{\partial L_t} = w_t - b$. Moreover, given the law of motion of employment $L_{t+1} = (1-\delta_t)[f(\theta_t)(1-L_t) + (1-s_t)]L_t$, we have

$$\frac{\partial L_{t+1}}{\partial L_t} = (1-\delta_t)(1 - s_t - f(\theta_t))$$

Putting it together, we have

$$S_t^H = c_t^{-\sigma}(w_t - b) + \beta(1-\delta_t)(1-s_t-f(\theta_t))\mathbb{E}_t S_{t+1}^H$$

The first-order condition of the Nash product (E.2) is

$$\phi\frac{\frac{\partial S_t^H}{\partial w_t}}{S_t^H} + (1-\phi)\frac{\frac{\partial J_t}{\partial w_t}}{J_t} = 0$$

We evaluate

$$\frac{\partial S_t^H}{\partial w_t} = \frac{\partial[u'(c_t)(w_t-b)]}{\partial w_t} + \beta(1-\delta_t)(1-s_t-f(\theta_t))\frac{\partial}{\partial w_t}\mathbb{E}_t S_{t+1}^H$$

The term $\frac{\partial}{\partial w_t}\mathbb{E}_t S_{t+1}^H = 0$ because the next-period wage is re-bargained based on next period conditions. Thus, $\frac{\partial S_t^H}{\partial w_t} = c_t^{-\sigma}$. Recall that the firm surplus is

$$S_t^J = w_t^{int,j} - w_t - K_t + (1-s_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right)$$

Note that $\frac{\partial S_t^J}{\partial w_t} = -1$, so we obtain the sharing rule

$$S_t^H = \frac{\phi}{1-\phi}c_t^{-\sigma}S_t^J$$

From (14),

$$\mathbb{E}_t\beta(1-\delta_t)\left(\frac{c_{t+1}}{c_t}\right)^{-\sigma}S_{t+1}^J = \kappa + \frac{K_t}{q(\theta_t)}$$

Nash bargaining at period $t+1$ corresponds to (E.3) forwarded one period. Hence,

$$\beta(1-\delta_t)\mathbb{E}_t S^H_{t+1} = \frac{\phi}{1-\phi} c_t^{-\sigma}\left(\kappa + \frac{K_t}{q(\theta_t)}\right)$$

We can rewrite the household surplus as

$$S^H_t = c_t^{-\sigma}(w_t - b) + (1 - s - f(\theta_t))\frac{\phi}{1-\phi} c_t^{-\sigma}\left(\kappa + \frac{K_t}{q(\theta_t)}\right)$$

Now, we wish to substitute out $S^H_t$. To do so, we combine the expressions for recruiter surplus and bargaining:

$$S^J_t = w^{int}_t - w_t - K_t + (1-s_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right)$$
$$S^J_t = \frac{1-\phi}{\phi} c_t^{\sigma} S^H_t$$

so that

$$w^{int}_t - w_t - K_t + (1-s_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right) = \frac{1-\phi}{\phi}\left(w_t - b + (1 - s - f(\theta_t))\frac{\phi}{1-\phi}\left(\kappa + \frac{K_t}{q(\theta_t)}\right)\right)$$

This expression can be rearranged explicitly for the wage:

$$w_t = \phi\left(w^{int}_t - K_t + f(\theta_t)\left(\kappa + \frac{K_t}{q(\theta_t)}\right)\right) + (1-\phi)b$$

*Appendix E.5. Log-linearized propagation equations*

This appendix collects log-linearized versions of the equations that govern impulse-response dynamics. Let a hat denote the log-deviation from the deterministic steady state: $\hat{x}_t \equiv \ln(x_t/\bar{x})$, so $x_t \approx \bar{x}(1+\hat{x}_t)$. For rates that are small at steady state ($\delta_e, s, \tau, u, v \ll 1$), we use the level deviation $\tilde{x}_t \equiv x_t - \bar{x}$ directly and note $\tilde{x}_t \approx \bar{x}\hat{x}_t$. We present equations in the order in which they feed into one another: the exit margin first, then the variety stock, the labor market, and finally the vacancy-value chain.

*1. Endogenous exit rate..* The survival probability is $\Lambda_t \equiv F(\chi^c_t) = (1-p_0) + p_0(\chi^c_t/\chi_m)^{\psi}$, and the total exit rate is (3):

$$\delta_{e,t} = 1 - (1-\delta_{t-1})\,\Lambda_t,$$

which in steady state is $\bar{\delta}_e = 1 - (1-\bar{\delta})\bar{\Lambda}$. Define $\varsigma \equiv \left(\frac{\bar{\chi}^c}{\chi_m}\right)^{\psi}$, so that $\bar{\Lambda} = (1-p_0) + p_0\varsigma$ and $1 - \bar{\Lambda} = p_0(1-\varsigma)$.

Totally differentiating around steady state:

$$d\delta_{e,t} \;=\; \bar{\Lambda}\, d\delta_{t-1} \;-\; (1-\bar{\delta})\,\frac{d\Lambda_t}{d\chi^c_t}\, d\chi^c_t.$$

The derivative of $\Lambda$ with respect to $\chi^c$ at steady state is

$$\left.\frac{d\Lambda}{d\chi^c}\right|_{\bar{\chi}^c} \;=\; \frac{p_0\,\psi\,\varsigma}{\bar{\chi}^c}.$$

Dividing through by $\bar{\delta}_e$ and writing $d\delta_{t-1}/\bar{\delta} = \hat{\delta}_{t-1}$, $d\chi^c/\bar{\chi}^c = \hat{\chi}_{c,t}$:

$$\hat{\delta}_{e,t} \;=\; \frac{\bar{\Lambda}}{\bar{\delta}_e}\,\bar{\delta}\,\hat{\delta}_{t-1} \;-\; \frac{(1-\bar{\delta})\,p_0\psi\varsigma}{\bar{\delta}_e}\,\hat{\chi}_{c,t}. \tag{E.4}$$

Define the *exit elasticity* $\varepsilon_x$ as the elasticity of the endogenous exit probability $(1-\Lambda_t)$ with respect to $\chi^c$:

$$\varepsilon_x \;\equiv\; \frac{d(1-\Lambda)}{d\chi^c}\,\frac{\bar{\chi}^c}{1-\bar{\Lambda}} \;=\; \frac{p_0\psi\varsigma}{p_0(1-\varsigma)} \;=\; \frac{\psi\varsigma}{1-\varsigma}.$$

Since $p_0(1-\varsigma) = 1-\bar{\Lambda}$, equation (E.4) becomes

$$\hat{\delta}_{e,t} \;=\; \frac{\bar{\Lambda}}{\bar{\delta}_e}\,\bar{\delta}\,\hat{\delta}_{t-1} \;-\; \frac{(1-\bar{\delta})(1-\bar{\Lambda})}{\bar{\delta}_e}\,\varepsilon_x\,\hat{\chi}_{c,t}.$$

Both coefficients are positive and $\hat{\chi}^c$ enters with a negative sign: a higher continuation threshold raises survival and lowers $\delta_{e,t}$. At the baseline calibration $\varepsilon_x = 5$, meaning a 1 % rise in $\chi^c$ raises the endogenous exit probability $(1-\Lambda_t)$ by 5 %.

*2. Continuation threshold.*. From the cutoff condition (26) with $\mu$ constant and $\rho_t = N_t^{1/(\varepsilon-1)}$:

$$\hat{\chi}_{c,t} = \underbrace{\alpha_\pi\Big(\hat{Y}_{c,t}-\hat{N}_t\Big)}_{\text{profit component}} \;+\; \underbrace{(1-\alpha_\pi)\,\frac{1}{\varepsilon-1}\hat{N}_t}_{\text{continuation-value component}}\quad,$$

where $\alpha_\pi \equiv \bar{R}^f/\bar{\chi}^c \in (0,1)$ is the steady-state share of per-firm gross profit in the threshold. The profit component (elasticity $\alpha_\pi \cdot \frac{\varepsilon-2}{\varepsilon-1} < 0$ in $N$ for $\varepsilon > 2$) rises when $N$ falls, reflecting the market-share gain to surviving firms under DS-CES demand. The continuation-value component (elasticity $(1-\alpha_\pi)/(\varepsilon-1) > 0$ in $N$) falls when $N$ falls, because the option value $\nu_f = \rho f_e/\mu$ shrinks with $\rho$. The two forces partially offset: whether $\hat{\chi}^c$ rises or falls in response to $\hat{N} < 0$ depends on calibrated $\alpha_\pi$ and $\varepsilon$.

*3. Firm stock (variety LOM).*. Log-linearizing the firm law of motion $N_{t+1} = (1-\delta_{e,t})(N_t + N_t^e)$ around steady state (using $\bar{N}^e/\bar{N} = \bar{\delta}_e/(1-\bar{\delta}_e)$):

$$\hat{N}_{t+1} = (1-\bar{\delta}_e)\,\hat{N}_t \;+\; \bar{\delta}_e\,\hat{N}_t^e \;-\; \hat{\delta}_{e,t}, \tag{E.5}$$

where the last term uses $(1+\bar{N}^e/\bar{N}) \approx 1/(1-\bar{\delta}_e)$ so the exit rate enters with unit coefficient in absolute log-deviation terms. The three forces are: (i) *mean reversion* at rate $\bar{\delta}_e$ — the survival coefficient $(1-\bar{\delta}_e)$ governs how long any $N$ deviation persists; (ii) *entry amplification* — the loading $\bar{\delta}_e$ on $\hat{N}^e$ is larger when steady-state churning is higher, so the same entry impulse produces a larger $N$ response; (iii) the *exit margin* $\hat{\delta}_{e,t}$ from (E.4), which feeds back through $\chi^c$. Combining (i) and (ii): models with higher $\bar{\delta}_e$ show larger amplitude but faster convergence; models with lower $\bar{\delta}_e$ show smaller but more persistent deviations. This is the common mechanism underlying Comparisons A and B.

*4. Interior wage (recruiter compensation).*. From $w_t^{int} = \rho_t z_t/\mu$ (22) with $\rho_t = N_t^{1/(\varepsilon-1)}$:

$$\hat{w}_t^{int} = \frac{1}{\varepsilon-1}\,\hat{N}_t + \hat{z}_t. \tag{E.6}$$

This is the *variety channel*: a contraction of the firm stock $N$ reduces the relative price $\rho$ (fewer varieties lower the CES price index), compressing the marginal revenue product of labor $w^{int}$ even holding productivity constant. The elasticity $1/(\varepsilon-1) \approx 0.30$ at $\varepsilon = 4.3$ is moderate but cumulates over the slow $N$ recovery.

*5. Aggregate separation rate.*. From $\tau_t = \delta_{e,t} + s_t(1-\delta_{e,t})$ (4), log-linearizing:

$$\hat{\tau}_t = \frac{\bar{\delta}_e}{\bar{\tau}}\,\hat{\delta}_{e,t} + \frac{\bar{s}(1-\bar{\delta}_e)}{\bar{\tau}}\,\hat{s}_t,$$

where the coefficients sum to one (Euler's theorem) since $\bar{\tau} = \bar{\delta}_e + \bar{s}(1-\bar{\delta}_e)$. With the calibrated ratio $\bar{\delta}_e/\bar{\tau} \approx 0.21$, firm exit contributes about one-fifth of aggregate separation variance; match-level separations account for the rest.

*6. Unemployment LOM.*. From (5), log-linearizing around $(\bar{u}, \bar{\tau}, \bar{f})$:

$$\tilde{u}_t = (1-\bar{f}-\bar{\tau})\,\tilde{u}_{t-1} + (1-\bar{u})\,\tilde{\tau}_t - \bar{u}(1-\bar{\tau})\,\tilde{f}_{t-1}, \tag{E.7}$$

where $\tilde{x}_t = x_t - \bar{x}$ denotes the level deviation and $\tilde{f}_t = f(\theta_t) - \bar{f}$. The autoregressive coefficient $(1-\bar{f}-\bar{\tau})$ governs unemployment persistence: with monthly $\bar{f} \approx 0.40$ and $\bar{\tau} \approx 0.031$, it equals approximately 0.57, giving an unemployment half-life of around two months in the absence of sustained $\tau$ or $\theta$ shocks. A $\delta$ shock is persistent precisely because it generates a sustained $\tilde{\tau}_t > 0$ through $\tilde{\delta}_{e,t} > 0$ and, via the variety channel (E.6), a sustained $\tilde{f}_{t-1} < 0$ through the depressed JCC.

*7. Vacancy LOM.* From (6), splitting into predetermined and new-entrant components:

$$\tilde{v}_t = (1-\bar{\delta}_e)\Big[(1-\bar{q})\,\tilde{v}_{t-1} + \bar{s}(1-\bar{u})\,\hat{s}_{t-1}\Big] - \bar{v}^{\text{pre}}\,\hat{\delta}_{e,t} + \tilde{e}_t, \tag{E.8}$$

where $\bar{v}^{\text{pre}} = \bar{v} - \bar{e}$ is the steady-state predetermined vacancy stock. The three terms capture: (i) *survival of existing vacancies*, which depends on $(1-\bar{\delta}_e)$ and amplifies the persistence asymmetry between high- and low-$\delta_e$ specifications; (ii) *destruction impulse* $\hat{\delta}_{e,t}$, which directly destroys the predetermined stock; and (iii) *new entrant vacancies* $\tilde{e}_t$, which respond contemporaneously to the vacancy-value signal $Q_t$. The sunk-cost predetermination means that $\tilde{v}_t^{\text{pre}}$ cannot adjust within the period, so the vacancy response to a $\delta$ shock is asymmetrically slow relative to unemployment, consistent with the empirical Beveridge curve dynamics in Section 4.4.

*8. Vacancy value and new postings..* From (35), log-linearizing under a one-shock approximation ($\mathbb{E}_t\hat{Q}_{t+1} = \rho_\omega\hat{Q}_t$, exact by undetermined coefficients for shock $\omega \in \{z,\delta\}$ with persistence $\rho_\omega$):

$$\hat{Q}_t = \underbrace{\frac{r+\bar{\delta}_e}{1+r-\rho_\omega(1-\bar{\delta}_e)}}_{\mathcal{D}(\bar{\delta}_e,\rho_\omega)}\hat{K}_t, \tag{E.9}$$

where $r$ is the steady-state discount rate. The duration multiplier $\mathcal{D}$ is strictly increasing in $\bar{\delta}_e$ (since $\partial\mathcal{D}/\partial\bar{\delta}_e = (1+r)(1-\rho_\omega)/\text{den}^2 > 0$ for $\rho_\omega < 1$) and decreasing in shock persistence $\rho_\omega$. Intuitively, $Q$ is the present value of the per-period dividend $K$: higher $\bar{\delta}_e$ shortens the effective duration of the vacancy asset, amplifying its proportional response to the same $\hat{K}$ innovation. New vacancy postings respond via (36):

$$\hat{e}_t = \xi\,\hat{Q}_t = \xi\,\mathcal{D}(\bar{\delta}_e,\rho_\omega)\,\hat{K}_t, \tag{E.10}$$

so the entry vacancy margin inherits both the duration amplification from $\mathcal{D}$ and the convexity of the posting cost through $\xi$.

*9. Entry elasticity and the $K$ decomposition..* Equation (E.9) is derived holding $\delta_{e,t}$ fixed and so applies exactly to $z$ shocks where $\hat{\delta}_{e,t} = 0$ on impact. For $\delta$ shocks, log-linearizing the full $K$

definition (35), $K_t = Q_t - \beta(1-\delta_{e,t})Q_{t+1}$, with $\delta_{e,t}$ treated as a dynamic variable gives

$$\hat{K}_t \;=\; \underbrace{\frac{\bar{\delta}_e}{r+\bar{\delta}_e}\,\hat{\delta}_{e,t}}_{(+)\text{ duration shortening}} \;+\; \underbrace{\frac{1}{\mathcal{D}(\bar{\delta}_e,\rho_\omega)}\,\hat{Q}_t}_{\text{vacancy value}}\,. \tag{E.11}$$

The *duration-shortening* term is positive: a higher $\delta_{e,t}$ shortens the expected life of a filled vacancy, concentrating the asset's value in the current-period dividend and raising $K$ for given $Q$. Solving (E.11) for $\hat{Q}_t$ yields the extended vacancy-value equation

$$\hat{Q}_t \;=\; \mathcal{D}(\bar{\delta}_e,\rho_\omega)\,\hat{K}_t \;-\; \underbrace{\frac{\bar{\delta}_e}{1+r-\rho_\delta(1-\bar{\delta}_e)}}_{\phi_D(\bar{\delta}_e,\rho_\delta)>0}\,\hat{\delta}_{e,t}, \tag{E.12}$$

which reduces to (E.9) when $\hat{\delta}_{e,t}=0$. The two terms act in opposite directions on $Q$: duration shortening raises $K$ and hence $Q$, while the direct discount effect lowers the present value of future dividends. At the baseline calibration the first channel dominates, so $\hat{K}^\delta > 0$ and $\hat{Q}^\delta > 0$. Combining (E.12) with (E.10), the entry response to a destruction shock is

$$\hat{e}^\delta_t \;=\; \xi\,\mathcal{D}\,\hat{K}^\delta_t \;-\; \xi\,\phi_D\,\hat{\delta}_{e,t}, \tag{E.13}$$

where the first term dominates, giving $\hat{e}^\delta > 0$: entry expands under a positive destruction shock. For a technology shock the second term is absent; a negative shock gives $\hat{K}^z < 0$ via the JCC surplus channel, so $\hat{e}^z < 0$: entry contracts. The signs therefore differ across shocks — $\hat{e}^\delta > 0$, $\hat{e}^z < 0$ — and the role this plays in shaping amplification is discussed in Appendix F.

*Transmission chain summary..* The equations above trace two distinct propagation chains from any aggregate shock. For a *destruction shock* ($\delta\uparrow$), the primary channel is direct vacancy and firm destruction, partially cushioned by an entry expansion driven by duration shortening:

$$\hat{\delta}\uparrow \xrightarrow{\text{(E.4)}} \hat{\delta}_e\uparrow \xrightarrow{\text{(E.7),(E.8)}} \tilde{u}\uparrow,\ \tilde{v}\downarrow\ \text{(direct)} \xrightarrow{\text{(E.11)}} \hat{K}\uparrow \xrightarrow{\text{(E.12)}} \hat{Q}\uparrow \xrightarrow{\text{(E.13)}} \hat{e}\uparrow \xrightarrow{\text{(E.8)}} \tilde{v}\uparrow\ \text{(partial offset)}.$$

The secondary channel prolongs the unemployment response through variety:

$$\hat{\delta}_e\uparrow \xrightarrow{\text{(E.5)}} \hat{N}\downarrow \xrightarrow{\text{(E.6)}} \hat{w}^{int}\downarrow \xrightarrow{\text{JCC}} \hat{\theta}\downarrow \xrightarrow{\text{(E.7)}} \tilde{u}\uparrow\ \text{(amplified)}.$$

For a *technology shock* ($z\downarrow$):

$$\hat{z}\downarrow \xrightarrow{\text{(E.6)}} \hat{w}^{int}\downarrow \xrightarrow{\text{JCC}} \hat{K}\downarrow,\ \hat{\theta}\downarrow \xrightarrow{\text{(E.9)}} \hat{Q}\downarrow \xrightarrow{\text{(E.10)}} \hat{e}\downarrow \xrightarrow{\text{(E.8)}} \tilde{v}\downarrow \xrightarrow{\text{(E.5)}} \hat{N}\downarrow \xrightarrow{\text{(E.6)}} \hat{w}^{int}\downarrow\ \text{(amplified)}.$$

In both chains, the $N \to w^{int}$ link (E.6) generates a second-round amplification absent in models without product variety (Comparison C). The entry sign is opposite across shocks — $\hat{e}^{\delta} > 0$ cushions the direct vacancy destruction while $\hat{e}^{z} < 0$ compounds the JCC-driven vacancy contraction — a distinction that drives the asymmetric role of $\xi$ documented in Appendix Appendix F.

## Appendix F. Impulse responses: additional comparisons

### *Appendix F.1. Comparison D: role of entry elasticity ($\xi$)*

Figures F.17 and F.18 compare impulse responses under the baseline ($\xi_{\text{inv}} = 1.0$, solid black) and a near-free-entry counterfactual ($\xi_{\text{inv}} = 0.1$, dashed blue). Both calibrations match the same empirical targets; $\xi_{\text{inv}}$ is the only structural difference, so all parameter changes — lower $Q$ and $K$, higher Nash share $\phi$ — are endogenous to the recalibration. Sign conventions follow Comparisons A–C: the $z$ figure shows a *negative* technology shock and the $\delta$ figure shows a positive destruction shock, placing both on a scale where unemployment rises. The log-linearized equations underlying the discussion are derived in Appendix Appendix E.5, equations (E.11)–(E.13).

The entry response $e_t = G(Q_t)$ is the mechanism that connects the model to Coles and Moghaddasi Kelishomi (2018). As shown in Remark 1, setting $s_t = 0$ and $\sigma = \kappa = 0$ recovers the Coles and Moghaddasi Kelishomi model exactly. In that limit, all vacancies arise from new postings, and $\xi$ governs the elasticity of entry to the vacancy value $Q$. For destruction shocks, duration shortening raises $K$ and $Q$, generating an entry surplus that partially cushions the direct vacancy destruction. The parameter $\xi$ controls the strength of this cushion: higher $\xi$ (lower $\xi_{\text{inv}}$) amplifies the entry response and weakens the net downward pressure on vacancies. Whether $u$ and $v$ comove positively or negatively after a destruction shock therefore depends on $\xi$ — and, as discussed below, on the level of $\bar{\delta}_e$. Empirically, $u$ and $v$ comove *negatively* following destruction shocks (Beveridge curve recessions), which requires the direct vacancy destruction to dominate the entry cushion. This places a lower bound on $\xi_{\text{inv}}$: too close to free entry, and the model generates the wrong sign of $u$-$v$ comovement for $\delta$ shocks.

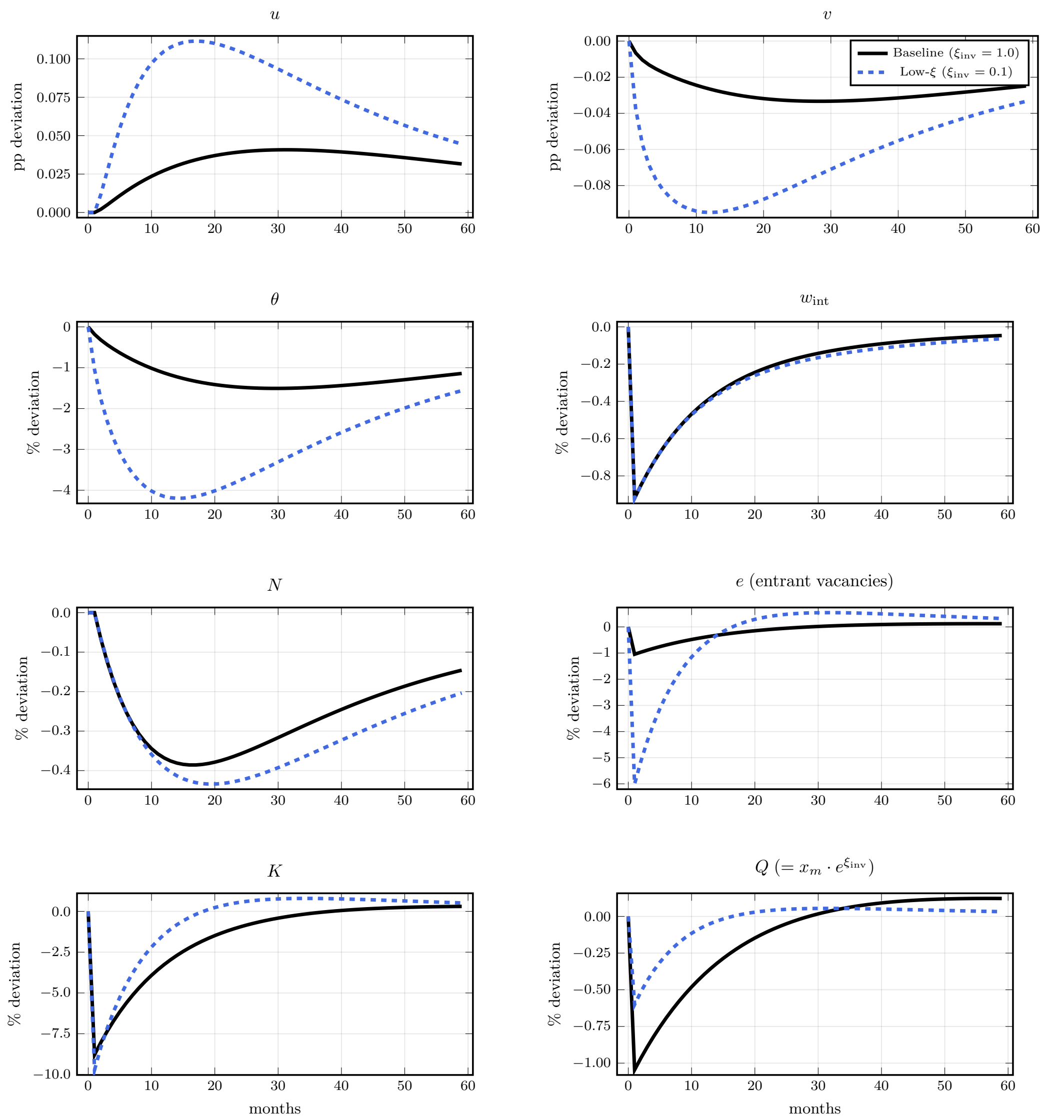


Figure F.17: Comparison D — impulse responses to a one-standard-deviation *negative* technology shock. Solid black: baseline ($\xi_{\text{inv}} = 1.0$). Dashed blue: near-free-entry ($\xi_{\text{inv}} = 0.1$, recalibrated). Row 1 in percentage-point level deviation from steady state; rows 2–4 in $100 \times \log$ deviation.

Under the *negative technology shock* (Figure F.17), lowering $\xi_{\text{inv}}$ substantially amplifies the labor-market response. The mechanism runs through entry. A negative $z$ shock depresses $w^{int}$, compressing the JCC surplus and lowering $K$. Via (E.9), $Q$ falls, and by (E.10) new vacancy postings contract: $\hat{e}^z < 0$. With the near-constant cost schedule ($\xi_{\text{inv}} = 0.1$, $\xi = 10$), the same $\hat{Q}$ movement generates a ten-fold larger $\hat{e}$ response, so entry collapses more sharply. The vacancy pool contracts further, $\theta$ falls more, and the $N \to \rho \to w^{int}$ feedback loop — $N$ falling faster, $\rho = N^{1/(\varepsilon-1)}$ contracting, $w^{int}$ falling further — amplifies unemployment substantially relative to the baseline.

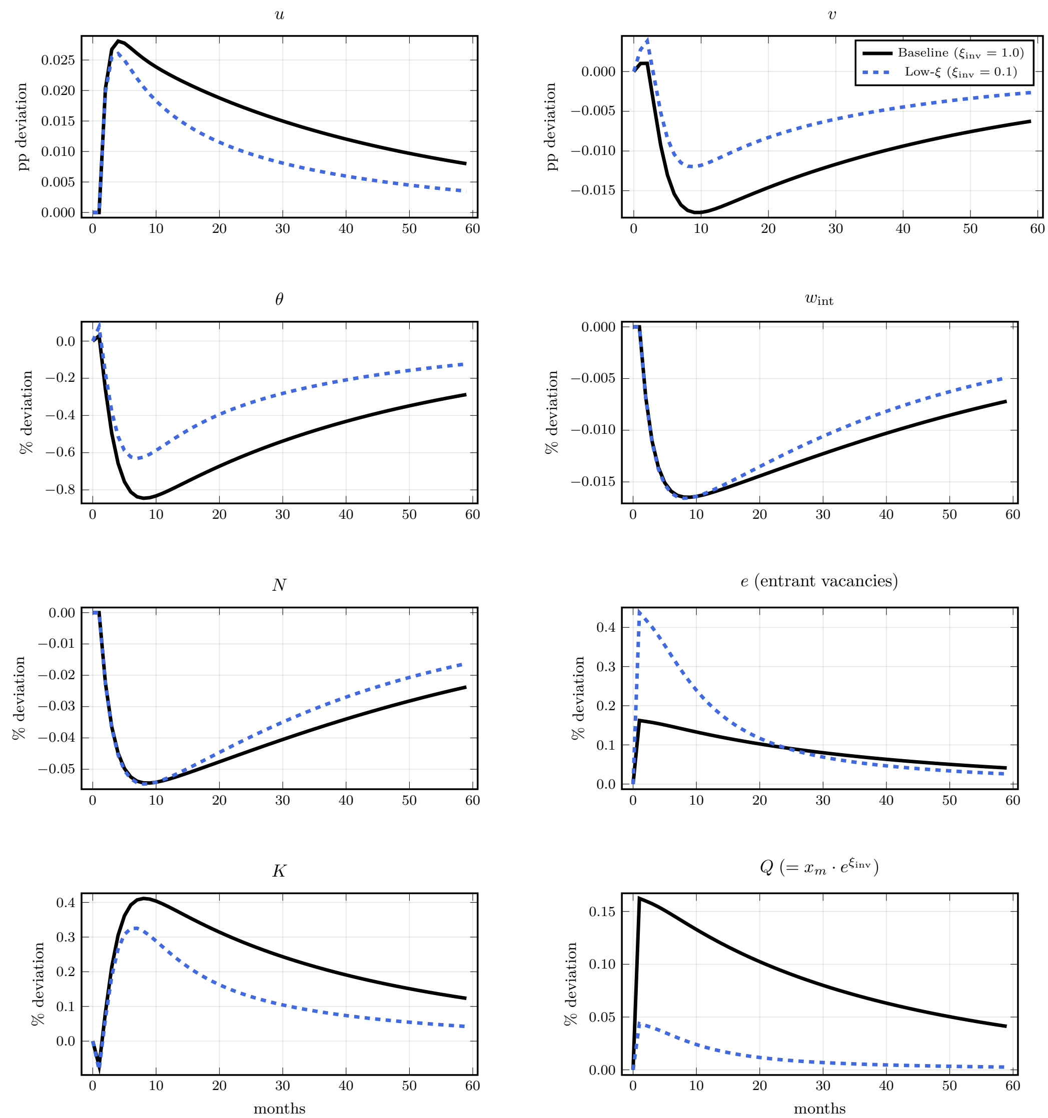


Figure F.18: Comparison D — impulse responses to a one-standard-deviation positive destruction shock. Lines and units as in Figure F.17.

Under the *destruction shock* (Figure F.18), the effect of $\xi$ is reversed. A positive $\delta$ shock triggers duration shortening: the vacancy asset's expected life falls, concentrating value in the current-period dividend and raising $K$ (E.11). Via the extended vacancy-value equation (E.12), $Q$ rises and entry *expands*: $\hat{e}^{\delta} > 0$. This entry expansion — the same $G(Q_t)$ channel as in Coles and Moghaddasi Kelishomi (2018) — partially backfills vacancies destroyed by the direct $\hat{\delta}_e \uparrow$ impulse, muting the decline in $\theta$ and the rise in $u$. A larger $\xi$ increases cushioning, so the near-free-entry calibration generates *less* unemployment amplification from destruction shocks.

Whether the entry cushion is large enough to reverse the sign of $u$-$v$ comovement is a joint function of $\xi$ and $\bar{\delta}_e$, not of $\xi$ alone. To see this, consider the free-entry limit $\xi_{\text{inv}} \to 0$. There

$Q \approx x_m$ is pinned and $K = x_m(r+\delta_e)/(1+r)$ rises with $\delta_e$ (duration shortening); the JCC then determines $\theta$, which *falls* — the LHS of the job-creation condition rises with $K$ while the RHS, proportional to $(1-\delta_e)(w^{int} - K - b)$, falls on both margins. Total vacancies $v = \theta u$ therefore have an ambiguous sign: positive $u$-$v$ comovement requires the $u$ rise to exceed the $\theta$ fall, which depends on how large a share of total separation flows through the $\delta$ channel.

At Coles and Moghaddasi Kelishomi's calibration, $\delta_e = \tau$ (all separation is destruction), so a $\delta$ shock raises $u$ sharply; the duration shortening simultaneously generates a large entry surplus. To prevent vacancies from comoving *positively* with unemployment, CK make vacancies partially inelastic.

The baseline calibration of the present model, disciplined by BED Deaths data, delivers $\bar{\delta}_e/\bar{\tau} \approx 0.21$: only one-fifth of total separation flows through firm destruction. At this low $\bar{\delta}_e$, the proportional $u$ rise from a $\delta$ shock is small — most separations are not of the $\delta$ type — and the duration shortening effect on $K$ is correspondingly weak. The $\theta$ decline therefore dominates, and vacancies fall following destruction shocks *even at free entry*. Figure F.18 confirms this: at $\xi_{\text{inv}} = 0.1$, vacancies decline after an initial one-period positive bump, and the baseline at $\xi_{\text{inv}} = 1.0$ shows a stronger decline still.

This is a key virtue of disciplining $\delta$ with data rather than setting $\delta_e = \tau$ as in Coles and Moghaddasi Kelishomi (2018): the empirically calibrated destruction rate endogenously preserves negative $u$-$v$ comovement across the entire range of $\xi$ shown here, including free entry. Convex entry costs ($\xi_{\text{inv}} > 0$) are not needed to prevent the CK Beveridge curve shift; the low empirical $\bar{\delta}_e$ already rules it out. The role of $\xi$ is then purely quantitative — governing the strength of the cushioning and hence the *degree* of negative comovement — rather than determining its sign.

In the limit $\xi \to \infty$, $\hat{e}^{\delta} = \xi \hat{Q} \to 0$: the entry cushion disappears and destruction shocks hit the vacancy stock with no replenishment. The central message of Comparison D is therefore the asymmetry across shock types together with this robustness result: the empirically calibrated $\bar{\delta}_e$ makes negative $u$-$v$ comovement for $\delta$ shocks robust to the entry-elasticity specification, while $\xi$ remains the key determinant of technology-shock amplification.

## Appendix G. Limiting cases

*Proof of Proposition 1*

We establish the if-and-only-if by showing necessity and sufficiency of each condition separately.

*Sufficiency.* Set $\sigma = 0$, $\zeta = 0$, $p_0 = 0$. Under $\sigma = 0$, the stochastic discount factor reduces to $m_{t+1} = \beta$, eliminating dependence on consumption $C_t$ from all intertemporal conditions. Under $\zeta = 0$, general CES preferences give $\rho(N_t) = 1$ for all $N_t$, so recruiter compensation $w_t^{int} = \rho(N_t)z_t/\mu(N_t) = z_t/\mu$ depends on $N_t$ only through $\mu(N_t)$; but under general CES with $\zeta = 0$, $\mu$ is a constant independent of $N_t$, so $w_t^{int} = z_t/\mu$ is exogenous to the labor market block. Under $p_0 = 0$, $\Lambda_t = F_\chi(\chi_t^c) = 1 - p_0 = 1$ for all $\chi_t^c$, so $\delta_{e,t} = 1 - (1 - \delta_{t-1})\Lambda_t = \delta_{t-1}$ depends only on the exogenous shock.

Under all three conditions, inspect the labor market block (34)–(36) and laws of motion (6)–(5). With $m_{t+1} = \beta$, $w_t^{int} = z_t/\mu$, and $\delta_{e,t} = \delta_{t-1}$, these equations form a closed system in $\{u_{t+1}, v_t, e_t, Q_t, K_t, w_t\}_{t=0}^{\infty}$ given exogenous states $\{z_t, \delta_t, s_t\}$ and initial conditions $\{v_{-1}, u_0\}$. No variable from the business formation block $\{N_t, C_t, Y_t^c, \chi_t^c\}$ appears. The labor market block is therefore self-contained.

Given the solution to the labor market block, the business formation block (30)–(33) determines $\{N_t, C_t, Y_t^c, \chi_t^c\}$ taking $\{u_t, X_t\}$ as given. This confirms the recursive structure.

*Necessity.* We show that relaxing any single condition breaks separability.

$\sigma > 0$: With $\sigma > 0$, $m_{t+1} = \beta(C_{t+1}/C_t)^{-\sigma}$ depends on $C_t$, which is determined by the resource constraint (32) jointly with $N_t$, $X_t^c$, and $Y_t^c$. The job creation condition (34) and flow vacancy value (35) both contain $m_{t+1}$, so the labor market block depends on $C_t$ and hence on the business formation block.

$\zeta > 0$: With $\zeta > 0$, $\rho(N_t) > 1$ for $N_t > 1$ and $\rho'(N_t) > 0$, so $w_t^{int} = \rho(N_t)z_t/\mu(N_t)$ depends on $N_t$. Since $N_t$ is determined in the business formation block, the job creation condition (34) depends on $N_{t+1}$ through $w_{t+1}^{int}$, breaking separability.

$p_0 > 0$: With $p_0 > 0$, $\Lambda_t = F_\chi(\chi_t^c) < 1$ for $\chi_t^c < \chi_m$, and $\chi_t^c$ satisfies the cutoff condition (33) which depends on $Y_t^c$, $N_t$, and $\mu(N_t)$ — all determined in the business formation block. Hence $\delta_{e,t} = 1 - (1 - \delta_{t-1})\Lambda_t$ depends on business formation variables. Since $\delta_{e,t}$ enters both laws of motion (6)–(5), the labor market block depends on the business formation block.

*Proof of Proposition 2*

Consider the DS-CES equilibrium with $p_0 = 0$ and parameters $(\varepsilon, f_e)$ satisfying $f_e = \bar{f}_e/\varepsilon$ for fixed $\bar{f}_e > 0$. Since $p_0 = 0$, $\Lambda_t = 1$ and $\delta_{e,t} = \delta_{t-1}$ throughout. We establish claims (iii), (iv), (i), and (ii) in that order, since (iii) is used in (iv) and (i) validates an assumption used in (iii).

*Claim (iii): $w_t^{int} \to z_t$.*. Under DS-CES, $\mu = \varepsilon/(\varepsilon - 1) \to 1$ and $\rho(N_t) = N_t^{1/(\varepsilon-1)}$. For any $N_t$ satisfying $N_t = O(e^{o(\varepsilon)})$ as $\varepsilon \to \infty$ — a condition weaker than boundedness, verified below — $\ln \rho(N_t) = \ln N_t/(\varepsilon - 1) \to 0$, so $\rho(N_t) \to 1$. Hence $w_t^{int} = \rho(N_t)z_t/\mu(N_t) \to z_t$ along any such equilibrium path. □

*Claim (iv): $N_t$ drops out of aggregate allocations.*. We show that $\{\theta_t, u_t, v_t, C_t, Y_t^c\}$ are determined independently of $N_t$ in the limit.

*Labor market block.* From claim (iii), $w_t^{int} \to z_t$. Since $p_0 = 0$ gives $\Lambda_t = 1$, the job creation condition (34), flow vacancy value (35), free entry (36), and laws of motion (6)–(5) converge to (37)–(38), which contain no reference to $N_t$.

*Resource constraint.* Since $p_0 = 0$, $X_t^c = 0$. Labor devoted to entry satisfies $l_t^e = f_e N_t^e/z_t = O(1/\varepsilon) \to 0$, established in claim (ii). Hence $L_t^c = L_t - l_t^e \to L_t = 1 - u_t$. Output per firm $y_t = z_t L_t^c/N_t \to z_t(1 - u_t)/N_t$, and $Y_t^c = \rho(N_t)N_t y_t \to 1 \cdot N_t \cdot z_t(1 - u_t)/N_t = z_t(1 - u_t)$. The resource constraint becomes $C_t = z_t(1 - u_t) - X_t$, independent of $N_t$.

Since all equations determining $\{\theta_t, u_t, v_t, C_t, Y_t^c\}$ are independent of $N_t$ in the limit, the normalization $\bar{N} = 1$ is a units convention for the measure of product lines — analogous to normalizing household measure to unity — and involves no loss of generality for aggregate allocations. No restriction on $\bar{f}_e$ is required to implement it. □

*Claim (i): steady-state value of $N_t$.*. Since aggregate allocations are independent of $N_t$ by claim (iv), the dynamic path of $N_t$ does not affect the proposition's main result. We nonetheless characterize the steady state and local dynamics of $N_t$ to confirm it remains well-defined in the limiting economy.

Substituting $f_e = \bar{f}_e/\varepsilon$ and $\rho(N_t) = N_t^{1/(\varepsilon-1)}$ into the firm LOM (30) with $p_0 = 0$ and taking $\varepsilon \to \infty$, using the approximation $N^{-1/(\varepsilon-1)} = e^{-\ln N/(\varepsilon-1)} \approx 1 - \ln N/(\varepsilon - 1)$:

$$N_t \to (1 - \delta_{t-1}) \left( N_{t-1} + \frac{z_{t-1}(1 - u_{t-1}) \ln N_{t-1}}{\bar{f}_e} \right) \tag{G.1}$$

This is a nonlinear predetermined variable equation driven by the exogenous processes $\{z_t, \delta_t, u_t\}$, with no forward-looking terms. Its unique positive steady state $\bar{N}$ satisfies:

$$\bar{\delta}\bar{N} = \frac{\bar{z}\bar{L}\ln\bar{N}}{\bar{f}_e} \tag{G.2}$$

where $\bar{L} = 1 - \bar{u}$. The right-hand side is zero at $\bar{N} = 1$, strictly positive and increasing for $\bar{N} > 1$, while the left-hand side is linear with positive slope $\bar{\delta}$. A unique solution $\bar{N} > 1$ exists for any $\bar{\delta}, \bar{f}_e, \bar{z}, \bar{L} > 0$ by the intermediate value theorem and strict monotonicity.

Linearizing (G.1) around $\bar{N}$ gives autoregressive coefficient:

$$\rho_N \equiv (1-\bar{\delta})\left(1 + \frac{\bar{z}\bar{L}}{\bar{f}_e\bar{N}}\right) = (1-\bar{\delta})\left(1 + \frac{\bar{\delta}}{\ln\bar{N}}\right)$$

where the second equality uses (G.2). Local stability requires $\rho_N < 1$, which holds if and only if $\bar{N} > e^{1-\bar{\delta}}$. For $\bar{\delta} \approx 0.05$, this requires $\bar{N} > e^{0.95} \approx 2.59$ — satisfied for sufficiently small $\bar{f}_e$ given other parameters. When the stability condition holds, $N_t$ follows a well-defined stationary process driven by $\{z_t, \delta_t, u_t\}$, confirming the path satisfies $N_t = O(e^{o(\varepsilon)})$ as required by claim (iii). □

*Claim (ii): rates of convergence..* *Firm value*: $\nu_t^f = \rho(N_t)f_e/\mu(N_t) \approx 1 \cdot (\bar{f}_e/\varepsilon) \cdot 1 = O(1/\varepsilon)$.

*Investment*: At steady state, $N_t^e = \bar{\delta}\bar{N}/(1-\bar{\delta}) = O(1)$, so $\nu_t^f N_t^e = O(1/\varepsilon) \cdot O(1) = O(1/\varepsilon)$.

*Labor to entry*: $l_t^e = f_e N_t^e / z_t = (\bar{f}_e/\varepsilon) \cdot O(1)/z_t = O(1/\varepsilon)$. □

*Proof of Proposition 3*

Under clone replacement, $N_t = 1$ and $\mu = 1$ are maintained by construction at all times. We verify that the business formation block drops out and the remaining system reduces to (37)–(38).

*Business formation block.* Substituting $N_t = 1$ and $\mu = 1$ into the baseline equilibrium definition:

The firm LOM (30): with $N_t = 1$ maintained by construction, this equation is satisfied trivially by the clone replacement assumption and imposes no restriction on other variables.

The business formation Euler (31): with $\rho(1) = 1$, $\mu(1) = 1$, $f_e\rho(N_t) = f_e$, and $(\mu(N_{t+1}) - 1) = 0$, the right-hand side reduces to $(1-\delta_t)\mathbb{E}_t m_{t+1}\Lambda_{t+1}\{f_e\} = f_e \cdot (1-\delta_t)\mathbb{E}_t m_{t+1}$. This is a condition on $f_e$ alone and is satisfied by appropriate choice of $f_e$ given $\beta$ and $\bar{\delta}$; it imposes no restriction on labor market variables.

The cutoff condition (33): with $\mu = 1$, $(\mu - 1)/\mu = 0$, so $\chi_t^c = f_e \rho(N_t)/\mu(N_t) = f_e$. Since clone replacement maintains $N_t = 1$ regardless of profitability, the exit decision is moot; $\chi_t^c = f_e$ is a constant that imposes no restriction on aggregate allocations.

The resource constraint (32): since $p_0 = 0$ under clone replacement (endogenous exit is absent by construction), $X_t^c = 0$. With $\rho(1) = 1$ and $\mu(1) = 1$, $Y_t^c = z_t(1 - u_t)$ and $l_t^e = 0$, giving $C_t = z_t(1 - u_t) - X_t$, which is (39).

*Labor market block.* Substituting $\rho(N_t) = 1$, $\mu(N_t) = 1$, $\Lambda_t = 1$, and $\delta_{e,t} = \delta_{t-1}$ into (34)–(5) yields (37)–(38) directly. The stochastic discount factor $m_t = \beta(C_t/C_{t-1})^{-\sigma}$ is unchanged. The stochastic processes (7)–(9) are unchanged.

The resulting system (37)–(38) together with $e_t = G(Q_t)$, $m_t = \beta(C_t/C_{t-1})^{-\sigma}$, (39), and stochastic processes (7)–(9) is well-posed. At $\sigma = 0$ the stochastic discount factor is $m_t = \beta$, the system reduces to a scalar expectational equation for $Q_t$ with characteristic root $\beta(1 - \bar{\delta}_e) < 1$, and the unique bounded solution follows directly by ruling out explosive paths. For $\sigma > 0$, local uniqueness is maintained as a standard regularity condition and verified numerically as part of the baseline model solution.

*Proof of Proposition 4*

Set $p_0 = 0$. Then $F_\chi(0) = 0$, so no firm pays a continuation cost: $X_t^c = 0$, $\Lambda_t = 1$, and $\delta_{e,t} = \delta_{t-1}$ for all $t$.

*Firm LOM* (30). With $\Lambda_t = 1$ and $\delta_{e,t} = \delta_{t-1}$, the baseline LOM becomes

$$N_t = (1 - \delta_{t-1})\left(N_{t-1} + \frac{z_{t-1}(1 - u_{t-1})}{f_e} - \frac{Y_{t-1}^c}{f_e \rho(N_{t-1})}\right),$$

where $\{1 - u_{t-1}\}$ is treated as an exogenous sequence by assumption. Using the gross-output identity $Y_{t-1}^c = z_{t-1}(1 - u_{t-1})\rho(N_{t-1})/\mu(N_{t-1})$ (from retail-firm optimality), the parenthesized term equals $N_{t-1}^e = z_{t-1}(1 - u_{t-1})/f_e \cdot (1 - 1/\mu(N_{t-1}))$, which is the BGM entry flow $N_t^e = G(Q_t)$ evaluated at $Q_t = f_e/\nu_t^f$. Hence (30) is the BGM firm LOM.

*Business formation Euler* (31). With $\Lambda_{t+1} = 1$ and $X_{t+1}^c = 0$, the baseline Euler simplifies to

$$f_e \rho(N_t) = (1 - \delta_t)\, \mathbb{E}_t m_{t+1} \frac{\mu(N_t)}{\mu(N_{t+1})} \left\{ f_e \rho(N_{t+1}) + \frac{Y_{t+1}^c}{N_{t+1}} (\mu(N_{t+1}) - 1) \right\}.$$

Dividing by $\rho(N_t)$ and using $\nu_t^f = f_e \rho(N_t)$ (free entry), this is the BGM asset-pricing equation for firm equity.

*Resource constraint* (29). With $X_t^c = 0$, gross output satisfies $Y_t^c = C_t + X_t$. Netting out $X_t$ on both sides gives the GDP identity $Y_t = C_t + \nu_t^f N_t^e$, which is the BGM resource constraint. The household budget constraint $c_t = w_t(1-u_t) + \pi_t - X_t$ closes the system identically in both models.

The two systems thus share the same three equations and the same exogenous sequences $\{z_t, \delta_t, 1-u_t, X_t\}$. Since $1-u_t$ and $X_t$ are endogenous in the baseline but exogenous in LR-BGM, the proposition asserts isomorphism for any realization of those sequences, not that the equilibrium distributions coincide.

## Appendix H. Proof of Proposition 5 and Lemma 1

We first establish the Lemma on pre-committed vacancy destruction, then prove each part of the Proposition.

**Lemma 1** (Pre-committed vacancy destruction)**.** *A positive $\delta_t$ shock unconditionally reduces the pre-committed vacancy stock at $t+1$:*

$$\frac{\partial\, v_{pre,t+1}}{\partial \delta_t} < 0.$$

*Proof.* The pre-committed vacancy stock is the component of $v_{t+1}$ determined before entry $e_{t+1}$ is chosen:

$$v_{pre,t+1} \;\equiv\; (1-\delta_{e,t+1})\big[(1-q(\theta_t))v_t + s_t(1-u_t)\big],$$

where $(1-\delta_{e,t+1}) = (1-\delta_t)\Lambda_{t+1}$ from (3). The bracket $[(1-q(\theta_t))v_t + s_t(1-u_t)] > 0$ collects unfilled vacancies from $t$ plus costless reposts from match separations at end of $t$; both are predetermined with respect to $\delta_t$ (which realizes at end of $t$, after $v_t$, $\theta_t$, $u_t$ are determined). Hence,

$$\frac{\partial\, v_{pre,t+1}}{\partial \delta_t} = -\Lambda_{t+1}\big[(1-q(\theta_t))v_t + s_t(1-u_t)\big] < 0.$$

□

*Proof of Proposition 5. Part 1: exogenous exit ($p_0 = 0$).* Under $p_0 = 0$: $\Lambda_t = 1$ and $\delta_{e,t} = \delta_{t-1}$.

*Unemployment.* The shock $s_t$ realizes at the end of period $t$. From (4),

$$\tau_{t+1} = \delta_{e,t+1} + s_t(1-\delta_{e,t+1}),$$

so $\partial \tau_{t+1}/\partial s_t = 1 - \delta_{e,t+1} > 0$. From the unemployment law of motion (5),

$$u_{t+1} = (1 - f(\theta_t))u_t + \tau_{t+1}\big[(1 - u_t) + f(\theta_t)u_t\big],$$

so $\partial u_{t+1}/\partial s_t = (1-\delta_{e,t+1})\big[(1-u_t) + f(\theta_t)u_t\big] > 0$, holding $\theta_t$ and other period-$t$ variables fixed. Unemployment strictly rises on impact.

*Vacancies.* From the vacancy law of motion (6),

$$v_{t+1} = \underbrace{(1 - \delta_{e,t+1})\big[(1 - q(\theta_t))v_t + s_t(1 - u_t)\big]}_{v_{pre,t+1}} + e_{t+1}.$$

The shock $s_t$ affects $v_{t+1}$ through two non-negative channels.

*Channel 1 (reposting):* $s_t$ appears directly in $v_{pre,t+1}$ via the costless-reposting term $(1 - \delta_{e,t+1})s_t(1 - u_t)$. Since $s_t$ does not destroy current vacancies $v_t$ (which were posted before $s_t$ realized) and does not reduce $(1 - \delta_{e,t+1})$ (which depends on $\delta_t$, not $s_t$),

$$\frac{\partial\, v_{pre,t+1}}{\partial s_t} = (1 - \delta_{e,t+1})(1 - u_t) > 0.$$

*Channel 2 (entry):* Entry $e_{t+1} = (Q_{t+1}/x_m)^{\xi}$ is chosen at the start of $t+1$; we show $\partial Q_{t+1}/\partial s_t \geq 0$.

*Sub-step (a): pre-entry tightness falls.* Comparing the two derivatives established above,

$$\frac{\partial u_{t+1}}{\partial s_t} - \frac{\partial v_{pre,t+1}}{\partial s_t} = (1 - \delta_{e,t+1})f(\theta_t)u_t > 0,$$

so unemployment rises strictly more than the pre-committed vacancy stock. Hence pre-entry tightness $\theta^{pre}_{t+1} \equiv v_{pre,t+1}/u_{t+1}$ falls below its steady-state value $\bar{\theta}$, implying $q(\theta^{pre}_{t+1}) > \bar{q}$.

*Sub-step (b): $K_{t+1}$ rises.* Under $p_0 = 0$, the product-line LOM gives $N_{t+1} = (1 - \delta_t)N_t$ (since $B_t = N_t$ from the business-formation block), so $N_{t+1}$ is predetermined with respect to $s_t$. Consequently $w^{int}_{t+1} = \rho(N_{t+1})z_{t+1}/\mu$ is also predetermined. At $\rho_s = 0$, future shocks revert to their mean, so the right-hand side of the JCC (19) at $t+1$—which involves expected future $w^{int}$ and $s$—is invariant to $s_t$ at leading order, pinning $C_{t+1} \equiv \kappa + K_{t+1}/q(\theta_{t+1})$ near its steady-state value $\bar{C}$. Therefore

$$K_{t+1} = \big(C_{t+1} - \kappa\big)q(\theta_{t+1}) \approx (\bar{C} - \kappa)\, q(\theta_{t+1}) > (\bar{C} - \kappa)\, \bar{q} = \bar{K},$$

where the strict inequality uses $q(\theta_{t+1}) > \bar{q}$, which holds because for $\xi < \infty$ entry does not fully restore tightness to $\bar{\theta}$.[21] It follows that $Q_{t+1} = K_{t+1} + \beta(1-\delta_{t+2})\mathbb{E}_{t+1}[Q_{t+2}] > \bar{Q}$, so $\partial e_{t+1}/\partial s_t > 0$.

Combining both channels: $\partial v_{t+1}/\partial s_t > 0$ and $\partial u_{t+1}/\partial s_t > 0$, so unemployment and vacancies comove positively at $h = 1$. The argument applies recursively: at every subsequent horizon $h \geq 2$, the $s$-shock does not enter the vacancy LOM with a negative sign, while it continues to raise $\tau$ and hence $u$. Positive $u$–$v$ comovement therefore holds for all $h \geq 1$.

*Part 2: endogenous exit ($p_0 > 0$).* With $p_0 > 0$, $(1-\delta_{e,t+1}) = (1-\delta_t)\Lambda_{t+1}$ where $\Lambda_{t+1} = F(\chi^c_{t+1})$ is the endogenous survival probability. The reposting channel becomes

$$\frac{\partial\, v_{pre,t+1}}{\partial s_t} = (1-\delta_t)\Lambda_{t+1}(1-u_t) + (1-\delta_t)\frac{\partial \Lambda_{t+1}}{\partial s_t}\big[(1-q(\theta_t))v_t + s_t(1-u_t)\big]. \tag{H.1}$$

The sign of the second term is $\partial\Lambda_{t+1}/\partial s_t = F'(\chi^c_{t+1}) \cdot \partial\chi^c_{t+1}/\partial s_t$ with $F' > 0$; signing $\partial\chi^c_{t+1}/\partial s_t$ requires accounting for the joint determination of $(\chi^c_{t+1}, N_{t+1})$.

*Joint system.* From (30):

$$N_{t+1} = (1-\delta_t)\, F(\chi^c_{t+1})\, B_t,$$

where $B_t \equiv N_t + z_t(1-u_t)/f_e - Y^c_t/(f_e\rho(N_t))$ is predetermined (all period-$t$ quantities; entry decisions $e_t$ committed before $s_t$ realized are included in $B_t$ and face the same exit draw $F(\chi^c_{t+1})$ at start of $t+1$). Together with (33) and $Y^c_{t+1} = \rho(N_{t+1})z_{t+1}(1-u_{t+1})$, the pair $(\chi^c_{t+1}, N_{t+1})$ solves

$$\chi^c = g(N,\, u_{t+1}), \tag{H.2}$$

$$N = (1-\delta_t)\, F(\chi^c)\, B_t, \tag{H.3}$$

where $g(N,u) \equiv \rho(N)z(1-u)(\mu(N)-1)/(\mu(N)N) + f_e\rho(N)/\mu(N)$.

*Role of the Jacobian.* The Jacobian of (H.2)–(H.3) is

$$J = \begin{pmatrix} 1 & -g_N \\ -(1-\delta_t)F'(\chi^c)B_t & 1 \end{pmatrix}, \quad \det J = 1 - (1-\delta_t)F'(\chi^c)B_t\, g_N.$$

The only channel through which $s_t$ enters is $u_{t+1}$; $g_u \equiv \partial g/\partial u = -\rho(N)z(\mu-1)/(\mu N) < 0$, so the exogenous shift to (H.2) is $-g_u\, \partial u_{t+1}/\partial s_t > 0$. By the implicit function theorem:

$$\frac{\partial \chi^c_{t+1}}{\partial s_t} = \frac{g_u}{\det J} \cdot \frac{\partial u_{t+1}}{\partial s_t}.$$

[21] Under free entry ($\xi \to \infty$), $Q_{t+1} = x_m$ is pinned, $K_{t+1} = \bar{K}$, and $\partial e_{t+1}/\partial s_t = 0$: Channel 2 vanishes but Channel 1 remains strictly positive.

Under DS-CES, $g_N \propto z(1-u)(2-\varepsilon)/(\varepsilon-1) + f_e N$, so condition (40) is equivalent to $g_N < 0$, giving $\det J = 1 - (1-\delta_t)F'(\chi^c)B_t\, g_N > 1$. This signs $\partial\chi^c_{t+1}/\partial s_t$ negative: it rules out the case in which feedback from lower $N$ to higher per-firm profitability reverses the direct employment channel $g_u < 0$. With $\det J > 0$:

$$\frac{\partial\chi^c_{t+1}}{\partial s_t} < 0, \qquad \frac{\partial\Lambda_{t+1}}{\partial s_t} = F'(\chi^c_{t+1})\,\frac{\partial\chi^c_{t+1}}{\partial s_t} < 0.$$

A positive $s$ shock raises endogenous exit, so the second term in (H.1) is negative: the endogenous exit channel opposes the reposting effect.

*Sufficient condition.* The drain is bounded by $F'(\chi^c_{t+1})\cdot|\partial\chi^c_{t+1}/\partial s_t|\cdot[(1-q(\theta_t))v_t + s_t(1-u_t)]$, where $|\partial\chi^c_{t+1}/\partial s_t| = |g_u/\det J|\cdot\partial u_{t+1}/\partial s_t$. This drain is increasing in $\rho_s$: the option value component $f_e\rho(N_{t+1})/\mu(N_{t+1})$ in (33) is determined by the business formation Euler (31), which depends on expected future $Y^c$. At $\rho_s = 0$, future employment reverts to $\bar{s}$ immediately, leaving the option value unaffected by $s_t$; only the current-period profit term $Y^c_{t+1}(\mu-1)/(\mu N_{t+1})$ falls. At $\rho_s > 0$, expected future $Y^c$ also falls (persistent lower employment), reducing the option value and amplifying $|\partial\chi^c_{t+1}/\partial s_t|$. The drain is therefore minimized at $\rho_s = 0$, so if the reposting channel dominates there, it dominates for all $\rho_s \in [0, \bar{\rho}_s)$ by continuity. The entry channel (Channel 2 from Part 1) is non-negative throughout.

*Part 3: $\delta$ shock.*

*Unemployment.* Consider a positive $\delta_t$ shock with $s_t$ fixed at its steady-state value $\bar{s}$ (a pure destruction shock). Then $\delta_{e,t+1}$ rises, so $\partial\tau_{t+1}/\partial\delta_t = \partial\delta_{e,t+1}/\partial\delta_t\cdot(1-\bar{s}) > 0$, and the same unemployment argument as Part 1 yields $\partial u_{t+1}/\partial\delta_t > 0$.

*Vacancies.* By Lemma 1,

$$\left|\frac{\partial v_{pre,t+1}}{\partial\delta_t}\right| = \Lambda_{t+1}\big[(1-q(\theta_t))v_t + \bar{s}(1-u_t)\big] \xrightarrow{\text{s.s.}} (1-\bar{q})\bar{v} + \bar{s}(1-\bar{u}) > 0,$$

a quantity bounded away from zero for any $\bar{\delta}_e/\bar{\tau} \in (0,1)$. The entry response partially cushions this: from the $K$-decomposition (E.11), the duration-shortening contribution to $\hat{K}^\delta$ carries coefficient $\bar{\delta}_e/(r+\bar{\delta}_e)$, so

$$\frac{\partial e_{t+1}}{\partial\delta_t} \propto \frac{\bar{\delta}_e}{r+\bar{\delta}_e} \to 0 \quad \text{as} \quad \bar{\delta}_e/\bar{\tau} \to 0.$$

For $\bar{\delta}_e/\bar{\tau}$ sufficiently small, the entry cushion is therefore strictly less than the pre-committed vacancy destruction, so $\partial v_{t+1}/\partial\delta_t < 0$, establishing negative $u$–$v$ comovement at $h = 1$. $\square$

## Appendix I. Steady-state equilibria: proof of Proposition 7

Throughout, restrict to DS-CES preferences: $\rho(N) = N^{1/(\varepsilon-1)}$ and $\mu = \varepsilon/(\varepsilon-1)$ constant. Set $p_0 = 0$ so that $\delta_e \in (0,1)$ is a parameter. Write $h(\theta) \equiv f(\theta)/q(\theta) = \theta$ for the tightness ratio, and recall $L(\theta) = (1-\delta_e)f(\theta)/[\tau + (1-\delta_e)f(\theta)]$, which is strictly increasing, concave, and satisfies $L(0) = 0$ and $L(\infty) = 1 - u_{\min} < 1$.

**Step 1: $N_{JC}(\theta)$ is well-defined and strictly increasing.**

Under DS-CES the JCC (45) can be written as

$$\frac{z}{\mu} N^{1/(\varepsilon-1)} = K(\theta) + b + \frac{1}{(1-\delta_e)(1-\phi)} \left(\kappa + \frac{K(\theta)}{q(\theta)}\right) \big(r + \tau + (1-\delta_e)\phi\,\theta q(\theta)\big) \equiv \Phi(\theta) \tag{I.1}$$

The right-hand side $\Phi(\theta)$ depends only on $\theta$: $K(\theta)$, $q(\theta)$, and $\theta q(\theta)$ are all functions of tightness alone. Since $1/(\varepsilon-1) > 0$, inverting gives a unique

$$N_{JC}(\theta) = \left(\frac{\mu\,\Phi(\theta)}{z}\right)^{\varepsilon-1}.$$

$\Phi(\theta)$ is strictly increasing in $\theta$ (each term is non-decreasing and $K(\theta)$ is strictly increasing because $e(\theta)$ is), so $N_{JC}(\theta)$ is strictly increasing.

**Step 2: $N_{RC}(\theta)$ is strictly increasing and bounded.**

Under DS-CES (46) gives

$$N_{RC}(\theta) = \frac{(\mu-1)z(1-\delta_e)}{f_e(\delta_e\mu + r)}\, L(\theta) \equiv \bar{N}\, \frac{L(\theta)}{L(\infty)},$$

where $\bar{N} = (\mu-1)z(1-\delta_e)L(\infty)/[f_e(\delta_e\mu + r)] < \infty$. Since $L(\theta)$ is strictly increasing, $N_{RC}(\theta)$ is strictly increasing, concave, with $N_{RC}(0) = 0$ and $N_{RC}(\theta) \uparrow \bar{N}$.

**Step 3: Boundary behavior.**

*At $\theta = 0$.* As $\theta \to 0$: $e(\theta) \to 0$, $K(\theta) \to 0$, and $q(\theta) \to q_0 > 0$ (the vacancy-filling rate at zero tightness). Hence

$$\Phi(0) = b + \frac{\kappa(r+\tau)}{(1-\delta_e)(1-\phi)},$$

so that

$$N_{JC}^0 \equiv N_{JC}(0) = \left(\frac{\mu}{z}\left(b + \frac{\kappa(r+\tau)}{(1-\delta_e)(1-\phi)}\right)\right)^{\varepsilon-1}.$$

If $b > 0$ or $\kappa > 0$ then $N_{JC}^0 > 0 = N_{RC}(0)$, so the JCC lies *strictly above* the RC curve at the origin.

*As $\theta \to \infty$.* $\Phi(\theta) \to \infty$ (since $K(\theta) \to \infty$), so $N_{JC}(\theta) \to \infty$, while $N_{RC}(\theta) \to \bar{N} < \infty$. The JCC therefore lies *strictly above* the RC curve for all sufficiently large $\theta$.

**Step 4: Even number of crossings; generically exactly two.**

Define $\Delta(\theta) \equiv N_{JC}(\theta) - N_{RC}(\theta)$. Steps 1–3 establish $\Delta(0) = N^0_{JC} > 0$ and $\Delta(\theta) \to +\infty$. If $\Delta(\theta) > 0$ for all $\theta$, the system has no equilibrium. For a non-degenerate parameterization in which at least one equilibrium exists, there must be some $\theta^* > 0$ with $\Delta(\theta^*) = 0$. Since $\Delta(0) > 0$ and $\Delta(\infty) > 0$, the intermediate value theorem requires $\Delta$ to cross zero an *even* number of times, yielding an even number of equilibria.

A crossing is non-degenerate (transversal) when $\Delta'(\theta^*) \neq 0$. A tangency $\Delta(\theta^*) = \Delta'(\theta^*) = 0$ is a codimension-two condition in the parameter space (it simultaneously imposes two equations on the model's parameters) and therefore fails for a generic open dense set of parameterizations. Excluding tangencies, the even-crossing structure collapses to generically *exactly two* equilibria, one with lower $(\theta_L, N_L)$ and one with higher $(\theta_H, N_H)$.

**Step 5: Uniqueness as $\varepsilon \to \infty$.**

As $\varepsilon \to \infty$, $\mu \to 1$ and $\rho(N) = N^{1/(\varepsilon-1)} \to 1$ for any fixed $N$. The JCC (I.1) becomes independent of $N$:

$$\frac{z}{\mu} = \Phi(\theta),$$

which determines a unique $\theta^*$ independently of $N$. The JCC is then a vertical line in $(\theta, N)$ space; it crosses the strictly increasing $N_{RC}(\theta)$ exactly once. Hence a unique equilibrium obtains.

*Appendix I.1. Details of comparative statics*

Comment out this section temporarily

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